

# PHILIPPINE ARMY MANUAL 3-0502

## ARMOR OPERATIONS

Proponent: Mechanized Infantry Division, Philippine Army

Issued: _____

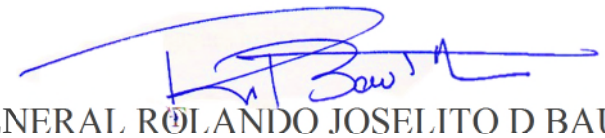
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Philippine Army Manual 3-0502 (PAM 3-0502) dated 12 June 2018 is promulgated on authority of the Commanding General, Philippine Army



LT GENERAL ROLANDO JOSELITO D BAUTISTA
Commanding General, Philippine Army

## PREFACE

1. The purpose of this manual is to provide guidance for armor operations. Specifically, this will provide guidance in the conduct of armor operations in the Philippine Army.

2. This manual contains information on the command and control; offensive operations; defensive operations; other tactical operations; combat support and combat service support of an armor unit. This is highly applicable in both conventional and internal security operations.

3. The proponent of this manual is Mechanized Infantry Division, Philippine Army. Users are invited to submit recommendations for amendments, comments, and suggested improvements to this publication to Director, Doctrine Center, TRADOC, PA at Camp O'Donnell, Sta Lucia, Capas, Tarlac 2315.

4. All publications, manuals and directives inconsistent with this manual are hereby rescinded.

5. Unless this publication states otherwise, masculine nouns and pronouns do not refer exclusively to men.

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# Chapter I
# Introduction

0101. The armor has been playing a significant role in the past and current operations of the Philippine Army. As a component of the combat arms, the armor is normally attached to the infantry battalions and brigades. Because of their mobility, firepower and flexibility, armor units are usually employed for maneuver and economy of force operations.

0102. The history of the armor in the Philippine Army traces its roots to what is known as the Regiment of Lancers, a component of the colonial army who served as one of the elite units that descended from a squadron of Dragoons of Luzon. Soon after, a unit known as the *Guardia a Caballo* (Guards on Horseback) was created in 1755 against the Spanish colonial forces. During the first Philippine revolutionary government, on 23 June 1898 the Cavalry Corps was formally created by Gen. Emilio Aguinaldo as one of the principal combat arms.

0103. During the American occupation in 1922, the 26th Cavalry, Philippine Scouts was organized at Fort Stotsenberg (later Clark Air Base and now Clark Special Economic Zone), Pampanga. However, the 26th Cavalry lost to a stronger force after persevering heroically during the defense of Bataan in World War II. In 01 March 1952, the postwar 1st Cavalry Squadron was organized under Col. Jacobo Zobel with the prewar Cavalry Regiment personnel recalled from the different units to comprise this fledgling organization. The first armored vehicle used by the Filipino was the World War I vintage 6.5-ton Renault Ft 17. The squadron was re-designated as the 1st Light Armor Battalion, 1ID and other similar units were activated in other divisions and separate brigades of the Army in 01 April 1974.

0104. Philippine Army Light Armor Regiment (PALAR) integrated all the existing Light Armor Battalions and Separate Companies of the Army on 16 August 1976. On 19 March 1986, the unit was upgraded and was officially designated as the Light Armor Brigade or LABde. The Light Armor Brigade became Light Armor Division on 11 September 2006 and then renamed into Mechanized Infantry Division on 01 April 2013.

## Operational Environment

0105. The country's strategic depth is determined by the land domain where the people, the government, its culture and socio-political and economic infrastructure-that defines it as a nation. As such, the conditions, circumstances, and influences that affect the employment of Army capabilities must be understood and viewed with the following physical and non-physical characteristics:

a. **Archipelagic.** The Philippines is an archipelagic country that consists of around 7,100 islands with an estimated 31,800 kilometers of coastline. It is divided into three main island groups, namely: Luzon, Visayas, and Mindanao.

Each island group has unique physical characteristics. Most of the islands are huge enough to accommodate large populations since their local resources allow for long-term habitation and development. These characteristics of the physical operational environment dictate that most military operations will potentially happen across a non-contiguous land domain with intervening variables such as political, economic, cultural, and other social factors.

b. **Disaster-prone.** The Philippines is prone to natural calamities due to its geographic location such as being situated along the "Pacific Ring of Fire" and "Pacific Typhoon Belt." This makes the Philippines a consistent witness to severe weather conditions as well as earth-shaking volcanic and seismic activities. Humanitarian Assistance and Disaster Relief (HADR) will continue to compete with the PA traditional task of warfighting.

c. **Highly Urbanized.** According to a recent population study, Metro Manila will be considered as one of the mega-cities in Asia by 2025 with a projected population of 14.8 million. Such will be the trend in other metropolitan areas of the country as well. Protecting population and infrastructures in a high-density urban terrain from the impact of war is a high probability in the future.

d. **Universally Connected.** Global economic, social, and communications systems are becoming highly interconnected. Information is disseminated immediately through various means. The transmission of information at high speed amplifies the interaction among people and organizations on a global scale. This highly connected environment makes the conduct of army operations equally advantageous and precarious.

e. **Politically, Economically, and Socially Constrained**. The political and social dynamics primarily define the direction of national priorities. The economic capacity of the nation influences what it may and may not do. Currently, economic pressures and other constraints curtail the nation's potentials.

f. **Collective Environment**. National security is a shared responsibility of government and non-government agencies, organizations, and partner nations. Internally, a collective environment would mean that both government and non-government sectors have a certain degree of shared interests, concerns, and responsibilities – which are manifested through joint efforts of various sectors of society. Externally, a collective environment is composed of the shared concerns of nations and how they forge partnerships for the common good. Therefore, a collective environment requires that all the sectors involved must have the capacity for interoperability.

g. **Culturally Diverse.** The diversity of the country's demography, as an effect of the Philippines being an archipelago and its long history of interaction with other countries, contributed to the uniqueness of the Philippine culture. In fact, local sub-cultures are distinct in every region in the country. The Philippine populace is composed of a multitude of ethnic groups that have their

respective cultural practices, norms and traditions. With these, the culture of the people in a particular region impacts on the conduct of military operations as far as establishing legitimacy and generating popular support is concerned.

# Chapter II
# Fundamentals of Armor Operations

## Roles of Armor

0201. The roles of armor units are the following:

a. Conduct offensive tasks

b. Conduct pursuit and mobile operations

c. Protection

d. Support of infantry and other branch

e. Reconnaissance

0202. **Offensive Tasks.** The armor can harass and raid a strong enemy. They can contain or destroy a lightly equipped guerilla or a disorganized enemy. They may be employed:

a. As part of a covering force, an advance guard, a screen force, a flank guard, or rearguard

b. As part of a counter penetration force

c. As part of a counter attack force

d. To secure obstacles and defiles ahead of an advancing force

0203. **Pursuit.** Because of its mobility, flexibility and firepower, armor units are well-equipped for pursuit, particularly in relatively open terrain.

0204. **Protection.** The armor may be employed to provide protection for a formation. This is achieved by deploying armor units to give early warning of the enemy's approach. They can also delay the enemy, if required, provided they have the necessary artillery and/or close air support. To impose a significant delay by night or in closed country, infantry support will be required.

0205. **Support of Infantry.** The armor can give close fire support to infantry. However, armored vehicles need covering fire to neutralize anti-tank weapons, and in closed terrain, an infantry escort to destroy anti-tank weapons may be needed to supplement the assault troopers.

0206. **Reconnaissance.** Armor units can undertake reconnaissance tasks ranging from medium reconnaissance in open terrain to close reconnaissance in closed terrain. They can perform both mounted and dismounted reconnaissance.

0207. **Counter-insurgency Operations.** The roles of armor in counterinsurgency operations are:

a. Ground and air reconnaissance operations

b. Offensive, defensive or retrograde operations in an economy of force role

c. Support infantry division/brigade/battalions engaged in stability operations

d. Force security

e. Rear area operations

f. Flexible and mobile reserve operations for the light infantry tasks

g. Force, brigade or division to provide rapid response to enemy mounted forces

h. Combat terrorism

i. Shows of force/demonstrations

j. Security of vital installations

0208. Said operations are conducted in line with the counter-insurgency (COIN) strategy of the Army. Armor forces may conduct Internal Security Operation (ISO) independently or as part of a task force or a joint headquarters with the police and/or other peacekeeping units.

0209. Other operations that armor units may conduct are:

a. Emergency relief operations

b. Noncombatant evacuation

c. Peacekeeping and peace enforcement

d. VIP security

e. Convoy escort/security

f. Log run/re-supply

g. Augmentation of military detachments and check points

0210. Said operations are conducted to support COIN operations involving civic-oriented activities, peacekeeping missions and other security tasks not necessarily supporting COIN operations.

## Organization of Armor

0211. The armor is a combined arms organization composed of tank, cavalry, and mechanized infantry. It can be organized into a tank unit or mechanized unit with cavalry. Moreover, both the tank and mechanized units can be designated/task organized to form a combined task force to address specific missions. Likewise, the armor units may be attached to an infantry unit to accomplish specific missions.

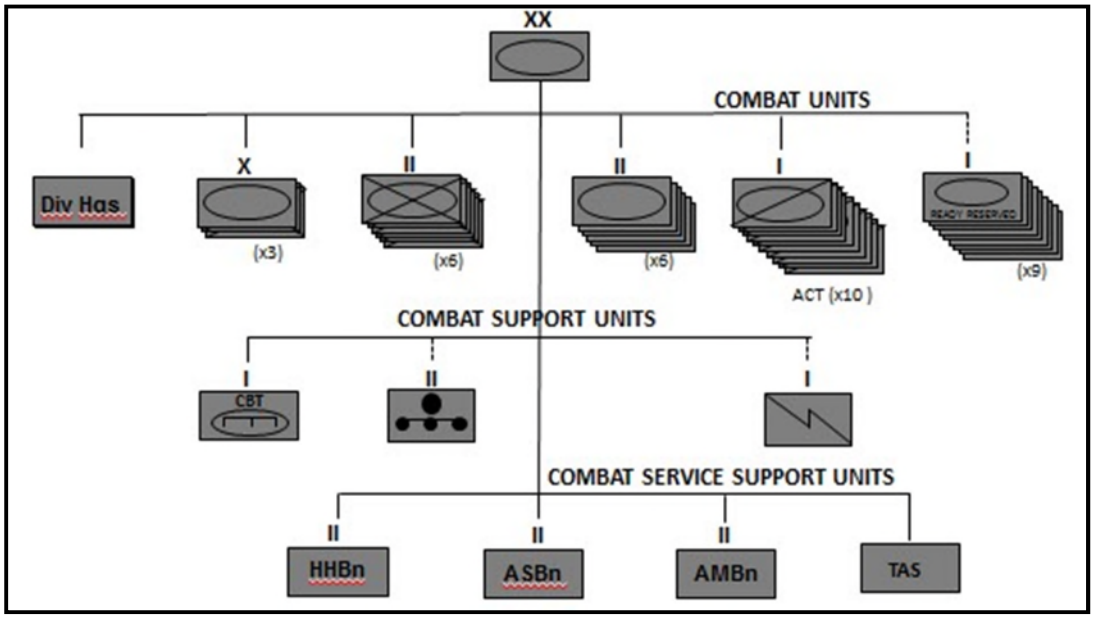



**Figure 2.1 – Armor Division**

0212. The organization of armor and its organizational structure varies with the armor equipment and its composition of troops.

0213. The mission of armor is to close with and neutralize enemy forces by fire, maneuver and shock effect in coordination with other arms. They may also conduct Internal Security Operations (ISO) activities, independently or as part of a task force in peacetime environment as required by higher headquarters.

0214. Capabilities of the armor are the following:

a. Deploy combined arms forces in the conduct of ground operations.

b. Win battles and engagements at operational and tactical level.

c. Conduct offensive and defensive operations across the full operational continuum, independently or as part of a larger force.

d. Provide operational commanders with increased operational and tactical flexibility; rapidly deployable (i.e., air assault and amphibious capability); dismount strength for close combat in urban and complex environments, mounted capability for rapid mobility, firepower, and shock effect; reconnaissance capability

e. Provide situational awareness and common operational picture down to the company level.

f. Conduct operations in support to civil authorities.

## Tank units

0215. The tank battalion can deploy as an organic unit, as companies, or platoons task organized to infantry task forces (TFs). The battalion may move into a theater where the majority of an infantry division is deployed. The tank units may be organized as a task force when a large enemy armor threat exists and/or escalation of hostilities is expected. When the battalion functions as a task force, its basic employment is to counter enemy armor threats until heavier forces arrive.

0216. The tank battalion is organized into three tank companies and a headquarters company.

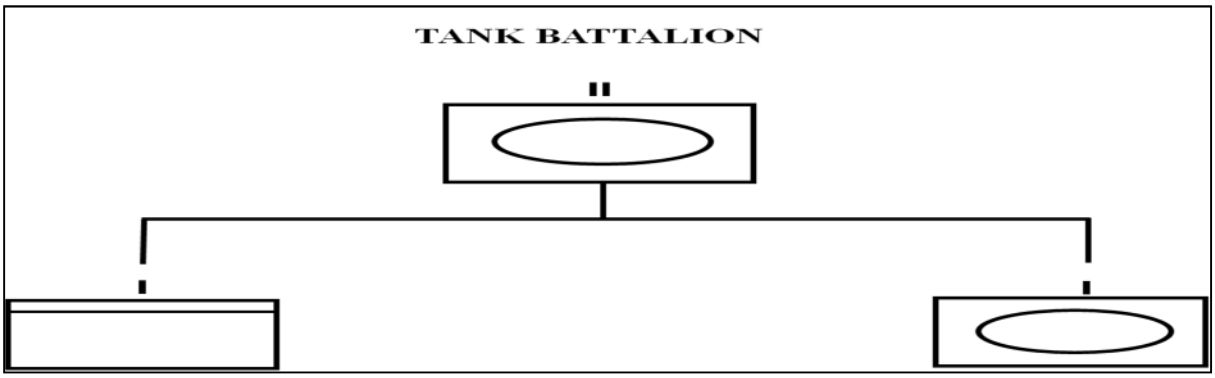



**Figure 2.2 - Tank Battalion**

0217. The tank battalion can operate as a pure armor or deployed in echelons as companies and platoons attached to infantry brigades and battalions, respectively. If the entire infantry division is deployed, a part of the battalion may consolidate once in theater. The battalion headquarters then forms the nucleus of a tank battalion TF.

0218. The tank battalion provides the commander with the following:

a. A highly mobile, protected potent ground combat force.

b. Shock effect.

c. Effective anti-armor capability.

d. Bunker- and building-busting capability and direct and indirect close-in fire support (FS) for dismounted infantry.

0219. The tank battalion may conduct operations as a battalion TF under the following conditions:

a. The enemy has a considerable mechanized or armor force.

b. A contingency mission has matured to a level in which the entire division has deployed to include the tank battalion.

c. Terrain favors the use of a larger maneuver force and/or supports long-range fires.

## Mechanized Infantry Battalion

0220. The Mechanized Infantry Battalion is composed of a headquarters company and three mechanized infantry companies. The battalion headquarters exercises control and supervision over the scout platoon, anti-armor platoon, mortar platoon, maintenance platoon, medical platoon, signal platoon, and armor support platoon.

0221. A Mechanized Infantry Battalion provides the combined arms of armor forces with a strength capable of fighting mounted or dismounted. It possesses high cross country mobility, light armor, and multiple means of communication, and complements and enhances the inherent capability of tank elements when employed as a part of a tank-infantry team.

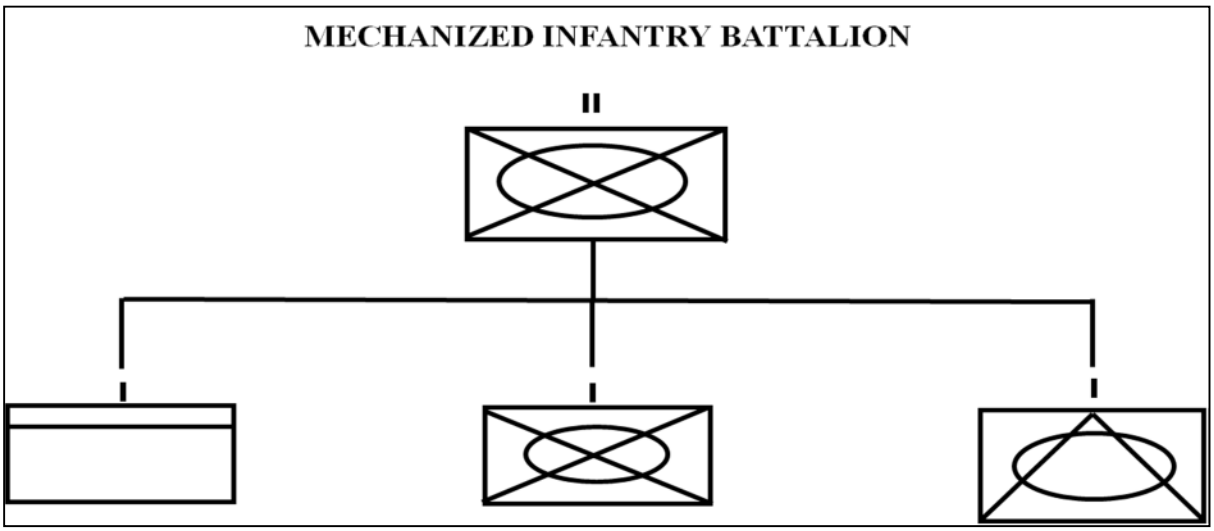



**Figure 2.3 – Mechanized Infantry Battalion**

## Cavalry Units

0222. An Armored Cavalry Troop consists of a troop headquarters, two mechanized platoons, and two armored platoons. The troop is organized, equipped, and trained to

protect and preserve the fighting ability of other combined forces. While the primary mission is reconnaissance and security, the cavalry troop may be called upon to execute attack, defend and delay missions as part of squadron and regimental missions. The troop accomplishes its missions by communicating, moving and shooting consecutively. The cavalry troop usually consists of the heavy troop and light troop.

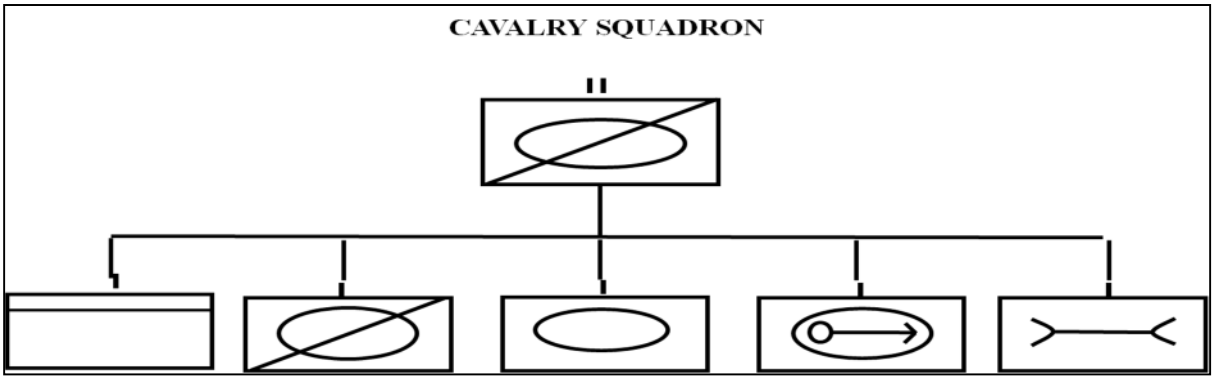



**Figure 2.4 - Armored Cavalry Troop**

0223. To determine the best organization of combat force available to accomplish an assigned mission, the troop commander considers the mission, enemy, terrain and weather, and troops available. It normally employs the organized armored cavalry platoons. In some situations, it may reinforce one platoon with elements of another or form provisional platoons. The armored cavalry troop normally operates without attachments.

# Types of Armor Operations

## Tank operations

0224. The land warfare strategy of the Philippine Army is evolving. The military forces have developed a strategy of rapidly projecting combat power to protect the national interest. The Army now focuses on deploying and fighting as part of contingency and reinforcing forces. Tank units give the Army a versatile, deployable, and lethal force structure that can operate with infantry-based forces nationwide. It can use its unique capabilities to conduct combat operations, often in support of contingency plans, across the operational continuum (peacetime, conflict, and war). It can also be required to operate in a wide range of political, military, and geographical environments. Its tactical missions include providing security, and anti-armor firepower to the infantry division or special operating troops, as well as standard armor operations to engage and destroy enemy forces using mobility, firepower, and shock effect in coordination with other combat arms. These missions may require rapid strategic and tactical deployment nationwide.

0225. Tank units may also be required to support operations other than war, such as:

a. Insurgency/counterinsurgency

b. Combating terrorism

c. Shows of force and demonstrations

d. Strikes and raids

e. Peacekeeping and peace enforcement

f. Rapid reinforcement of forward-deployed forces

g. Wartime contingency operations

0226. Tanks units have the capability to:

a. Support the close fight as part of a combined arms team

b. Use strategic and tactical mobility to advantage

c. Provide armor protection against small arms, machine gun, and overhead artillery fire

d. Detach quickly from their parent unit and be employed during the initial stages of contingency or reinforcing operations

e. Accept routine attachment of operational control of engineer assets and light infantry or anti-armor companies, field and air defense artillery batteries;

f. Conduct operations with light infantry and special operations units

g. Accomplish rapid movement and limited penetrations

h. Exploit success and pursue defeated enemy elements as part of a larger force

0227. The limitations of tank units include the following:

a. Mobility and firepower are restricted in extremely close and non-trafficable terrain

b. Tank assets are vulnerable to enemy infantry with anti-armor weapons and may not withstand the impact of enemy tank and artillery fires and anti-tank mines

c. Consumption of supply items is moderate to high in Classes III, V and IX

## Mechanized Infantry Operations

0228. The mechanized infantry will often fight in conjunction with tank units to form a combined arms team. The armored vehicle provides the infantry with mobility but the armored vehicle for mechanized infantry has less firepower and protection compared to tank units.

0229. Mechanized infantry complements tanks by providing a dismount capability that enables the unit to reduce obstacles and to hold ground.

0230. Mechanized and tank units are organized to fight successful engagements in conventional and various operations. They are subordinate commands of a division and perform major tactical operations as part of a division operation. Regardless of the threatening environment, the key to victory is to impose the will on the enemy by forcing him to conform to the unit’s desired end state. This requires the commander and staff to identify the decisive point(s) and synchronize the efforts of subordinate maneuver battalions, combat support, combat service support, and available higher headquarters' combat power in support of the unit effort.

0231. Mechanized units can be employed as follows:

a. A combination of tank and mechanized infantry

b. Oftentimes organized as a task force

c. Occasionally composed of a light infantry battalion and other supporting units grouped under the command of a higher headquarters

d. Task-organized as directed

0232. Mechanized infantry capabilities are the following:

a. Conduct sustained combat operations with proper augmentation

b. Accomplishes rapid movement and deep penetrations

c. Exploits success and pursues a defeated enemy as part of a larger formation

d. Conducts security operations for a larger force

e. Conduct defensive, retrograde, and other operations

f. Conduct offensive operations

g. Conduct operations with light and special operations forces

h. Deploys rapidly into pre-positioned equipment

i. When mounted: 1) Suppress and kill dismounted elements 2) Secure terrain 3) Maneuver with tanks 4) Assault unprepared, suppressed defense

j. When dismounted: 1) Fix the enemy to allow maneuver 2) Infiltrate enemy positions 3) Secure obstacles to defeat enemy breaching efforts 4) Move over terrain not trafficable by vehicles 5) Defend versus dismounted attack 6) Seize terrain 7) Destroy tanks

0233. Mechanized Infantry limitations are the following:

a. Mobility and firepower are limited by: 1) Urban areas 2) Dense jungles and forests 3) Very steep and rugged terrain 4) Significant water obstacles

b. Strategic mobility is limited by the availability of heavy lift equipment

c. Consumption of classes III, V, and IX supplies is extremely high

d. Armored Infantry Fighting Vehicles (AIFVs), M113s, Simba Fighting Vehicles (SFVs) and Commando V-150 are vulnerable to high-powered armaments

e. Dismounted infantry are vulnerable to small arms and indirect fire

f. Infantrymen need sufficient time to mount and dismount

## Cavalry Operations

0234. The fundamental purpose of the cavalry is to perform reconnaissance and to provide security in close operations. In doing so, the cavalry facilitates the division or brigade commander's ability to maneuver brigades and battalions and to concentrate superior combat power and apply it against the enemy at the decisive time and point.

0235. The cavalry is, by its role, an economy of force. The flexible capabilities of the cavalry allow the commander to conserve the combat power of divisions or brigades for engagement where he desires. The combat power of Cavalry units, in particular, makes them ideal for offensive and defensive missions as an economy of force.

0236. The cavalry has served historically as a flexible multipurpose force. Capitalizing upon a significant mobility advantage over infantry, the cavalry performed long-range reconnaissance and security for commanders. These missions give commanders the ability to maneuver and concentrate forces on a battlefield for decisive battle.

0237. Cavalry is organized into four (4) basic designs:

a. Armored Division Cavalry Squadron

b. Infantry Division Cavalry Squadron. (Current Setup: OPCON to Infantry Divisions/Area Commands)

c. Air Cavalry Squadron. (Current Setup: Aviation Battalion – Refer to Aviation Battalion's Reference Manuals/ SOP's/ TOE's/ Task Organizations)

d. Separate Cavalry Troop. (Current Setup: Armor Cavalry Troop (S) – Refer to Reference Manuals/ SOP's/ TOE's/ Task Organizations)

0238. The cavalry serves as a catalyst that transforms the concepts of maneuver warfare into a battlefield capability. Maneuver is the essence of Army fighting doctrine. The successful execution of maneuver warfare continues to be the product of thorough reconnaissance and continual security. As the "eyes and ears" of the commander, the cavalry provides the commander with situational awareness and enhances his ability to maneuver successfully. Once on the chosen field, the cavalry continues to play key roles such as: close security to protect the flanks or rear of the infantry line; countering enemy cavalry; counterattacking enemy infantry attacks; reserve; administering the decisive blow to a faltering enemy; covering retreat; pursuing a retreating enemy; and, close reconnaissance to detect enemy weaknesses.

0239. The cavalry has decisive advantages over other intelligence resources because it, works through and counters enemy deception efforts better than any sensor system; provides the fastest, most reliable means of assessing terrain that the enemy is trying to configure to his advantage; and is not a passive source. The cavalry does not only finds the enemy but can further develop the situation and force the enemy to reveal more information; and, can more effectively disseminate information to commanders with an immediate need (eavesdrop, liaison officer).

0240. At the decisive point, Cavalry guides maneuver units into engagements with the enemy, assists in rapidly massing and dispersing maneuver units, controls routes and choke points, and monitors the movement of combat support and combat service support units.

0241. **Cavalry Capabilities.** In terrain suitable for employed of cavalry units, commanders can apply all their inherent combined arms mobility, firepower and shock effect to:

a. conduct reconnaissance and surveillance

b. conduct security operations

c. operate in an economy of force role without being reinforced in offensive, defensive, or delay operations

d. conduct continuous operations

e. move to contact independently, or as part of the advance covering force, or as the flank or rear guard of a larger force

0242. **Cavalry Limitations.** Mobility can be restricted by dense forest, steep, rugged terrain and major water obstacles, whether natural or man-made. Heavy logistics support is required.

## Armored Vehicle (AV) Fighting Positions

0243. Armored Vehicle (AV) commanders choose from the following fighting positions that offer the best possible observation and fields of fire. These positions use any cover and concealment available.

0244. **Hull Down.** This is the basic fighting position. Weapons that are mounted on the vehicles can be fired up front. The vehicle's hull is protected from direct fire from the front.

0245. **Turret Down.** This position is used by AVs when observing. Only the tank commander's head is exposed. The tank is buttoned up.

0246. **Periscope Up.** This is a variation of the turret down where the AV commander or gunner can see only over the crest through his periscope. He can lay on the target and fire the moment the tank has moved forward to allow crest clearance. This technique allows maximum protection and concealment for as long as possible.

0247. **Trunnion Cant.** This position may be a problem to hull down tanks if they don't have the cant correcting devices. One side of the AV may be raised with stones and logs to level the gun trunnions.

0248. **Hide Positions.** Hide position conceals armored vehicle before battle. It covers the AV completely from enemy direct fire and ground observation.

0249. Any position is relative to the enemy's point of view. It is possible to be turret down to an enemy observer at 1,000 meters, but almost hulled up to a position 3,000 meters away viewed from a high ground. The enemy could be anywhere.

0250. **Occupying a Position.** Armored vehicle commanders must learn to recognize suitable hull down and turret down positions from a distance, and to move in unnoticeably without delay.

0251. When time and earth-moving equipment are available, firing positions should be improved. Where there is scarce natural cover, sandbags may be placed to provide protection. In wide open terrain, hull down positions can be scooped out of the ground.

0252. **Giving away positions.** An AV is more likely to be noticed and engaged if: The armored vehicle commander chooses a position near a landmark which may be adjusted as an artillery or mortar task which is noted on the enemy range card. An armored vehicle gives away positions by making dust, or through exhaust smoke, or by poor use of cover and concealment. Even small movements attract attention and

could result in the enemy making a more detailed search. Slow movement is better for concealment than fast movement.

0253. **Tactical positions at night.** In general Positions at night must be chosen for the same reason as during daytime. However two, factors may dictate some modifications to this general rule.

0254. **Type of engagement**. To avoid backscatter, it is better to move to a hull up position for a search light engagement, whereas a hull down vehicle is less likely to be silhouetted during a flare engagement.

0255. **Range of engagement.** Ranges may not be as long by night as by day, and this may enable a vehicle to be positioned in low ground.

0256. **Tone and background.** A vehicle positioned close to good background will not be silhouetted in enemy white light. Likewise, a vehicle stands out at night as a dark mass if it's positioned in front of a light tone background.

0257. **Jockeying.** Once a vehicle has fired, it will be seen more likely by other enemy elements from a flank or further back. After an engagement, or a maximum of three rounds, the commander must "jockey" in his position by reversing out of his fire position and coming up in another. He must see that the tank moves down behind cover before moving to its new position.

## Armored Vehicle Movement Formations

0258. Formations are used to establish AV positions and sectors of responsibility during operations. Formations facilitate control; increase protection, speed, and fire effectiveness; and alleviate confusion. Formations are not intended to be rigid; vehicles are not expected to be a specific distance apart. Position of each AV in the formation depends on the terrain and the wingman driver's ability to see the lead armored vehicle.

0259. Individual armored vehicle should move in the same relative position within the formation. This will ensure that each crew knows where to move, who is behind them, and where to observe and direct fires. Gun tube orientation for rear and flank vehicles should be modified to ensure 360-degree security based on the position of the unit within the parent unit formation. The seven basic formations are:

0260. **Wedge.** The wedge is employed when an armor platoon is provided to overwatch by another armor unit/s and the terrain is open or rolling. It is normally used when the enemy situation is vague and contact is imminent. The wedge has these advantages:

a. It permits excellent fire to the front and good fire to each flank when leading armor formations.

b. It allows platoon leaders excellent observation up front while being covered by their wingmen.

c. It enhances control since leaders are in close contact and can easily communicate with each other and any following armor formations.

0261. The wedge has these disadvantages:

a. It requires lateral space for movement; therefore, it is difficult to use in closed terrain with dismounted infantry.

b. It may expose the entire platoon to enemy fire simultaneously.

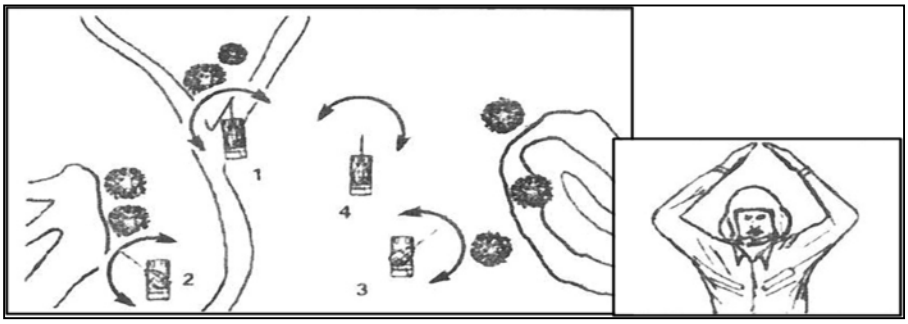



**Figure 2.5 - Wedge Formation**

0262. **Vee.** This formation is used when the weather or terrain restricts movement or when overwatch within the platoon is required.

0263. The vee has these advantages:

a. It provides excellent protection.

b. It provides excellent control.

c. It facilitates rapid deployment to any other formation.

0264. The vee has these disadvantages:

a. It limits fires to the front.

b. It is more difficult to maintain orientation than in a wedge.

c. It provides less control in wooded areas.

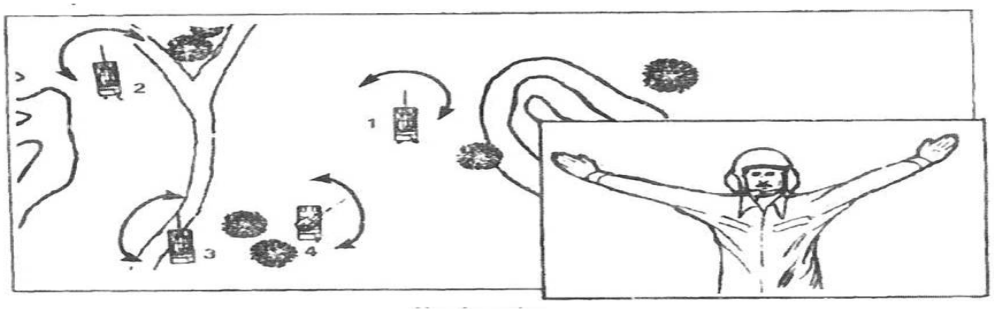



**Figure 2.6 - Vee Formation**

0265. **Echelon.** The echelon is used to screen an exposed flank of a larger moving force.

0266. Echelons have these advantages:

a. They provide excellent firepower to the front and one flank.

b. They provide the best security to the higher unit formation.

c. They facilitate rapid deployment perpendicular to the direction of movement.

0267. The echelon has these disadvantages:

a. It is difficult to control in wooded terrain.

b. It is difficult to integrate into an infantry formation.

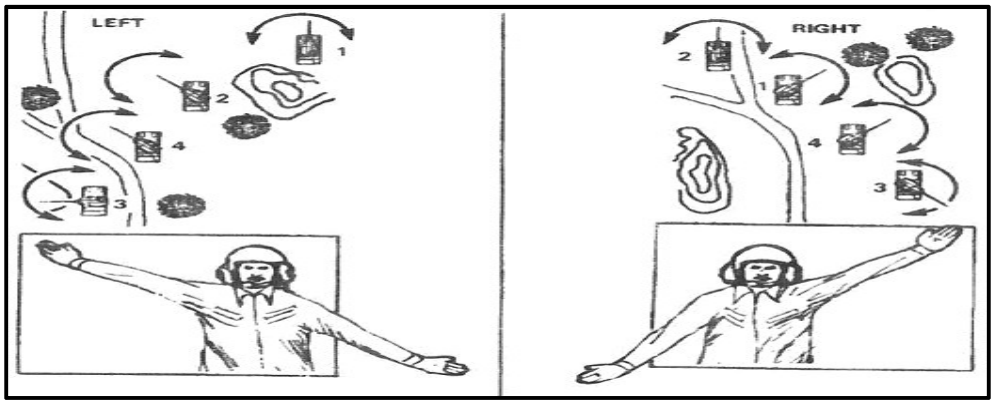



**Figure 2.7 - Echelon Formation**

0268. **Line.** A line formation is used to cross dangerous areas and assault a position. It also facilitates mutual support when emerging from limited visibility conditions such as smoke or heavy woods.

0269. The line has these advantages:

a. It provides excellent firepower forward.

b. It provides protection to dismounted infantry.

c. Maximum vehicles can close on an objective in minimum time.

0270. The line has these disadvantages:

a. It provides minimum fire to flanks.

b. It is less secure than other formations because of lack of depth.

c. It is the most difficult to control.

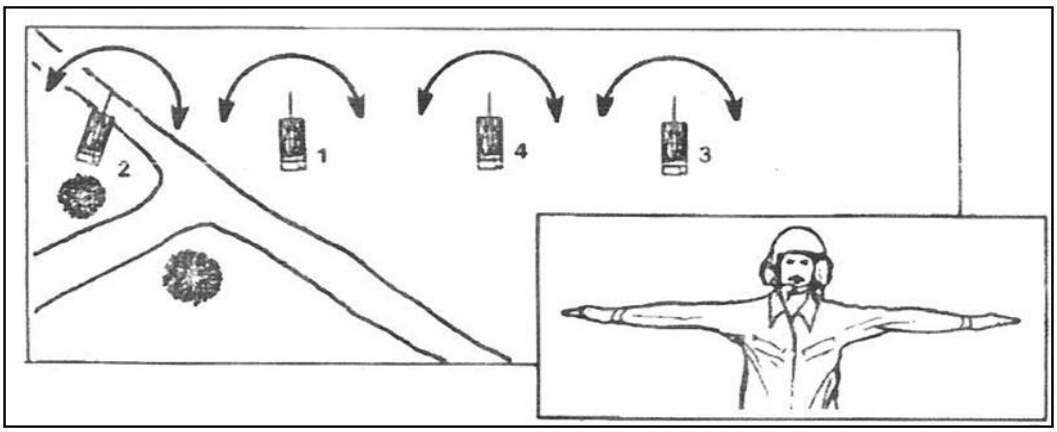



**Figure 2.8 - Line Formation**

0271. **Column/Staggered Column.** In these formations, the unit commander positions himself where he can best control his elements. If he does not lead, he must ensure the lead vehicle commander is thoroughly familiar with the route of march and direction of travel. The column is used in night movements, in fog, when passing through defiles or dense woods, and during road marches when speed is required. The staggered column is used when terrain allows for dispersion.

0272. The column or staggered column has these advantages:

a. It provides excellent control.

b. It provides excellent firepower to the flanks.

c. It facilitates rapid deployment to other formations.

d. It facilitates rapid movement.

e. The dispersion increases security against enemy air and artillery attack.

0273. A disadvantage of the column and staggered column is that they allow little fire to the front.

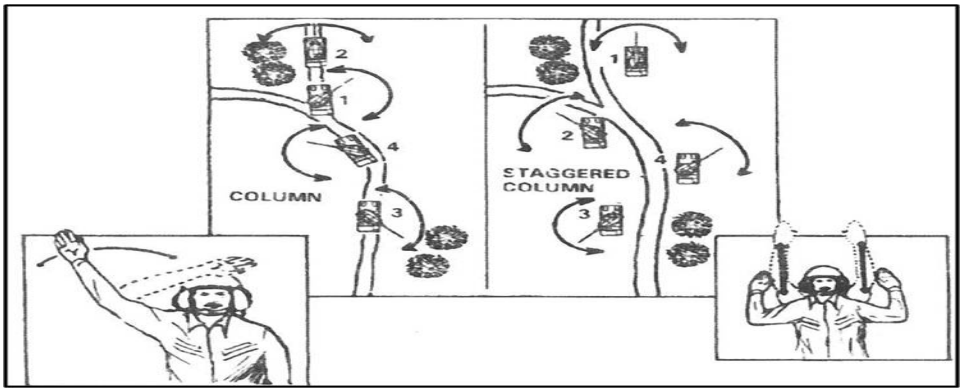



**Figure 2.9 - Column and Staggered column**

0274. **Coil.** The armor commander can employ this formation when it is operating independently and experiences extended halts or lulls in combat. The armor commander positions his AVs, and the remaining vehicle commander position their AVs based on the terrain.

0275. The coil has these advantages:

a. It provides good all-around security.

b. It facilitates expanding to a perimeter defense.

c. It provides protection to dismounted elements in open terrain.

0276. A disadvantage of the coil is that without dismounted elements it offers only limited security in heavily wooded terrain.

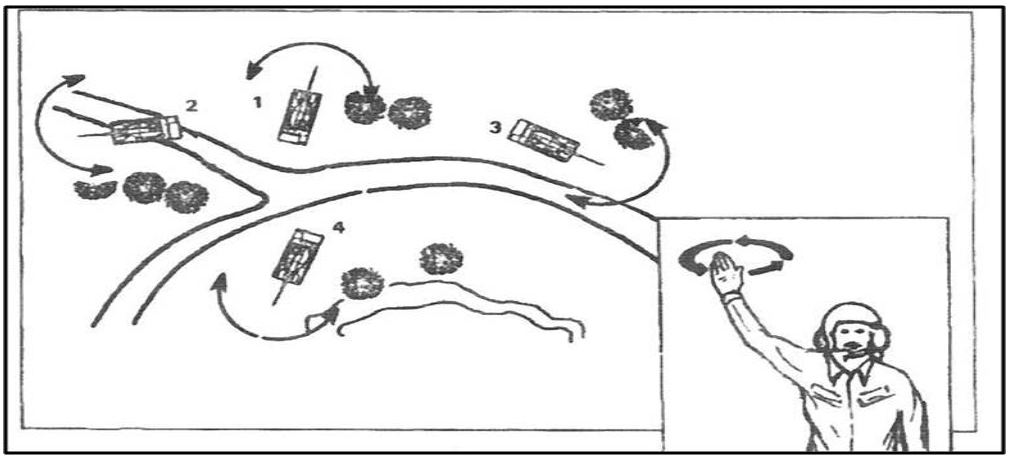


**Figure 2.10 - Coil Formation**

0277. **Herringbone.** An armor unit uses this formation when it assumes a hasty defensive posture or temporary halt on a road where terrain does not allow adequate

off-road dispersion. Armored vehicles move off the road if the terrain permits. Mechanized Infantry should dismount and seek cover and concealment while providing additional security and observation.

0278. The herringbone has these advantages:

a. Vehicles can pass through the center of the formation.

b. It facilitates dismounting of mechanized infantry prior to further deployment of the mechanized platoon.

0279. The herringbone has these disadvantages:

a. Vehicles may be vulnerable to enemy air attack.

b. Security is limited (without mechanized infantry) and terrain does not allow for dispersion.

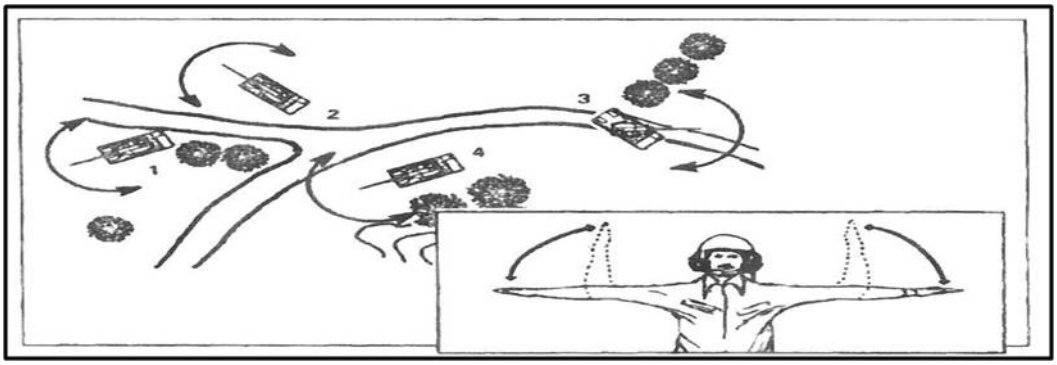



**Figure 2.11 - Herringbone Formation**

0280. **The wingman concept** facilitates command and control of an armor unit for dispersion. It requires that one AV orients on another AV, on either the left or the right. In a platoon setting, the second AV orients the platoon leader's vehicle, the third AV on the platoon sergeant's, armored vehicle and the platoon sergeant's AV on the platoon leader's armored vehicle. In the absence of specific instructions, wingmen move, stop, and shoot along with their leaders. A basic guideline on distances between vehicles should not be less than 50 meters or more than 100 meters. This is dependent on Mission, Enemy, Troops, Terrain and weather, and Time available (METT-T) and may be difficult in very close (jungle) or very open (desert) terrain.

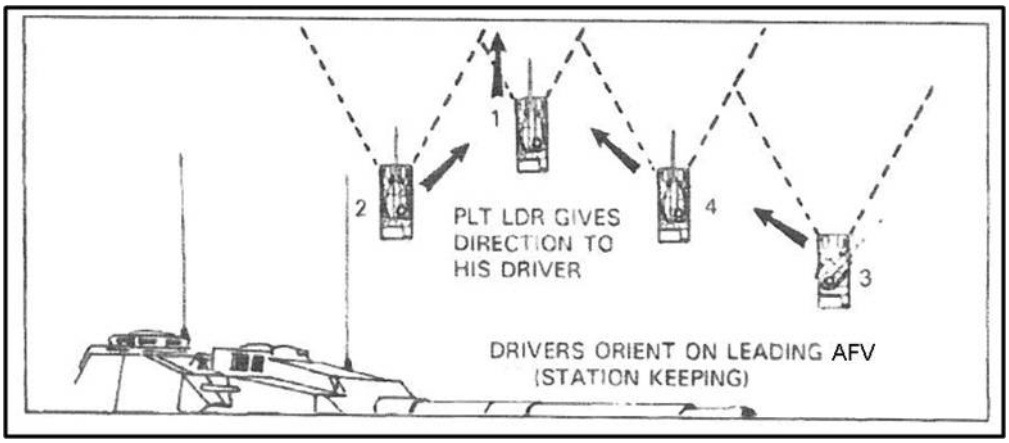



**Figure 2.12 - Wingman Concept**

## Movement Techniques

0281. **Traveling**. Traveling is characterized by continuous movement of all elements and is best suited to situations where enemy contact is unlikely and speed is important. This is the most likely used technique when the mechanized infantry rides on the AV.

0282. **Traveling overwatch.** Traveling over watch is an extended form of traveling that provides additional security when contact is possible but speed is desired. The lead AV moves continuously while the trail AV moves at various speeds and halts periodically to overwatch the movement of the lead AV. The trail AV maintains a minimum distance of 500 meters, depending on terrain, to permit movement in case the lead AV is engaged.

0283. **Bounding overwatch.** Bounding overwatch is used when contact is expected. It is the most secure and slowest armor movement technique. There are two methods of bounding: 1). Alternate bounds. The lead AV halts and assumes overwatch positions. The rear AV advances past the lead AV and assumes overwatch positions. The initial lead AV then advances past the initial rear AV and assumes overwatch positions. Only one AV moves at a time. This method of bounding is usually more rapid than successive bounds. 2). Successive bounds. In this method, the lead AV, covered by the rear AV, advances and assumes overwatch positions. The rear AV advances to a position abreast of the lead AV, halts, and occupies over watch positions. The lead AV then moves to the next position. Only one AV moves at a time, and the rear AV does not advance beyond the lead AV. Successive bounding is easier to control and more secure than alternating bounds.

0284. **Overwatch.** When armor elements are given the task to overwatch, they should occupy positions that offer cover and concealment, good observation positions, and clear fields of fire. Elements occupying over watch positions should 1) Visually check the security of the positions; 2) Occupy hull-down firing positions; 3) Assign sectors of fire; 4) Orient weapons on likely or suspected enemy positions; 5) Search for targets.

0285. Because armor works in proximity with infantry, the armor commander must understand the basic movement techniques of the infantry units that will move forward, behind, and on the flanks of his own unit.

0286. **Methods of Armor/Infantry Movement.** When infantry and armor move together in any operation, the infantry moves using one of the following three methods: dismounted, truck-mounted, or AV-mounted.

0287. **Dismounted.** There are two ways dismounted infantry and armor can move together: armored vehicles lead, followed by the dismounted infantry, or infantry leads and armored vehicles follow. Movement with armored vehicles followed by infantry is used when the terrain is relatively open and the infantry has little or no cover and concealment. Moving behind the armored vehicles provides some protection and concealment from enemy small-arms fire. Dismounted infantry followed by armored vehicle is used in restricted terrain when visibility is limited. The infantry provides security for armored vehicles, clearing lanes or zones in front when fields of fire and observation are limited. Considerations for using dismounted movement include the following:

a. Speed is reduced to that of dismounted elements. If AVs lead, leaders must guard against leaving infantrymen too far behind.

b. Communications between infantry and armor elements can be accomplished through radio or hand-and-arm signals.

c. Pre-arranged signals (flags, smoke, panels, or hand-and-arm) are coordinated and practiced to ensure close team cooperation.

d. The infantry does not move in front of the vehicles unless told to do so.

0288. **Truck Mounted.** In some cases, the infantry/armor team may have trucks, jeeps and other infantry troop carriers from internal or external sources to transport light infantry. This method may be used in several situations:

a. When traveling over long distances to an objective Line of Departure (LD).

b. When speed is important and it outweighs the risk of exposure to enemy fire. Examples include penetration, breakout from encirclement, exploitation, or pursuit.

c. When armor and truck-mounted infantry move together, the armored vehicles usually move in front to provide protection and firepower. One platoon or section may also be placed at the rear of the infantry formation for protection.

0289. **Armored Vehicle (AV) Mounted.** Some of the infantrymen in the infantry/armor team may be transported through the armored vehicles. This occurs mostly during a march, exploitation, or pursuit. This method has the advantage of

maintaining speed while keeping the infantry and armor together. It also saves infantrymen from dismounted marches.

0290. The decision to carry infantry on an armored vehicle requires careful planning. Tactical unity of the infantry and armor teams must be maintained. Infantry leaders should mount armor leaders' vehicles. Usually, an infantry squad (approximately nine (9) infantrymen) can fit in one AV. During planning, each squad should link up with the vehicle it will ride on to allow time for the squad and AV crew to work together and rehearse mounting, dismounting, and action drills.

0291. Infantry platoon leaders should ride on the armor platoon leaders' vehicles, and infantry platoon sergeants should mount on the armor platoon sergeants' vehicles. The infantry leaders should be located next to the vehicle commanders.

0292. The lead armored vehicle should not carry mounted infantrymen. This will allow them to remain free to scan left and right during movement and to return fire immediately if enemy contact is made.

0293. The following are considerations for infantry and armor leaders when mounting infantrymen on the armored vehicle:

a. Always notify the armored vehicle commander before mounting or dismounting.

b. Follow the commands of the armored vehicle commander.

c. Infantry platoons should be broken down into squad-sized groups, similar to airmobile chalks, with the infantry platoon leader on the armor platoon leader's vehicle and the infantry platoon sergeant on the armor platoon sergeant's vehicle.

d. Whenever possible, mount and dismount over the left front slope of the vehicle. This ensures the driver sees infantry and that the infantry does not pass in front of the machine guns.

e. Passengers must always have three points of contact with the armored vehicle and watch for low-hanging objects like tree branches.

f. Passengers should take the following actions on contact: 1) Wait for the armored vehicle to stop. 2) Dismount immediately on the vehicle commander's command (one fire team on each side). Do not move forward of the turret. 3) Move at least 5 meters away from the vehicle.

g. Practice mounting, dismounting, and actions on contact so that their team will be trained.

h. If possible, the lead vehicle should not carry infantry because it restricts turret movement. Initial contact may also cause casualties among infantrymen mounted on the lead vehicle.

i. Infantrymen should search in all directions. They may be able to spot a target the armored vehicle commander does not see.

j. Do not move in front of the AV unless ordered to do so.

k. Do not move off an AV unless ordered to do so.

l. Do not dangle legs, equipment, or anything else on any side of the vehicle; they could get caught in the tracks, causing death, injury, or equipment damage.

m. Do not over-crowd the armored vehicle. This may result in falls, bums, and clogged air intakes.

n. Do not fall asleep. The warm engine may induce drowsiness, and a fall could be fatal.

o. Do not smoke when mounted in an armored vehicle.

p. Do not stand near vehicles during refueling and rearming.

q. Do not stand near a moving or turning armored vehicle at all times.

r. Stay clear of the AV's canister ejection device. Canisters ejected from main gun rounds can cause serious injury or death.

s. Do not mount inside the vehicle with loaded firearms.

## Armor Battle Drills

0294. These drills are basic guides to combined arms tactics. Armor platoons and the supporting infantry should be able to execute the drills upon command. Repetition in training, conducted along standards, is the key to the proper execution of drills in combat. When habitual relationships are maintained, the execution of such drills is enhanced. If habitual task organization is not possible, rehearsals are keys to proper performance.

0295. This discussion describes the drills that each armor commander must become familiar with and be able to execute, whether working alone or with light infantry. Each drill shows light infantry moving dismounted with the platoon, as well as mounted whenever applicable. The unit must practice drills to prepare for the C2 problems inherent in battle. These drills teach unit members virtually automatic responses to combat situations, outlining actions to be taken immediately upon contact or in response to brief oral commands or visual signals. Crews and units gain

proficiency only through practice before the battle. The drills can be carried out from any formation or movement technique.

0296. **Actions on Contact.** When enemy fire is encountered, the armor commander should execute the following actions on contact:

a. Return fire and alert the rest of the unit.

b. Initiate a battle drill (action drill or contact drill). If no drill is specified, the unit should seek cover and concealment.

c. Send a contact report to the unit commander.

d. Develop the situation through fire and movement to determine the size, type, and location of enemy forces.

e. Send a spot report to the Armor Commander. The engaged unit may destroy the target if initial fire is effective, or it may continue fire and movement to fix or destroy the enemy (based on instructions from the higher commander).

0297. The armor units are expected to execute any of the following standard battle drills:

a. Change of Formation Drill

b. Action Drill

c. React to Indirect Fire Drill

d. React to Air Attack Drill

0298. **Change of Formation Drill.** To accomplish a rapid change of formation, each armor vehicle commander must know the following information:

a. The new formation.

b. The relative position of each armored vehicle in the new formation.

c. The position of infantrymen if they are moving dismounted within the formation.

0299. The armor commander can use hand-and-arm signals or the radio to inform his elements of the new formation. He should always use hand and arm signals when dismounted infantrymen are in the vicinity, regardless of the method used to communicate to the other vehicles. Each vehicle commander will know his position in the formation by following an SOP that specifies armor vehicle positions when assuming each formation. The key to a successful change is practice.

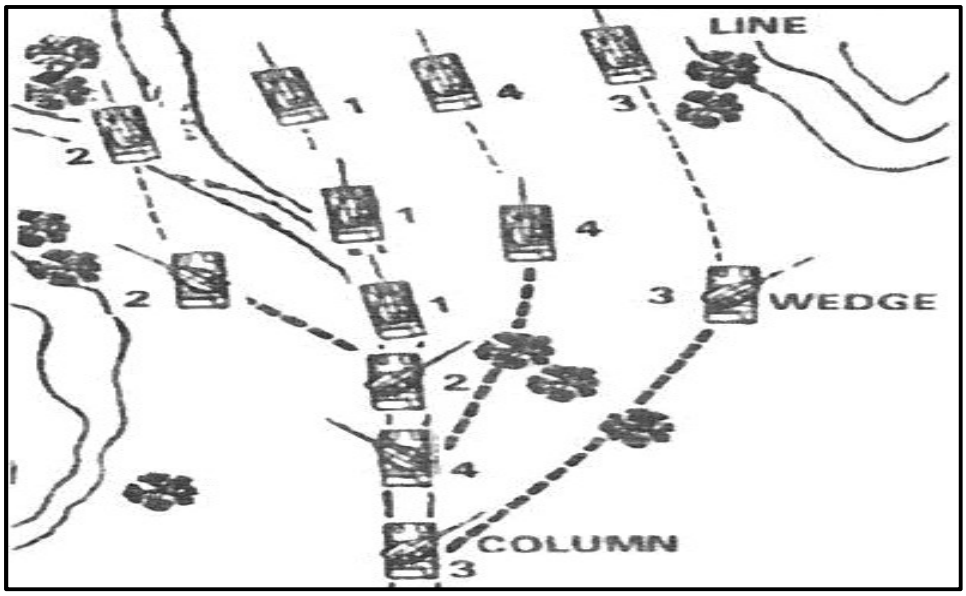



**Figure 2.13 - Change of Formation Drill**

02100. **Contact Drill.** Contact drills teach the armor unit how to orient weapon systems and engage an enemy without changing the direction or speed of movement along the axis of advance. This can be used when contact is made with small arms fire or when the platoon sights the enemy without being detected and does not want to stop or slow its momentum. The armor commander can initiate a contact drill by radio or hand-and-arm signals. Even when using the radio, he should also use hand-and-arm signals when moving with dismounted infantrymen.

02101. **Action Drill.** Action drills permit the armor unit to change direction when reacting to changes in terrain or during enemy contact.

a. **Changes in Terrain.** Using the wingman concept, this change occurs automatically when the lead vehicle changes direction. To speed up a change in direction, the commander can direct an action drill using hand-and-arm signals or the radio. After executing the change of direction, the elements in a unit automatically come on line and continue to move. To return to the original formation, a hand-and-arm signal or radio command can be given.

b. **Enemy Contact**. Following a contact report involving anti-tank weapon systems, the armor commander can direct an action drill to orient the unit's frontal armor toward the anti-tank fire while moving to cover and concealment.

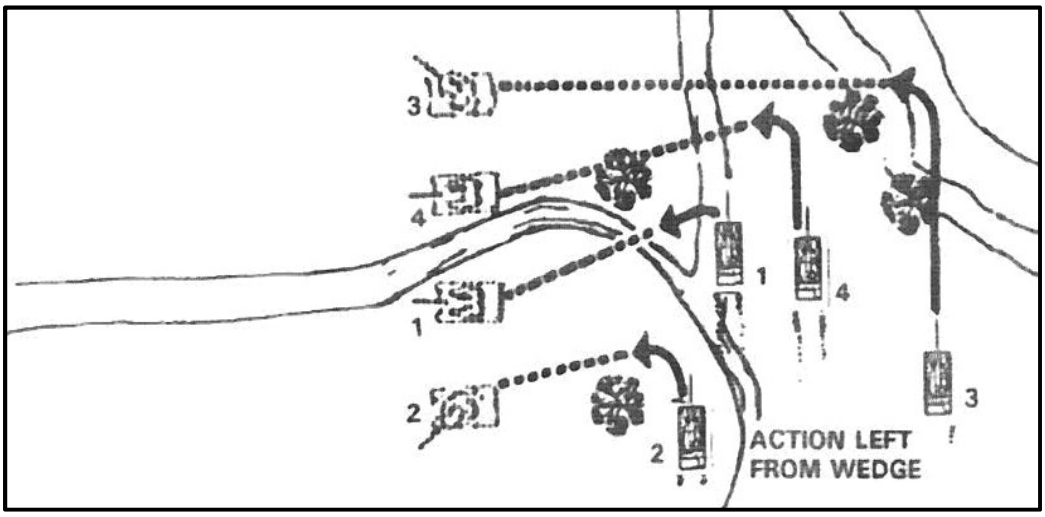



**Figure 2.14 - Action Left Drill from Wedge**

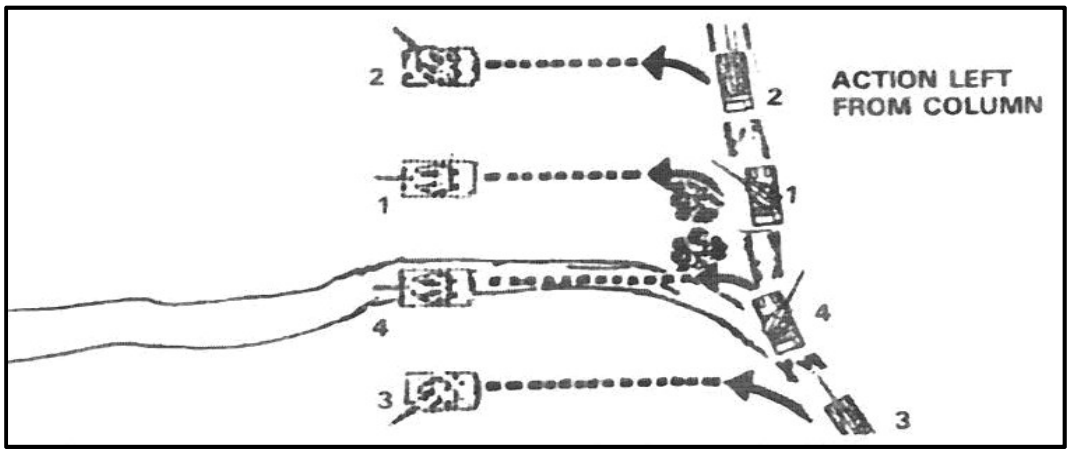



**Figure 2.15 - Action Left Drill from Column**

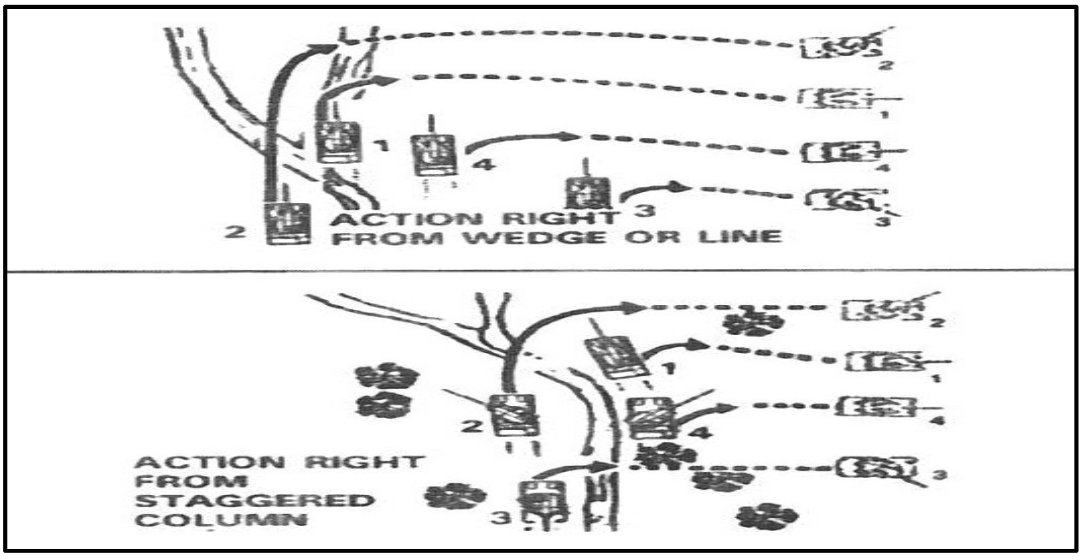



**Figure 2.16 – Action Right**

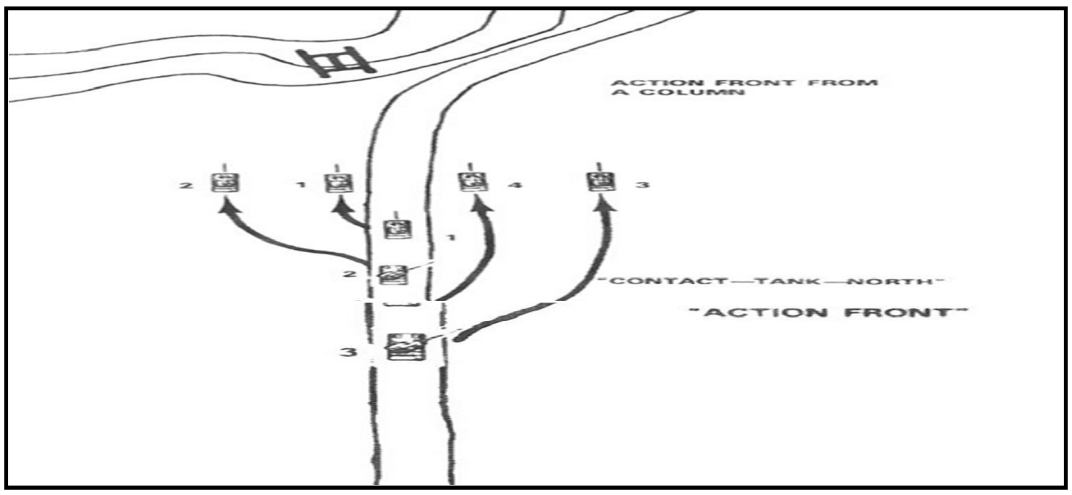



**Figure 2.17 - Use of action drill to react enemy contact (action front from a column)**

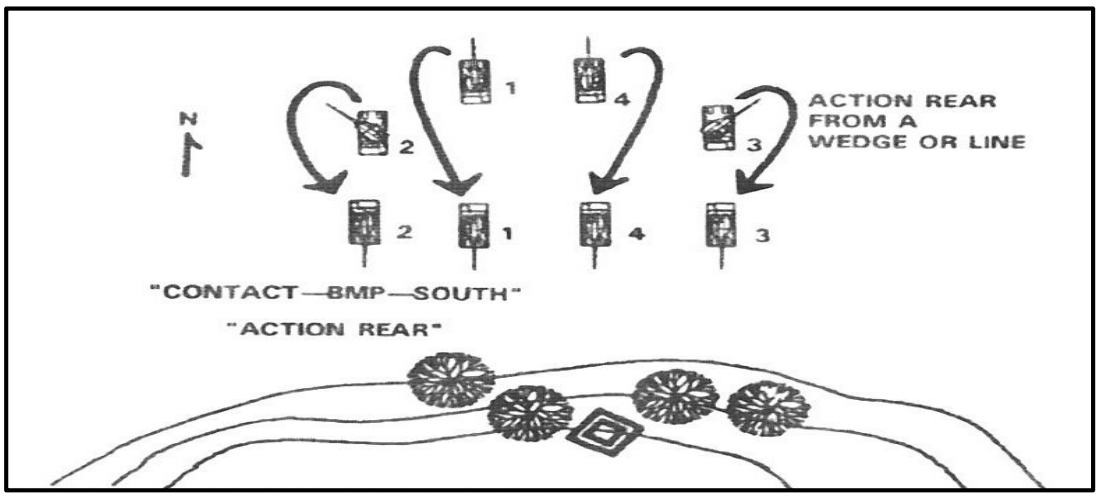



**Figure 2.18 - Use of Action drill to react enemy contact (action rear from a wedge or line)**

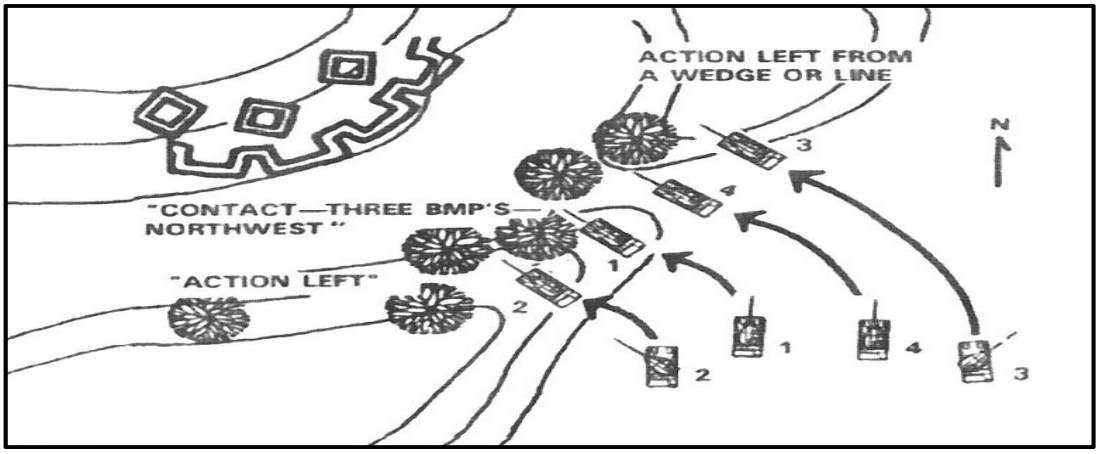



**Figure 2.19 - Use of action drill react to enemy contact (action left from a wedge or line)**

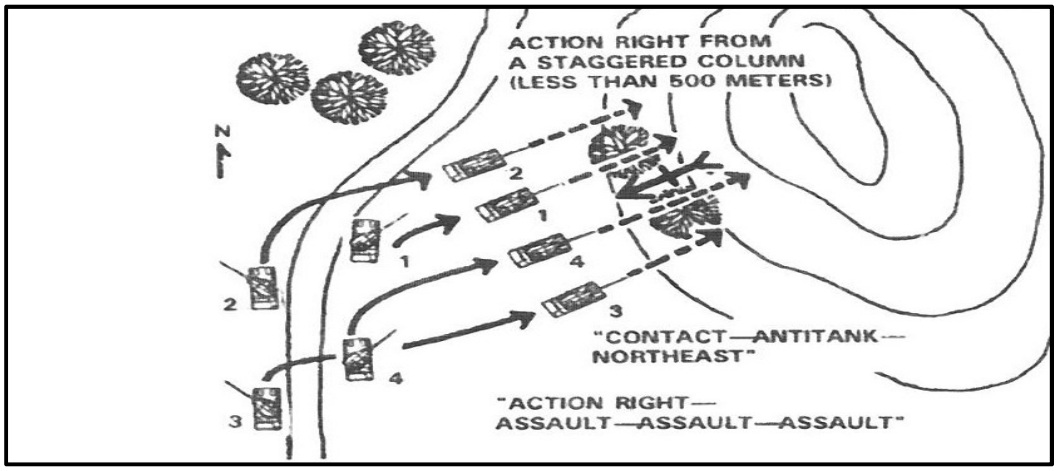



**Figure 2.20 - Use of action drill to react enemy contact (action right from a staggered column)**

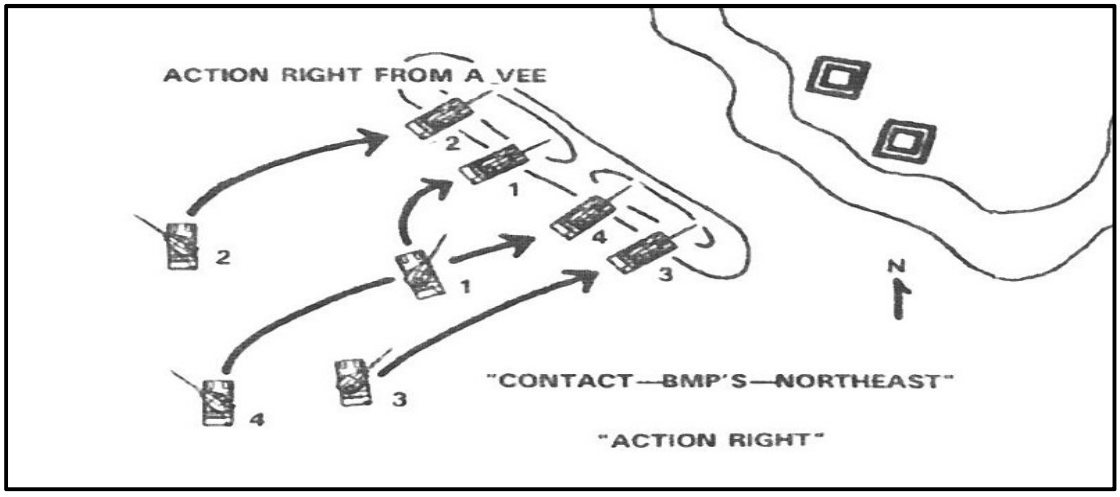



**Figure 2.21 - Use of action drill to react enemy contact (action right from a vee)**

02102. **React to Indirect Fire Drill.** When the armor unit receives indirect fire, crewmen close their hatches and ballistic doors. If on the move, the unit maintains its speed and direction. With infantry mounted, the armor commander may issue the command for an action drill to avoid casualties to the infantry. If maneuvering with light infantry, a common rally point is desirable. Once clear of the indirect-fire effects, crews can open hatches and necessary ballistic doors.

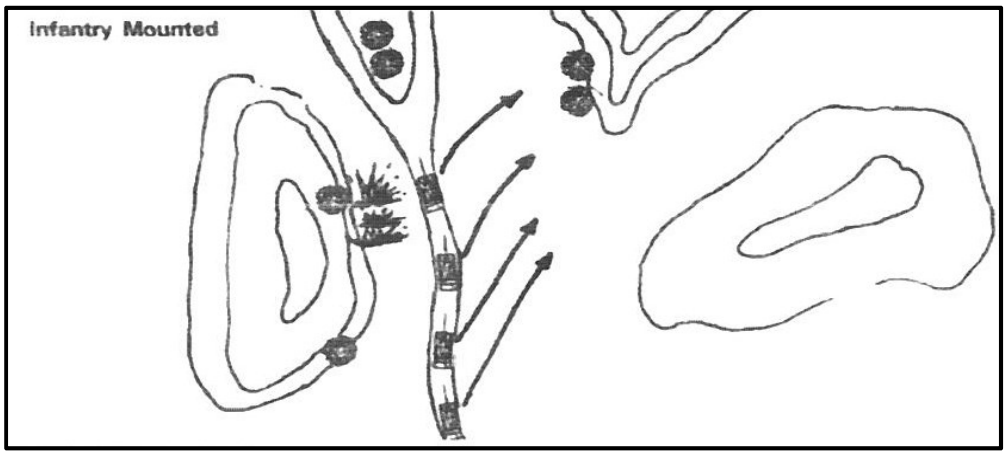



**Figure 2.22 - React to Indirect Fire Drill**

02103. **React to Air Attack Drill.** The armor unit should practice passive defense against air attack. The use of cover and concealment can frequently prevent high-performance aircraft and helicopters from detecting and attacking the unit. The air attack drill involves four steps discussed in the following paragraphs.

a. Alert the Platoon

b. Seek Cover and Concealment

c. Prepare to Engage

d. Report

02104. **Alert the Platoon.** (over the radio or using hand-and-arm signals)

02105. **Seek Cover and Concealment.** When moving, AVs seek immediate cover and concealment. If concealment is not available, moving vehicles should stop. A stationary vehicle is harder to see than a moving vehicle. If the enemy aircraft detect the vehicles and initiates an attack, the commander announces “ENEMY AIR” and exposed vehicles immediately move at a 45-degree angle toward or away from the attacking aircraft. Vehicles should maintain a 100-meter interval and avoid presenting a linear target in the direction of attack.

02106. **Prepare to Engage.** Vehicle commanders should prepare to engage aircraft with a high volume of machine gun fired upon the order of the platoon leader. The overall commander must be sure that the aircraft are attacking, since firing machine guns could give away their positions. Volume is the key to effectiveness. The idea is to throw up a wall of fire and let the aircraft fly into it. In some cases, the main gun can be used against hovering helicopters.

02107. **Report.** The armor leader sends the higher headquarters a contact report as soon as possible. If the unit is engaged by bombs or spray, the reaction to indirect fire drill is used. Enemy aircraft operate in pairs, with two to eight, or more, aircraft in each flight. After the first aircraft passes overhead, another may follow. Vehicles

should remain in covered and concealed positions for at least 60 seconds after the first aircraft leaves.

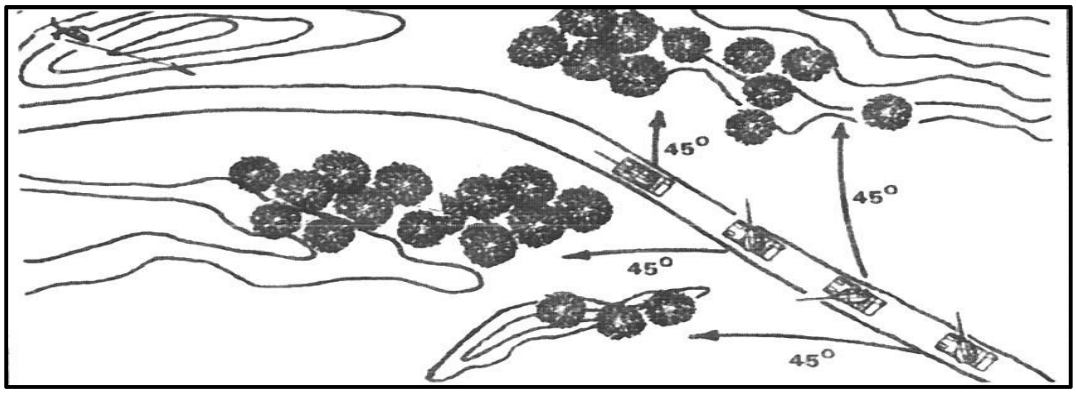



**Figure 2.23 - React to Air Attack Drill**

02108. **Break Contact Drill.** When the armor unit makes contact with a superior force, the commander may give the order to break contact. The voice command to break contact is "BREAK CONTACT" followed by a direction (clock method, or front, rear, left, or right). The visual signal is a red flag raised then dropped in the direction where the break is to be conducted. The following are steps in successfully executing a break contact fire and movement drill:

02109. **Make the initial break**. The commander designates a support by fire element to provide overwatch and supporting fire, if needed. Dismounted infantry moves to a designated covered position in the direction of the break (clock method). If infantrymen are mounted on the vehicles not in contact (not the support-by-fire force), they may remain mounted as the vehicles move to break contact.

02110. **AFVs disengage.** The support-by-fire element uses successive bounds to join infantry. These vehicles may employ smoke as needed. The bounds continue until contact is broken.

02111. **Report.** The commander sends a spot report to the higher headquarters and prepares to continue the mission.

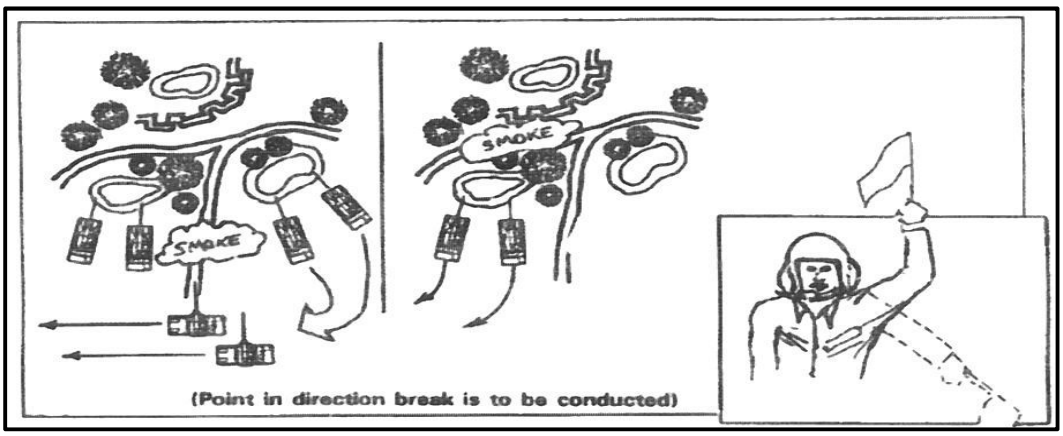



**Figure 2.24 - Break Contact Drill**

02112. **Attack an Objective Drill.** This drill can be used during a hasty attack as a reaction to enemy contact. The steps for execution of this drill are as follows:

02113. The commander must determine and announce how the basic maneuver will be conducted.

a. Armor attacks by fire while infantry assaults.

b. Armor attacks by fire while infantry assaults, then armor joins in the assault.

c. Armor and infantry assault the objective on different axes.

d. Armor and infantry assaulting together. 1) If necessary, infantrymen dismount the vehicles and move to the flanks. 2) The designated element establishes a base of fire and suppresses the objective. Deception smoke is used to confuse the enemy as to the true location of the assault force. This support by fire position may later serve as the rally point for all elements if needed. 3) The assault element maneuvers to the assault position (nearest covered and concealed position). 4) The support by fire element shifts tires when the prearranged signal is given by the assault element as it reaches the final coordination line. 5) The assault element conducts fire and movement onto the objective, secures it, begins consolidating its position, and reports to the commanders

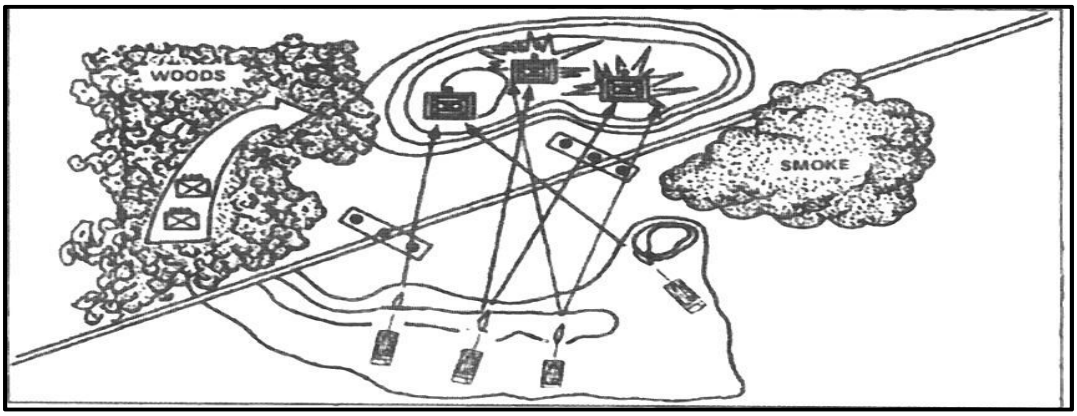



**Figure 2.25 - Attack an Object Drill**

02114. **Action at Danger Areas.** A danger area is any place the armor unit, or any force maneuvering with it, might be exposed to enemy observation, fire, or both. If a danger area cannot be avoided, the armor crosses it with great caution and as quickly as possible. Armor and infantry maneuvering together coordinate their efforts using the armor's overwatch capabilities and infantry's ability to clear areas where hidden enemy infantry may engage friendly forces.

02115. **Types of Danger Areas.** The following paragraphs discuss some examples of danger areas and crossing procedures.

a. Alert the Platoon

b. Seek Cover and Concealment

c. Prepare to Engage

d. Report

02116. **Open Areas.** The infantry on the near side observes and provides local security, including the near side flanks. Using an infantry guide, armor maneuvers to the flanks to provide overwatch while the infantry moves across the open area to secure the far side. When cleared, the remaining infantry moves across at the shortest exposed distance as quickly as possible. The armor unit then bounds by section across the open area and reassumes the previous designated formation. Different techniques could include the use of Dragon and TOW systems as overwatch to either supplement armor, or to remain in overwatch as armor bounds across the open area and clears the far side with the infantry. After clearing the far side, the armor would provide flank security.

02117. **Linear (Roads and Trails).** Crossroads or trails at or near a bend, a narrow spot, or on low ground use the same techniques for linear areas as those used for open areas, except that the armor unit may orient down the road or trail while providing near side flank security.

02118. **Defiles.** Crossing defiles requires the infantry to clear the flanks of the defile (often high ground), the far side of the defile, and then the defile itself. If engineers are available, they help locate and neutralize mines in the defile. Once the defile is cleared, the armor moves through quickly and augments the infantry on the far side security until the remaining infantry moves through or past the defile.

02119. **Trenches, Gullies, Wadis, and Tunnels.** Typically in operations other than war, the enemy will not have the resources to build fortifications and will maximize use of below/ground protection. The armor provides overwatch while infantry fights the belowground battle. Recognition signals such as smoke grenades, flaxes, and flags on antennas can indicate the location of friendly infantry to AFV crews. The armor must be prepared to engage with enemy reinforcements or any retreating enemy as they go aboveground. The armor can also advance to engage the enemy with machine guns along the length of a trench, gully, or wadis.

02120. **Planning Considerations for Crossing Danger Areas.** The following planning considerations apply for armor and infantry when crossing danger areas:

a. Plan rally points, the use of hand-and-arm signals (for example, signals for danger areas or bypasses), and the use of indirect fire targets around suspected danger areas.

b. Rehearse avoiding danger areas by practicing maneuvering around them.

c. Rehearse actions to be taken at unavoidable danger areas; upon enemy contact (both direct and indirect fires); to control friendly direct and indirect fires; when consolidating; and upon reassuming the formation.

02121. **Breach an Obstacle.** The armor/infantry team may not be able to bypass a small or unmanned obstacle, so the commander determines to conduct a breach. The voice command for this exercise is "BREACH." The steps for this drill are as follows:

02122. **Organization.** The commander should organize the team into three elements as described below:

a. Support force. This is usually an armor platoon or section with an infantry weapons section (M60 machine gun and/or 60-mm mortar). It leads on the movement to the obstacle.

b. Breach force. This consists of the infantry and engineers if available. It follows the support force to the obstacle.

c. Assault force. This is usually the infantry, but the commander may be able to add armor section. It follows the breach force organization when moving to the obstacle.

d. Suppression. The support force establishes a base of fire and suppresses enemy direct-fire defenses. It must be able to suppress the enemy by direct and indirect fires, includes providing air defense coverage.

e. Obscuration**.** The support force adjusts artillery-projected smoke on the far side of the obstacle to prevent enemy observation of the breach and assault forces.

f. Security**.** The breach force secures the near side of the obstacle while the assault force maneuvers to a better position to prepare for the assault. Ideally, the obstacle is breached after securing the far or enemy side. Friendly forces from the support force can reach the far side of the obstacle by infiltration, bypass, air assault, or minor breach.

g. Reduction. The breach force clears, marks, and secures the lane. The support force continues to suppress and shift indirect fires and smoke while the assault force is poised to move quickly through the breach once it is cleared and marked. The assault force moves through the breach, conducts below ground battle, and provides far-side security by adding the shoulders against enemy counterattacks. The remainder of the breach force, followed by the support force, moves through the breach and continues the designated mission.

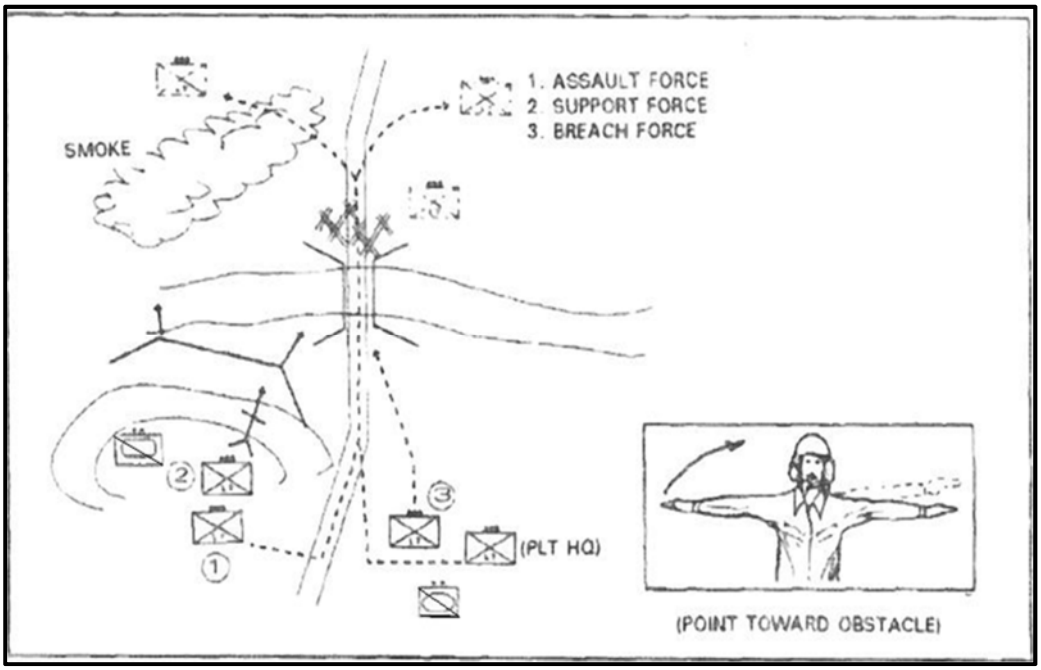



**Figure 2.26 - Breach an obstacle (in stride)**

# Chapter III
# Mission Command in Armor Operations

0301. This chapter discusses the importance of mission command in the conduct of armor operations. As well as the principles and fundamentals of mission command. It then elaborates the processes that serve as a guide for the commanders to execute their mission. Lastly, this chapter will discuss the facilities and techniques of command and control that can be executed by the commander.

0302. Mission Command is the conduct of military operations through centralized planning and decentralized execution based on mission orders. Armor units get its mission order from its supported units. This is the Armor's preferred method of exercising command and control in full spectrum operations.

0303. Command and control is the process through which the activities of the armor units are planned, directed, coordinated, and controlled to accomplish a mission. This process encompasses the personnel, equipment, communications, facilities, and procedures necessary to gather and analyze information used to plan, issue directives and instructions, and supervise the execution of military operations. The armor commander is responsible for command and control of organic, assigned, attached, and supporting elements, and for the synchronization of their tasks and purposes into his operation.

## Principles and Fundamentals of Mission Command in Armor Operations

0304. Mission command gives subordinates the greatest possible freedom of action. Commanders focus their orders on the purpose of the operation rather than on the details of performance of assigned tasks. They delegate most decisions to subordinates. This minimizes detailed control and empowers subordinates' initiative. Mission command emphasizes timely decision-making, understanding the higher commander's intent, and clearly identifying the subordinates' tasks necessary to achieve the desired end state. It improves subordinates' ability to act effectively in fluid, chaotic situations.

0305. Mission command requires an environment of trust and mutual understanding. Respect and full familiarity with the armor commander's intent and concept of operations are very essential. Mission command applies to all operations across the spectrum of conflict. The elements of mission command are:

a. Armor commander's intent.

b. Subordinates' initiative.

c. Mission orders, which include: a brief concept of operations, minimum necessary control measures, and resource allocation.

0306. Chaos and uncertainty dominate land conflict. Predictability is rare, making centralized decision-making and orderly processes ineffective. In addition, commanders must contend with thinking, adaptive hostile force or an unpredictable natural phenomenon. Under such conditions, leaders of forces in contact can often see and act on immediate opportunities and threats better than their superiors can. Delegating the greatest possible authority to subordinates helps the force adapt the operation to the situation quickly and retain the initiative.

0307. While mission command restrains higher level commanders from micromanaging subordinates, it does not remove them from the fight. Rather, mission command frees these commanders to focus on accomplishing their higher commander's intent and on critical decisions they are authorized to make. Higher commanders anticipate developments, allocate resources to exploit successes, and intervene to shape the necessary operations. Mission command is ideally suited to an environment of complexity and uncertainty.

0308. Mission command tends to be decentralized, informal, and flexible. Orders and plans are as brief and simple as possible. The fundamental basis of mission command is trust and mutual understanding between superiors and subordinates, an atmosphere that the senior commander must cultivate. Mission command counters the uncertainty of war by empowering subordinates at the scene to make decisions quickly. Commanders rely on their subordinates' ability to coordinate with one another using the human capacity to understand with minimum verbal information exchange.

0309. In any operation, the situation may change rapidly. The speed and violence of modern operations add to combat's inherent chaos and disorder. Operations that the commander envisions may bear little resemblance to actual events. Even with the most advanced information systems, higher headquarters often have difficulty understanding the situation on the ground. Perhaps most important, opportunities for exploitation or counterattack may suddenly occur. Subordinate commanders need maximum latitude to take advantage of such situations and meet the higher commander's intent when the original orders no longer apply. Full spectrum operations require leaders schooled in independent decision-making, aggressiveness, and risk taking in an environment of mission orders and mission command at every level.

0310. Mission command applies to all operations across the spectrum of conflict. Often the operational environment encountered during support to civil governance operations proves to be more complex than that encountered in offensive and defensive operations. The continuous, often volatile, interaction of brigades and smaller units with the local populace during support to civil governance operations requires that leaders willing to exercise initiative. They must be able and are willing to solve problems without constantly referring to higher headquarters.

0311. Mission command encourages commanders to act promptly, consistently, and decisively in all situations. Under mission command, commanders explain not only

the tasks assigned and their immediate purpose but also the higher commander's intent. Doing this helps junior commanders and their soldiers understand what is expected of them and what constraints to apply. Most importantly, they understand the mission's purpose and context. The commander's intent also guides subordinates working with agencies not under military control.

# Command and Control in Armor Operations

0312. Command and Control is the process through which the activities of the armor units are planned, directed, coordinated, and controlled to accomplish the mission. This process encompasses the personnel, equipment, communications, facilities, and procedures necessary to gather and analyze information, to plan, to issue instructions, and to supervise the execution of operations. The armor commander is responsible for command and control of organic, assigned, attached, and supporting forces, and for their synchronization into his operation.

## Purpose

0313. Command and Control (C2) comprises coordinating, planning, directing and controlling all unit activities. The C2 process serves two purposes:

a. It gives the armor commander the means to communicate his intent to his staff and maneuver and support forces;

b. It enables the staff to quickly and effectively assist their commanders in planning and executing operations faster than the enemy can react.

0314. To be effective, C2 must be properly organized and the staff must be well trained. Armor commanders and staffs must practice C2 process so that procedures become instinctive. The armor commander must have a reliable, secure, fast and durable Command and Control (C2) system used to communicate, coordinate and provide direction to his forces.

## Leadership

0315. Effective leadership is a primary dynamic element of combat power that affects the success of C2. Armor leaders must be willing to take responsibility and use initiative, guided by their commander's intent. Armor leaders must be able to think clearly and quickly. Speed and decisive action are critical.

0316. One of the keys to success of C2 is the ability of leaders at all levels, to issue effective orders. These orders must place restrictions on subordinates to assure the coordinated action of the command. They must use terminologies widely understood throughout the command. Above all, they must accurately communicate the commander's intent, which guides the subordinate in the pursuit of the common goal in the absence of communications.

0317. The command and control system must be faster and more effective than the enemy's system. This allows the commander to receive and process information and to make and execute decisions faster than the enemy. The overriding goal of this system is to implement the commander's will be in pursuit of an objective.

0318. Armor units are most effective when massed. However, they are often decentralized and separated by means of task organization with infantry troops. This can occur down to section level. In such an organization, commander-subordinate relationships must be characterized by mutual trust and respect. Armor commanders and subordinates must know how each other thinks. Armor commanders must teach subordinates not what to think, but how to think. Conversely, when armor units are decentralized and attached to infantry units, armor commanders must be assertive in advising the infantry commander and his staff on the best means for armor employment and logistical requirements.

### Organization

0319. The armor commander determines the command and control organization that best supports his method of operations. He organizes his staff, determines the succession of command, and assigns responsibilities.

### Chain of Command

0320. The armor commander issues orders and receives information through the chain of command. He issues all orders to the commander of the next lower unit. Bypassing commanders should occur only in urgent situations. In such instances, the bypassed commander should be notified by the commanders involved as soon as possible.

## Types of Staff

0321. The unit staff is composed of personal, coordinating and special staffs. The unit staff are outlined below:

### Personal Staff

0322. The personal staff consists of the deputy commander or executive officer (Ex-O) and the Command Sergeant Major (CSM) who works under the immediate control of the commander and directly assists him in the exercise of command. Some staffs have personal staff officers who work under the immediate control of the commander and therefore have direct access to the commander. The armor commander establishes guidelines or gives specific guidance when the personal staff officer should inform, or coordinate with, the chief of staff or other members of the staff on issues. Most personal staff officers also perform duties as special staff officers, working with a coordinating staff officer. This is done case by case, depending on the guidance of the commander or the nature of the task. Personal staff officers also may work indirectly under the supervision of the Ex-O or Chief of Staff (CS).

## Coordinating Staff

0323. The coordinating staff officers are the principal staff assistants to the commander. Every staff has coordinating staff officers who coordinate actions for the commander and special staff officers. Coordinating staff responsibility includes:

a. Ensuring that the special staff officer or section has personnel, logistics, facilities, and proper support.

b. Coordinating actions and tasks of special staff officers across the entire staff, as necessary.

c. Informing the Chief of Staff on the special staff officer's actions.

## Special Staff

0324. Every staff has special staff officers. The special staff aids the commander in professional, technical, or other functional areas. The specific number and duties of special staff officers vary at each level of command based on the table of organization and equipment (TOE) authorizations, desires of the commander, and the size and level of command. If, at a given echelon, a special staff officer is not assigned, the corresponding coordinating staff officer assumes those responsibilities as necessary.

# Command Post (CP) Facilities

0325. The command and control facilities consist of the vehicles and locations from which the commander, assisted by his staff, directs the battle and sustain the force. The exercise of command and control is focused on the following facilities; CP types for brigades and higher headquarters : the main CP, the tactical CP (TAC CP), alternate CP, rear CP and assault CP; CP types for Battalion/ Task Force (BN/TF): main CP, tactical CP (TAC CP) combat trains CP (CTCP), and the field trains CP.

## Main Command Post

0326. The main CP has a broader and more future-oriented focus than the TAC CP. Its staff does not only control deep operations but also general activities, such as conducting the overall battle, allocating resources, and planning future operations. Most importantly, the main CP serves as the primary synchronization point for the entire battlefield. Its functional staff is mainly on triad (Operations-Intelligence-CMO), it also includes Maneuver Support and Sustainment. Divisions, brigades and battalions establish main CPs. The main CP locates to the rear of forward-deployed forces. At higher levels, the main CP is a large organization. It displaces less frequently than the TAC CP, although in the requirements of battle, the threat or enemy action and the desires of the commander dictates its movement.

## Tactical Command Post

0327. The TAC CP is the forward echelon of a combat brigade or higher-level maneuver headquarters. It focuses on close operations while monitoring deep and rear operations (for their effect on close operations). Its organization is simpler, smaller, and more austere than the main CP operating as one integrated cell that provides intelligence for current operations, and fire support. Normally, only the maneuver command uses a TAC CP and it will not necessarily be deployed; its use is METT-TC dependent. Divisions and combat brigades usually establish a TAC CP that locates forward on the battlefield. Although it is limited in physical size and electronic signature. It must be capable of displacing rapidly and frequently. The flow of battle, the threat of enemy action, and the desires of the commander dictate its movement.

## Rear Command Post

0328. The rear CP at division focuses on rear operations. Its main functions are sustainment, transportation, protection, and terrain management in the rear area in support of combat operations. It is mainly composed of Sustainment Support (SS) units. The rear CP is large. It is difficult to conceal and even more difficult to reposition. The commander should strategically place it well behind forward units.

## Alternate Command Post

0329. The alternate CP normally assumes the responsibilities of the main CP if the latter is moving, damaged, or destroyed. The alternate CP retains these duties until the main CP is reconstituted. The commander shall establish an alternate CP with a clear understanding of its purpose and roles, which enables the unit to sustain continuous C2 during catastrophic loss. The alternate CP is not used to support CP displacement operations. The alternate CP does not have to be a mirror image of the main CP; it does not have to be able to perform all main CP C2 functions. Initially, it must be capable of serving as a life-support function, being located where surviving personnel and equipment of a destroyed CP can rally and begin to rebuild. The reasons for establishing an alternate CP are to:

a. Ensure continuous support of C2 operations.

b. Identify another location where surviving elements of a destroyed CP can rally.

c. Assess casualties and damage, reorganize, and reestablish critical C2 functions.

## Assault Command Post

0330. The assault CP is an austere, ad hoc CP. It performs critical C2 functions that support division and higher command echelons in tactical operations for special purposes (such as entry, deployment, and retrograde operations). During deployment,

the commander normally sequences the assault CP to arrive as soon as possible after the initial assaulting brigade secures the airhead or beachhead. Its role then becomes that of fighting the current fight with tactical forces on the ground, synchronizing the flow of follow-on forces into the Area of Operation (AO), and fazing them into the fight to expand and secure the airhead or beachhead.

0331. Early in the deployment, the commander assault CP serves as the division C2 link between the division forces on the ground. It facilitates the future establishment of the main CP and continues to function until the remainder of division C2 systems arrives. Normal doctrinal functions then resume at TAC, main, and rear CPs. There is no set design for the assault CP. Each mission demands different capabilities. Therefore, the commander tailors the assault CP for specific missions, when the TAC is insufficient or when the main CP is too large.

## Main Command Post

0332. The battalion or task force main CP is the control, coordination and communications center for combat operations. The main CP is composed of the Intelligence (S2) and Operations (S3) sections, the Fire Support Element (FSE), representatives from other attached elements, and the tactical CP (when not forward). The main CP vehicles and personnel must be as few as possible to allow for rapid displacement, but large enough to accomplish command and control functions in support of the commander.

a. **Functions.** The functions of the main CP are to monitor and assist in command and control by maintaining contact and coordination with higher and adjacent units, continuously updating the enemy situation, planning operations, analyzing and disseminating tactical information, maintaining situation maps, and requesting and synchronizing additional Maneuver Support (MS) and Sustainment Support (SS) for the battle. Factors that have immediate operational impact must be monitored by the main CP and communicated to the commander.

b. **Operations.** The primary considerations in positioning the main CP are communications, accessibility, and survivability. The personnel who operate the main CP must be organized to provide both security and continuous operations on a 24-hour-a-day basis. A sleep plan must be enforced to preserve the ability of the main CP personnel to perform continuous operations. The main CP internal Standard Operating Procedures (SOP) establishes this internal organization. Coordination between the main CP, the combat trains CP, and the field trains CP must be continuous to ensure that SS is integrated into the mission effort. One technique is to monitor the administrative/logistics (A/L) net on a remote. When possible, a landline link is established with the combat trains CP. The security of the main CP is enhanced by its capability to displace rapidly and provide a reduced electronic signature. Displacements are planned to ensure that the main CP is stationary during critical phases of the battle.

## Alternate Main CP

0333. The combat trains CP is normally designated as the alternate main CP. Aid Station and Unit Maintenance Collection Point (UMCP) assets may be collocated to enhance communications capability. A mortar Fire Direction Center (FDC) track may also be used as an alternate main CP. If required to function as the main CP, the primary functions of these elements will suffer, and the effectiveness of the new main CP will also be degraded.

## Tactical Command Post

0334. A tactical CP may be formed during fast-moving offensive or retrograde operations to maintain communications and facilitate the movement of the main CP. In such circumstances, the commander may designate one of the command post vehicles from the main CP to act as the tactical CP. Some or the entire command group may locate at the tactical CP at various times.

## Command Group

0335. The command group consists of the commander and those he selects to go forward to assist him in controlling maneuver and fires during the battle. There is no requirement for these people to collocate; for example, the commander may be in one part of a unit's sector while the S3 officer may be in a separate part of the sector. The composition, nature, and tasks of the command group are determined by the commander to permit the optimum command and control of his unit during the battle.

## Combat Trains Command Post

0336. The combat train CP is the coordination center for sustainment support of the battalion or task force and the control element of the combat trains. It is positioned forward in the field trains. The Supply (S4) officer is responsible for operations, movement, and security of the combat trains and is assisted by the Personnel (S1) Officer. The S1, S4, and Battalion Maintenance Officer (BMO) must continually assess the situation, anticipate the needs of units, and prepare to push support forward. Anticipating requirements is the key to successful sustainment support.

0337. The combat train CP maintains the SS status of the battalion or task force. In preparation to assume its functions as alternate main CP, the combat trains CP monitors the battalion or task force command net and maintains charts and tactical situation maps identical to those at the main CP.

0338. Any change in the main effort of the unit should be reported to the combat trains CP by the main CP or tactical CP. Similarly, a major change in the ability of the SS system to support an operation must be immediately reported to the main CP by the combat trains CP. The combat trains CP relays information to the field trains CP.

## Field Trains Command Post

0339. The field trains are under the control of the Headquarters and Headquarters Company (HHC) commander whose headquarters is the field trains CP. When the battalion or task force commander co-locates his field trains with the Brigade Support Area (BSA), the HHC commander coordinates with the forward support battalion for positioning and the defensive sector for the battalion or task force field trains elements. When the battalion or task force commander deems it necessary, the field trains may be positioned outside of the BSA in the battalion or task force area of operations; in this case, the S3 officer designates the general location for the field trains or directs that unit trains be formed under the control of the S4 officer.

0340. The field trains CP is composed of the HHC commander, and the remaining elements of the S1 and S4 sections. It coordinates the collection and movement of the SS from the battalion or task force field trains and the forward support battalion to the forward elements of the unit. It controls and coordinates the activities of the battalion or task force field trains including operations of the support platoon, elements of the maintenance platoon in the field trains, company and attached units' supply sections, and the Personnel and Administration Center (PAC). The field trains CP monitors the battalion or task force A/L net and maintains a communications link with the forward support unit CP.

0341. Supplies, personnel, and documents going forward from the field trains are grouped together into Logistics Packages (LOGPACs) under the control of the support platoon leader or a company supply sergeant. The field trains CP organizes and dispatches LOGPACs based on the instruction from the combat trains CP. The unit trains are normally echeloned with both a combat trains CP and a field trains CP. The alternative configuration is to form unit trains with a single rear CP which is operated by the S4 officer. In this case, there is no field trains CP.

# Command and Control Process

0342. The unit's command and control process involves planning, coordinating, and executing combat operations. While higher level headquarters give broad missions and allocate assets to fight the close and deep battle, task forces directly control and synchronize the actions of company teams, supporting fires and obstacles on the ground and against the enemy. Key command and control considerations for the battalion/task force commander include:

a. Making maximum use of time.

b. Conducting physical reconnaissance.

c. Planning for and maintaining flexibility.

d. Instilling and requiring initiative from all leaders.

e. Decentralizing execution.

f. Providing clear, concise missions.

g. Synchronizing all assets.

h. Designating and sustaining main effort.

# Planning Process

0343. **Planning.** Planning is the means by which the commander envisions a desired outcome, lays out effective ways of achieving it, and communicates his vision, intent and decisions to his subordinates focusing on the results he expects to achieve. Plans forecast but not predict. A plan is a continuous, evolving framework of anticipated actions that maximizes opportunities. It guides subordinates as they progress through each phase of the operation. A good plan facilitates effective action in the face of unforeseen events; good plans foster initiative.

0344. Scope, complexity, and length of planning horizons differ between operational and tactical planning. Campaign planning coordinates major actions across significant periods. Planners mesh service capabilities with those of joint and multinational formations as well as interagency and non-governmental organizations. Tactical planning has the same clarity of purpose as operational planning, but has a shorter planning horizon. Comprehensive, continuous, and adaptive planning characterizes successful operations both at the operational and tactical levels.

0345. Plans specify what commanders will decide personally. In the offense, for example, commanders normally decide when to commit the reserve. Regardless of echelon, commanders identify those information requirements they consider most important to their decisions referred to as the Commander's Critical Information Requirements (CCIR). The CCIR help the commanders confirm their vision of the battlefield or identify significant deviations from it. The staff incorporates CCIR into the appropriate parts of the plan and passes them to subordinate units.

## Operational and Tactical Planning

0346. Planning is dynamic and continuous. Operational-level planning focuses on developing plans for campaigns, subordinate campaigns, and major operations. Combatant commanders develop theater campaign plans to accomplish multinational, national, and theater strategic objectives. Since campaign plans generally set a series of long-term objectives, they often require phases. Therefore, a campaign plan normally provides a general concept of operations for the entire campaign and a specific operation order for the campaign's initial phase.

0347. Operational and tactical planning complements each other but have different aims. Operational planning prepares the way for tactical activity on favorable terms; it continually seeks to foster and exploit tactical success. Major operations depend on creatively using tactical actions to accomplish strategic or operational purposes in specific contexts against adaptive opponents. Tactical planning emphasizes flexibility and options. Planning horizons for tactical actions are relatively short. Comprehensive

planning may be feasible only for the first engagement or phase of a battle; succeeding actions depend on enemy responses and circumstances. The art of tactical planning lies in anticipating and developing sound branches and sequels.

0348. Brevity is essential, so is speed. Staff must avoid consuming too much time developing lengthy plans that contain irrelevant details. When plans arrive late, subordinate units can only react afterwards. To save time and shorten plans, commanders and staffs anticipate support requirements and forecast options. Headquarters on each level plan alongside higher and lower headquarters. Parallel planning expedites the exchange of information among headquarters and should be used as much as possible. Commanders exploit technology to increase situational understanding and speed of planning.

## Doctrinal Planning Procedures

0349. There are two doctrinal planning procedures. In units with a formally organized staff, the military decision-making process helps commanders and staffs develop estimates, plans, and orders. It provides a logical sequence of decision and interaction between the commander and staff. The military decision-making process provides a common framework for all staff that supports the maximum use of parallel planning. At the lowest tactical echelons, commanders do not have a staff. Consequently, commanders and leaders follow the Troop Leading Procedures (TLP). Both procedures are hinged upon the commander's ability to visualize and describe the operation. Both are means to an end/ their value lies in the result, not the process.

0350. **Phasing.** A phase is a specific part of an operation that is different from those that precede or follow it. A change in phase usually involves a change of task. Phasing assists in planning and controlling. Considerations of time, distance, terrain, resources, and important events contribute to the decision to phase an operation. A phase is a period when a large portion of the force conducts similar or mutually supporting activities. Operations link successive phases. Individual phases gain significance only in the larger context of the campaign or major operation. Each phase should strive for simultaneity in time, space, and purpose. In this way, commanders combine simultaneous operations within phases while sequencing operations to achieve the end state. Links between phases and the requirement to transition between phases are critically important. Commanders establish clear conditions for how and when these transitions occur.

0351. **Branches and Sequels.** Operations never proceed exactly as planned. An effective design places a premium on flexibility. Armor commanders incorporate branches and sequels into the operational design to gain flexibility. Visualizing and planning branches and sequels are important because they involve transition changes in mission, in type of operation, and often in forces required for execution. Unless planned and executed efficiently, transitions can reduce the tempo of the operation, slow its momentum, and cede the initiative to the adversary.

a. **Branch.** A branch is a contingency plan or course of action (an option built into the basic plan or course of action) for changing the mission, disposition,

orientation, or direction of movement of the force to aid the success of the current operation, based on anticipated events, opportunities, or disruptions caused by enemy actions. Units prepare branches to exploit success and opportunities, or to counter disruptions caused by enemy actions. Commanders anticipate and devise these counters actions/attacks. Although anticipating every possible threat action is impossible, branches anticipate the most likely ones. Commanders execute branches to rapidly respond to changing conditions.

b. **Sequels.** Sequels are operations that follow the current operation. They are future operations that anticipate the possible outcomes - success, failure, or stalemate of the current operation. A counter offensive, for example, is a logical sequel to a defense; exploitation and pursuit follow successful attacks. Executing a sequel normally begins another phase of an operation, if not a new operation. Commanders consider sequels early and revisit them throughout an operation. Without such planning, current operations leave forces poorly positioned for future opportunities, and leaders are unprepared to retain the initiative. Both branches and sequels should have execution criteria, which are carefully reviewed before their implementation and updated based on the assessment of current operations.

0352. **Concept of Operations.** The concept of operations describes how commanders see the actions of subordinate units fitting together to accomplish the mission. As a minimum, the description includes the scheme of maneuver and concept of fires. The concept of operations expands the commander's selected course of action and expresses how each element of the force will cooperate to accomplish the mission. Where the commander's intent focuses on the end state, the concept of operations focuses on the method by which the operation uses and synchronizes the War Fighting Functions (WFF) to translate vision and end state into action. Commanders ensure that the concept of operations is consistent with both their intent and that of the next two higher commanders.

0353. **Planning process.** The planning process is a systematic approach to formulating tactical plans. Processes used are military decision making process, the estimate of the situation (staff estimates), the factors of METT-TC, and the Intelligence Preparation of the Battlefield (IPB). These processes are interrelated which will be discussed on separate succeeding sections. Figure 3.1 shows the flow of the planning process. The planning process is accomplished based on the time and other resources available.

## Military Decision Making Process

0354. The Military Decision Making Process (MDMP) is a single, established and proven analytical process. MDMP helps the commander and his staffs examine a battlefield situation and reach a logical decision. The process helps apply thoroughness, clarity, sound judgment, logic, and professional knowledge to reach a decision. The full MDMP is a detailed, deliberate, sequential and time consuming process used when adequate planning time, operations, movement rates, fuel

consumption, weapons effect are quantifiable and, therefore, part of the science of war. Other aspects, like the impact of leadership, complexity of operations and uncertainty regarding enemy intentions, belong to the art of war.

0355. This section will only discuss the MDMP in an armor operation's perspective. The whole process is still the same but there are some areas that need focus in planning an armor operation. The whole MDMP will be discussed in a separate manual.

0356. Armor unit commanders and staffs focuses on how to maintain the momentum of the fight using the available assets of the unit. This entails a detailed plan on sustainment which relies heavily on the G4 (S4) of the unit. In order to be effective, armor assets need to be sustained at a regular rate to achieve massing and maintain momentum throughout the whole operation.

0357. Staff needs to update their staff estimate on a regular basis throughout the MDMP in order to give the commander a clearer picture of the operation. The G3 (S3), with constant coordination with the G2 (S2) and the G4 (S4), should be able to maximize the effectiveness of the armor assets in conducting the operation and minimizing the risks for the said assets. They should have a thorough understanding of the capabilities and limitations of their assets to be able to come up with a sound and effective plan.

0358. The G3 (S3) should be able to produce, at the minimum, three courses of action to give the commander numerous options on how to conduct the operation. It is also implied that other staff should be able to give other options for their respective specialties to these courses of actions. Staff should not be restricted to only one option to make the operation react to the constantly changing situation on the ground.

0359. During the war gaming, all the staff should be involved and be able to share their inputs in order to produce a more refined course of action. G4 (S4) should be able to come up with a plan to effectively sustain the fight in accordance with the G3 (S3) concept of operation. Likewise, the G2 (S2) should be able to identify threats to armor assets to mitigate risk during the operation.

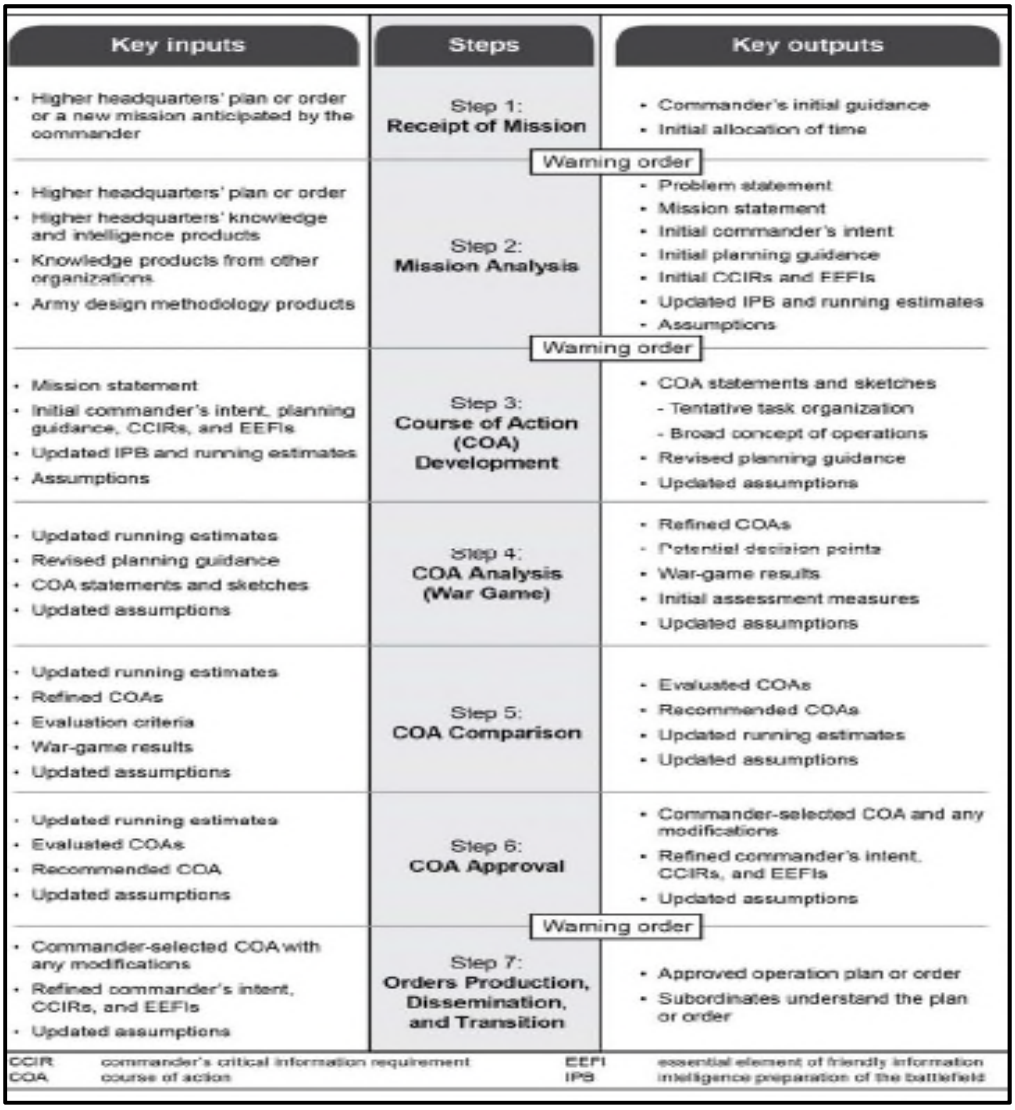



**Figure 3.1 - Military Decision Making Process (ADRP 5-0)**

## Staff Estimates

0360. Mission analysis, facts and assumptions, and the situation analysis (of the area of operations, area of interest, and enemy, friendly, and support requirements) furnish the structure for the staff estimate. The estimate consists of significant facts, events, and conclusions based on analyzed data. It recommends how to best use available resources. Adequate, rapid decision making and planning are hinged on good, timely command and staff estimates. They are the basis for forming viable courses of action. Failure to make estimates can lead to errors and omissions when developing, analyzing, and comparing Courses of Actions (COAs).

0361. **Essential Qualities of Estimates.** Commanders control battlefield tempo by making and executing decisions faster than the enemy. Therefore, commanders must always strive to optimize available time. They must not allow estimates to become overly time-consuming. However, they must be comprehensive and continuous and

must visualize the future. Staff estimates should be:

a. **Comprehensive.** Comprehensive estimates consider both the quantifiable and the intangible aspects of armor operations. They translate friendly and enemy strengths, weapons systems, training, morale, and leadership into combat capabilities. The estimate process requires a clear understanding of weather and terrain effects and, more important, the ability to visualize the battle or crisis situations requiring armor forces. Estimates must provide a timely, accurate evaluation of the unit, the enemy, and the unit's area of operations at a given time.

b. **Continuous.** The demand on the C2 system is continuous as opposed to cyclical. Estimates must be as thorough as time and circumstances permit. The commander and staff must constantly collect, process, and evaluate information. They update their estimates: 1) when the commander and staff recognize new facts. 2) when they replace assumptions with facts or find their assumptions invalid. 3) when they receive changes to the mission or when changes are indicated. Estimates for the current operation can often provide a basis for estimates for future missions as well as changes to current operations. Technological advances and near-real-time information estimates ensure that estimates can be continuously updated.

c. **Visualize the future.** Estimates must visualize the future and support the commander's battlefield visualization. They are the link between current operations and future plans. The commander's vision directs the end state. Each subordinate unit commander must also possess the ability to envision the organization's end state. Estimates contribute to this vision.

## Types of Estimates

0362. The commander and his staff make estimates that apply to any operational situation and all levels of command. They use estimates to look at possible solutions to specific operational missions and requirements. These estimates can form the cornerstone for staff annexes to orders and plans. The coordinating staff and each staff principal develop facts, assessments, and information that relate to their functional field or operating system.

0363. Normally, commanders and staff do not produce written estimates. Units usually do staff estimates by exception, graphically representing data and statistics on charts to assist the commander in decision making. The types of estimates are:

a. **Commander's Estimate.** The commander's estimate, like the operations estimate, is an analysis of all the factors that could affect a mission. The commander integrates his personal knowledge of the situation, his analysis of METT-T factors, the assessments of his subordinate commanders, and any relevant details he gains from his staff. Estimate analysis includes risk assessment, force protection, and effective utilization of all resources. The estimate also includes visualizing all reasonable COAs and how each COA

would affect friendly forces. The estimates of the commander's and operations generally follow the same format. However, the commander's estimate deals more with assessing the intangibles of training, leadership, and morale, and it results to a decision. The commander uses his personal estimate as a crosscheck of his staff's estimates.

b. **Operations Estimate.** The G3 (S3) prepares the operations estimate, which considers all elements that can influence the unit's current operations and feasible future courses of action. It results in a recommendation to the commander. To prepare this estimate, the G3 (S3) must understand: 1) the commander's intent (one and two echelons above), 2) the risk assessment, 3) the current task organization (two echelons below), 4) the unit's status, such as locations, combat capabilities (including level of training, effectiveness, degree of mobility, type of equipment, radiation exposure, and limitations), and current missions, 5) the availability and capabilities of joint assets, such as air support, naval gunfire, or amphibious assets. 6) other information, such as location, status, and mission of flank and supporting units.

c. **Personnel Estimate.** The G1 (S1) prepares the personnel estimate, which is an analysis of how all human resources and personnel factors impact soldier and unit effectiveness before, during, and after the mission. It includes a current overall personnel status of the organization, its subordinate units, and any attached or supporting elements. Personnel status includes assessments of the following tangible and intangible factors 1) medical evacuation and hospitalization, 2) unit-strength maintenance, 3) replacements, 4) soldiers' readiness, 5) organizational climate, 6) cohesion, 7) discipline, law and order. The personnel estimate predicts losses where and when losses could occur and if such losses cause the culmination of an operation. It contains the G1's (S1's) conclusions and recommendations about the feasibility of supporting major operational and tactical missions.

d. **Intelligence Estimate.** The G2 prepares the intelligence estimate. Both the G2 (S2) and the G3 (S3) examine the area of interest to identify intelligence collection needs. The intelligence estimate should focus on threats to armor assets of the units in order for the G3 (S3) to devise a plan to mitigate those risks.

e. **Logistics Estimate.** The G4 prepares the logistics estimate, which provides an accurate and current assessment of the SS situation of the organization, its subordinate units, and any attached or supporting elements. The logistics estimate is an analysis of how service support factors can affect mission accomplishment. It contains the G4's (S4's) conclusions and recommendations about the feasibility of supporting major operational and tactical missions. This estimate includes how the functional areas of supply, transportation, services, maintenance, labor, facilities, and construction affect various COAs.

f. **Civil-Military Operations Estimate.** The G3 (S3) prepares the civil-military

operations (CMO) estimate in relation to the situation and his functional responsibilities.

g. **Signal Estimate.** The G6 (S6) prepares the communications estimate in relation to the situation and his functional responsibilities.

h. **Special Staff Estimates.** Each special staff officer creates his own staff estimate in relation to the situation and his functional responsibilities. A staff officer may also make such estimates when performing dual roles, for instance, when he functions both as a subordinate unit commander and a special staff officer.

0364. **Estimate of the Situation Process.** The estimate of the situation incorporates analysis of the factors of METT-T and analysis of friendly courses of action into a process that allows the commander to select the best course of action as the basis for the overall plan. The following are the steps to be used in the estimate of the situation process (detailed discussion of the steps is discussed in other references):

a. Step 1 – Detailed mission analysis,

b. Step 2 – Develop the situation and courses of action,

c. Step 3 – Analyze courses of action,

d. Step 4 – Compare courses of action,

e. Step 5 – Recommendation or Decision,

0365. **Factors of METT-T.** The factors of METT-T are considered during the development of all estimates, including IPB. Each factor must be evaluated in conjunction with the other factors in order to gain a true appreciation of the battlefield. A significant change in one or more of the factors of METT-T will usually cause change in the OPLAN or OPORD.

0366. **Mission.** The task force mission is given to the brigade OPORD or Fragmentary Order (FRAGO). The mission contains key requirements which are who, what, when, where, and why. The commander conducts a detailed mission analysis to determine the role of the task force in the higher commander's operation, and the specified and implied tasks necessary for mission accomplishment. The "why" in the mission statement is the basis of the task force commander's intent for the operation. He develops his intent in terms of the overall effect on the enemy, resulting task force positioning, and any activities that are otherwise critical to accomplish the higher commander's plan.

0367. **Enemy: Enemy** forces are evaluated to determine their capabilities, likely courses of action, and impact on task force courses of action. At the start of hostilities, or when the task force is assigned a new area of operations, information on the enemy force is initially provided by higher headquarters. Later, the task force may have first-

hand information on the enemy, but will still require information from higher headquarters on enemy flank and second echelon forces.

0368. IPB includes a detailed analysis of the enemy and provides the commander with templates of likely enemy positions and enemy ground and air avenues of approach, as well as other information. The S2 begins the IPB process. In conjunction with the S3 and the commander, he develops probable time guidelines, templates of enemy positions or formations, Named Areas of Interest (NAIs) and Target Areas of Interest (TAIs). NAIs are areas that will most likely give indicators of enemy intentions; TAIs are areas to be attacked that will most likely have a damaging effect on enemy forces or plans.

0369. The task force uses its scouts, infantry patrols, and any supporting ground surveillance radars to verify enemy information. Additionally, maneuver companies' reports, calls for fire monitored by the Fire Support Element (FSE), and adjacent unit reports are reviewed to verify information and to disclose the enemy's intent.

0370. **Terrain and Weather:** Terrain analysis is the process of examining the military aspects of terrain and their effects on friendly and enemy capabilities to move, shoot, and communicate. The military aspects of terrain include the following five factors (OCOKA): Observation and fields of fire, Cover and Concealment, Obstacles and movement, Key terrain, and Avenues of approach. The military aspects of terrain at battalion-level are considered in terms of platoon positions and actions.

0371. Weather affects equipment, terrain (traffic ability), visibility, and soldiers. Generally, adverse weather favors the attacker even though mobility and command and control are degraded. Task force plans and orders must accommodate adverse weather (including the effects of smoke) and take advantage of them.

0372. **Troops available.** The commander evaluates troop strength and combat status in terms of the seven battlefield operating systems and human dimension factors such as fatigue, training, morale, and leadership. Maneuver companies, fire support, and engineer assets are initially the key elements considered.

0373. **Time Available***:* Time is the critical factor in all operations. The commander gets his first indication of time available from the higher headquarters order. Specific considerations of time (and timing) include: a). coordination and planning time, b). time operation is the commencement, and therefore the time available for preparation and reconnaissance. c). movement times from present positions to sectors, battle positions, or lines of departure,

0374. Other timing considerations are considered during the war-gaming process and include: a) rate of closure of a moving enemy force, b) movement times during the operation (moving from one battle position to another; time from one phase line to the next), c) timing and duration of preparation fires or smoke, d) time to complete specific actions during the operation.

## Intelligence Preparation of the Battlefield (IPB)

0375. IPB is a systematic, continuous process of analyzing the threat and environment in a specific geographic area. It is designed to support staff estimates and military decision making. Applying the IPB process helps the commander selectively apply and maximize his combat power on the battlefield at critical points in time and space by:

a. Determining the threat's likely COA.

b. Describing the environment where a unit is operating within and the effects of the environment on your unit.

0376. IPB is a continuous process which consists of four steps which are performed each time an IPB is conducted. The steps are:

a. Define the battlefield environment.

b. Describe the battlefield's effects.

c. Evaluate the threat.

d. Determine threat COAs.

0377. IPB is conducted prior to and during the command's initial planning for an operation, but the IPB can be continued during the conduct of the operation. Each function in the process is performed continuously to ensure that:

a. The products of IPB remain complete and valid.

b. Support is provided to the commander as well as direction to the intelligence system throughout the current mission and into the preparation for the next.

0378. This section will only discuss IPB specifically applicable to armor operations. IPB is the same throughout the whole army operations and the IPB process will be discussed in detail in a separate manual (FC 2-00231).

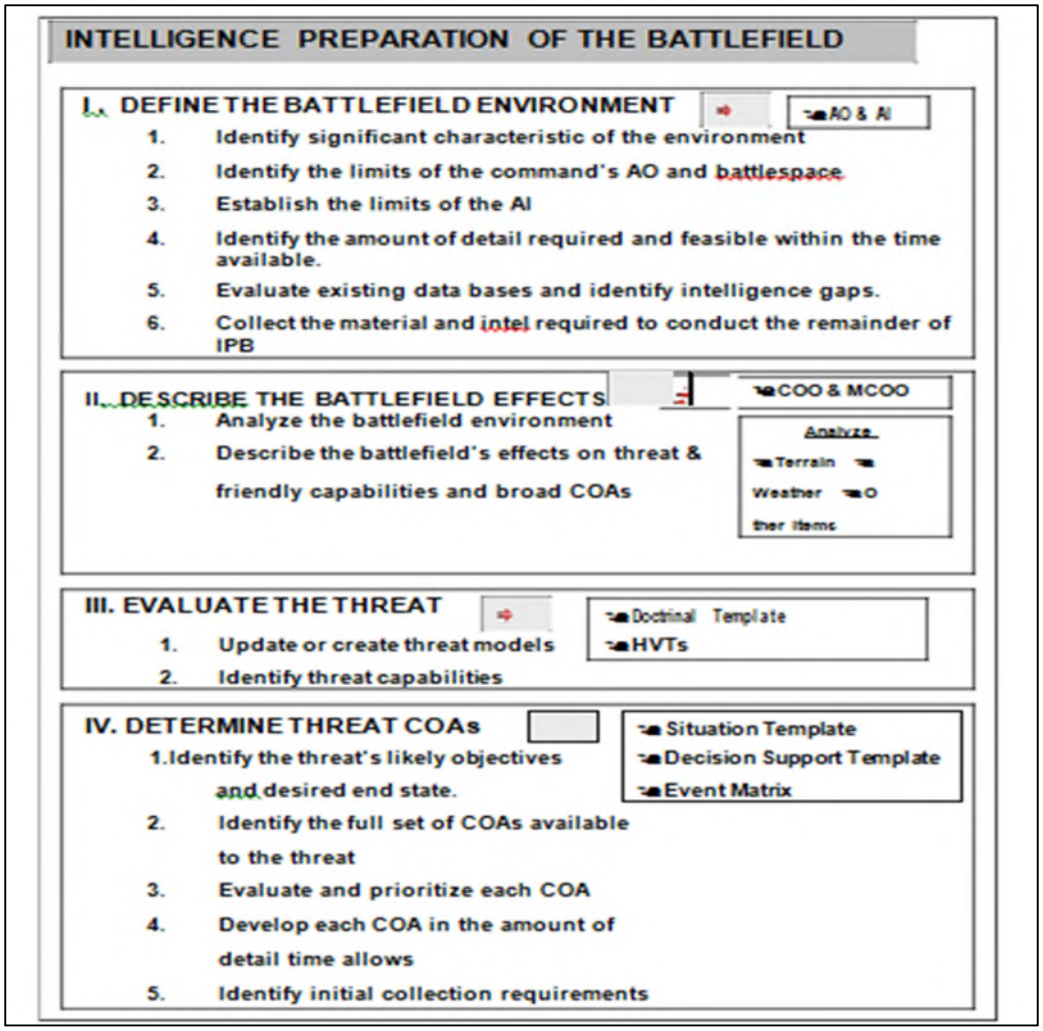



**Figure 3.2 - Intelligence Preparation of the Battlefield**

0379. G2 (S2) analyzes the terrain and determines if the armor assets can maneuver their AO effectively. Detailed analysis of terrain is important because the terrain not restricted to infantry can be severely restricted to armored vehicles, thus minimizing the effectiveness of an armor operation.

0380. Weather also greatly affects the effectiveness of armored vehicles deployment. The effectiveness of its armaments could be degraded if the target mechanism is affected by humidity and temperature. However, if the G2 (S2) were able to analyze the weather effectively, the G4 (S4) will be able to prepare instruments, like NVG or Thermal Imaging System. This avoids the degradation of the armored vehicle's target acquisition system.

0381. A detailed analysis of the enemy's capability and limitation is also important in planning armored operations. The greatest threat to any armored vehicles is the anti-tank capabilities of the enemy. Being able to successfully locate these enemy assets on the ground and determine their most likely COA for these assets could greatly reduce the risk to the armored vehicles.

# Operation Security

0382. Operation Security (OPSEC) is all measures taken to maintain security and achieve tactical surprise. Units must deny the enemy information about planned, ongoing, and after-operation activity until it is too late for the enemy to react effectively. This section will summarize OPSEC in terms of counter surveillance, information security, signal security, and physical security. Armor units are extremely vulnerable to surveillance. The Area of Responsibility (AOR), especially in COIN operations, will consist predominantly of dismounted infantry and wheeled utility vehicles.

## Counter-surveillance

0383. Counter-surveillance involves taking measures to protect friendly activities from being observed or detected (visually, electronically, or seismologically) by the enemy. The following are examples of counter-surveillance measures and guidelines that apply to them:

a. The guidelines for noise and light discipline are :1) Shield all light sources from enemy view 2) Move only when necessary 3) Use headsets or Combat Vehicle Crewman (CVC) helmets to avoid radio noise. 4) Avoid slamming hatches or doors 5) Use the short count method to start engines simultaneously 6) Perform resupply and maintenance in areas that are masked by terrain 7) Generate sandbag or shield generators 8) Use hand-and-arm signals or flag signals when possible 9) Use extra precaution at night when noise and light carry farther 10) Prohibit smoking outdoors at night

b. The guidelines for camouflaging are: 1) Place tree branches or other vegetation on vehicles; hold them in place with camouflage wire. Drape camouflage nets over turrets 2) Park vehicles in natural concealment and shadows 3) Cover all headlights, mirrors, and optics when possible 4) Consider the effects of dust and smoke when moving. 5) When possible, ensure vehicles drive in the tracks created by the vehicles ahead of them or in previously created tracks 6) Blend vehicles with other objects having a thermal emission.

c. The guidelines for concealment are: 1) Disperse vehicles and personnel under foliage or inside structures when possible 2) Conceal vehicles behind objects that can block the thermal line of sight 3) Keep vehicles traveling on existing tracks or roadways in heavily used areas, such as CPs and trains.

d. Ensure that challenges and passwords are used and enforced.

e. Consider Air Defense Artillery (ADA) coverage against enemy air reconnaissance.

f. The procedures for jamming are: 1) Consider jamming to disrupt enemy communications 2) Destroy enemy jamming, direction finding, and intercept

equipment.

g. Screen enemy observers and friendly movement with smoke.

h. In most instances, it is impossible to keep armored vehicles from being detected or observed by enemy forces because of their large signature. Deception plans should be employed so that friendly routine actions are conducted with greater uncertainty. Deception can play a significant role in making the movement of formations, and inducing the enemy to miscalculate friendly objectives or weaknesses.

## Information Security

0384. Information security is the protection of friendly information. Leaders can help prevent compromising sensitive information by:

a. Ensuring that soldiers do not send critical information in the mail.

b. Policing areas to ensure nothing of value is left behind.

c. Destroying overlays, orders, and other documents after use or when they are no longer necessary.

## Signal Security

0385. Signal security is a vital component of OPSEC procedures. Take the following steps to protect friendly communications:

a. Use secure communication means or operational and numerical codes.

b. Use low-power transmissions and terrain to mask signals from enemy direction-finding equipment.

c. Keep radio transmissions short. Messengers or wire should be used for lengthy messages.

d. Units must practice using signal operation instructions (SOI), SOPs, and operational terms. The battalion establishes priorities for issue of SOI and extracts.

e. Protect cryptographic systems and classified documents from capture or loss. Before an area is vacated, inspect it for any materials that could provide friendly information to the enemy.

f. Patrol wire lines to prevent enemy tapping.

g. When SOI codes or cryptographic equipment is lost or captured, report the facts promptly to the next higher command. The unit SOP must contain

instructions for destruction of equipment and classified documents to prevent capture or use by the enemy.

### Physical Security

0386. Physical security is the protection of material and equipment. Some examples of physical security include:

a. Employing guards, Observation Posts (OPs), and patrols at all unit and CP sites.

b. Employing anti-intrusion devices, such as trip flares, when stationary.

## Rehearsals

0387. A rehearsal is the act or process of practicing an action in preparation for the actual performance of that action. Rehearsing key combat actions allows participants to become familiar with the operation and to translate the relatively dry recitation of the tactical plan into visual impression. This visual impression helps them orient themselves to both their environment and to other units during the execution of the operation. Moreover, the repetition of combat tasks during the rehearsal leaves a lasting mental picture of the sequence of key actions within the operation.

0388. Rehearsals allow subordinate units and leaders to analyze the tactical plan to ascertain its feasibility, its common sense, and the adequacy of its C2 measures before it is too late. For units to be effectively and efficiently employed in combat, rehearsals need to become habitual in training. All units at every level should routinely train and practice a variety of rehearsal types and techniques. Local SOPs should identify appropriate rehearsal type technique, and standards for their execution.

0389. Time is the key to conducting rehearsals and time is probably the most precious resource available to commanders and units. The time required for rehearsal varies with the complexity of the task to be rehearsed, the type and technique of rehearsal, and the level of participation. Therefore, the emphasis on rehearsals should be at the lowest level possible, using the most thorough technique possible given the available time (Figure 3.3). When time is at a premium, the staff conducts reduced rehearsals, focusing only on critical events using the backward planning cycle.

0390. During offensive operations, the staff should focus on the following actions in order:

a. the assault and action on the objective,

b. breaching operations,

c. passage of lines,

d. movement to the objective

e. other phases of the operation as needed.

0391. During defensive operations, the staff should focus on: 1) security, 2) battle handover, 3) commitment of counterattack forces on the striking force, and 4) the other phases of the operation as needed.

0392. Each unit will have different critical events based on its training status and the unit commander's evaluation of training needs.

## Rehearsal Types

0393. The five types of rehearsals are the: 1) confirmation brief, 2) back brief, 3) combined arms rehearsal, 4) support rehearsal, and 5) battle drill or SOP rehearsal. Each of the five types achieves a specific result and has a specific place in the MDMP time line.

0394. **Confirmation Brief.** The confirmation brief is routinely performed by a subordinate leader immediately after receiving any instructions, such as an OPORD, a FRAGO, and so forth. Subordinate leaders brief the higher commander on their understanding of his intent, their specific task and purpose, and the relationship between their unit's missions and the other units in the operation.

0395. **Back brief.** The back brief is normally performed throughout the MDMP. This rehearsal allows the commander to clarify his intent early in the subordinates' tactical estimate procedure. It allows the higher commander to:

a. Identify problems in his concept of operations.

b. Identify problems in a subordinate unit commander's concept.

c. Learn how subordinates intend to accomplish their mission.

0396. **Combined Arms Rehearsal.** The combined arms rehearsal is normally conducted by a maneuver unit headquarters and performed after subordinate units have issued their OPORD. This rehearsal ensures that:

a. The subordinate units' plans are synchronized with those of the other units in the organization.

b. All subordinate commanders' plans will properly achieve the intent of the higher commander.

0397. **Support Rehearsal**. Support rehearsals are normally performed within the framework of a single or limited number of operating systems. They are referred to by the primary operating system being rehearsed, for example, the Fire Support (FS) rehearsal or the SS rehearsal. Support rehearsals are performed throughout the MDMP time-line. Although these rehearsals differ slightly by operating system, they achieve the same results which are:

a. Ensure the proponents of a particular operating system that they can support the higher commander's plan and accomplish all missions assigned to them.

b. Ensure that the particular operating system support plan is synchronized with the maneuver plan.

0398. **Battle Drill or SOP Rehearsals.** A battle drill or SOP rehearsal ensures that all participants understand a technique or a specific set of procedures. These rehearsals are used by all echelons, but most extensively at platoon, squad, and section levels. They are performed throughout the MDMP time-line. This type of rehearsal is not limited to published battle drills. It could be the rehearsal of a Tactical Operations Center (TOC) shift change, an obstacle breach lane marking SOP, or the actions a Petroleum, Oils, and Lubricants (POL) section takes at a Refuel On the Move (ROM) site or in the defense of the Brigade Support Area (BSA).

## Rehearsal Techniques

0399. Techniques for performing rehearsals are limited only by the resourcefulness of the unit. Generally, six techniques are used: full dress, reduced force, terrain model, sketch map, map, and radio. These six techniques range from requiring extensive preparation time and resources to minimal requirements. As the techniques are listed, each successive technique takes a decreasing amount of time and resources to prepare and conduct. Each rehearsal technique provides different degrees of understanding to participants. Each has different security risks. Figure 3.3 shows the rehearsal techniques relative to time, resources, operational security (OPSEC), participation, and understanding.

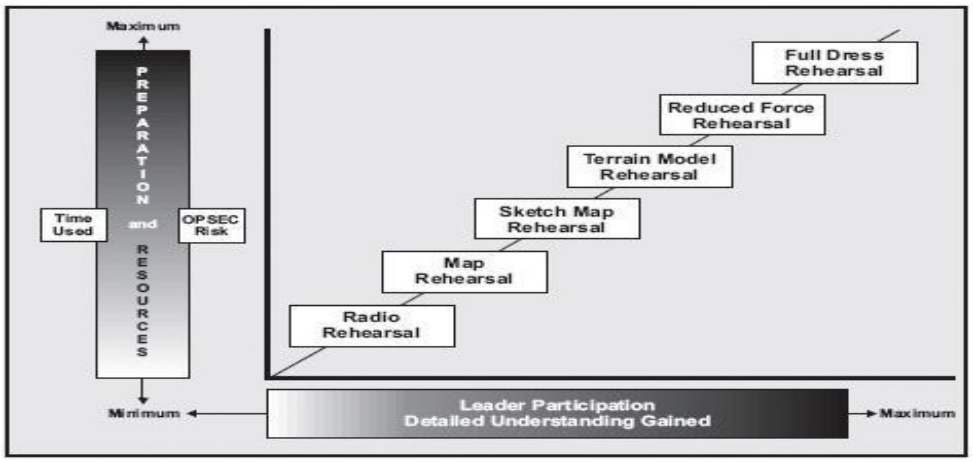



**Figure 3.3 - Rehearsal techniques relative to time, resourcing, OPSEC, participation and understanding**

03100. **Full Dress Rehearsal.** The full dress rehearsal produces the most detailed understanding of the mission. It involves every soldier and system participating in the operation. If possible, units should conduct full dress rehearsal under the same conditions: weather, time of day, terrain, and the use of live ammunition that the force will encounter during the actual operation. The full dress rehearsal is the most difficult to accomplish at higher command levels. A second rehearsal plan must be

developed that mirrors the actual plan but fits the terrain the rehearsal is on. Full dress rehearsal considerations are:

a. **Time.** Full dress rehearsals are the most time consuming of all the rehearsal techniques. The commander must not encroach on subordinate units timetables by scheduling a full dress rehearsal at his convenience. For smaller units, company and below, the full dress rehearsal is the most effective technique for ensuring everyone in the operation understands his part of the mission being performed.

b. **Multi-echelon.** A subordinate unit can perform a full dress rehearsal as part of a larger unit's reduced force rehearsal.

c. **OPSEC.** The movement of a large body of the force will attract attention from the enemy. Units must develop a plan to protect the rehearsal from the eyes of the enemy. One method is to develop a second plan to include graphics and radio frequencies. This must be done carefully to avoid confusing subordinates or giving the plan away if the enemy observes the rehearsal.

d. **Terrain.** Terrain management for the full dress technique can be difficult if it is not integrated into the initial array of forces. The rehearsal area must be identified, secured, cleared, and maintained throughout the rehearsal process.

03101. **Reduced Force Rehearsal.** The reduced force rehearsal normally takes fewer resources than a full dress rehearsal because it involves only the unit's and subordinate unit's key leaders. Terrain requirements can similar with a full dress rehearsal, but there are fewer participants. The commander first decides the level of leader involvement he desires. His selected leaders then rehearse the plan while traversing the actual or like terrain. Commanders often use this technique to rehearse the fire control measures in an engagement area during defensive operations. The reduced force rehearsal allows the leadership to rehearse the mission before moving to the full dress rehearsal. A second rehearsal plan must be developed that mirrors the actual plan but fits the terrain the rehearsal is on. Reduced force rehearsal considerations are:

a. **Time.** The reduced force rehearsal normally requires less time than the full dress technique. Again, the commander must not encroach on the timetables of subordinate units by scheduling the rehearsal at his convenience.

b. **Multi-echelon.** A small subordinate unit can perform a full dress rehearsal as part of a larger unit's reduced force rehearsal.

c. **OPSEC.** This rehearsal is less likely to become an OPSEC problem than the full dress rehearsal because the unit is smaller, but the number of radio transmissions stays the same as the full dress rehearsal and remains a consideration.

d. **Terrain.** Terrain management for the reduced force rehearsal can be just as difficult as for the full dress rehearsal. The rehearsal area must be identified,

secured, cleared, and maintained throughout the rehearsal process.

03102. **Terrain Model Rehearsal.** The terrain model rehearsal takes less time and fewer resources than the reduced force or key leader rehearsal, and is the most popular technique. An accurately constructed terrain model can help subordinate leaders visualize the battle in accordance with their commander's intentions. When possible, the commander should place the terrain model where it overlooks the actual terrain of the area of operations. However, if the situation requires more security, he places the terrain model on the reverse slope of a vantage point within walking distance of a point overlooking the area of operations.

03103. The model's orientation should coincide with the actual orientation of the terrain to help participants be oriented to the actual area of operations. The size of the terrain model can vary from the use of icons to represent units to a large model on which the participants can walk. A large model helps reinforce the participants' perception of unit positions on the actual terrain. Terrain model considerations are:

a. **Time.** The most time-consuming part of the technique can be the terrain model's construction. Units must have a clear SOP stating how it will be built to ensure the model is accurate, large enough, and sufficient in detail to rehearse the mission. The SOP must also state who shall build it and when it shall be built.

b. **Multi-echelon.** Because the terrain model rehearsal takes less time than the real operation, multi-echelon rehearsals are difficult.

c. **OPSEC.** This rehearsal can become an OPSEC problem if the area around the rehearsal site is not secured. The collection of commanders and their vehicles can draw the enemy's attention. Upon completion of the rehearsal, the terrain model must be sanitized.

d. **Terrain.** Terrain management is less difficult than with the previous techniques. The location of the site must be easy to find for the friendly commanders, yet invisible to the enemy. The optimum location is overlooking the terrain on which the mission will be performed.

03104. **Sketch Map Rehearsal.** Units can use the sketch map technique almost anywhere, day or night. The procedures are the same as for a terrain model rehearsal, except the commander uses a sketch in place of a model. Sketches must be large enough for all participants to see as each subordinate "walks" through the interactive verbal execution of the operation. Units move symbols to represent their maneuver and location on the sketch. Sketch map considerations are:

a. **OPSEC**. This rehearsal, like the terrain model rehearsal, can become an OPSEC problem if it is performed outside and the area around the rehearsal site is not secured. Another concern is that the collection of commanders and their vehicles that can bring attention from the enemy.

b. **Terrain.** The optimum location is overlooking the terrain on which the

mission will be performed.

03105. **Map Rehearsal.** The map rehearsal procedures are similar to the sketch map rehearsal, except the commander uses a map and operation overlay of the same scale as he used to plan and control the operation. Map rehearsal considerations are:

a. **Time.** The most time-consuming part is the rehearsal itself. The setup for this rehearsal is normally the easiest because it only requires the maps and the current operational graphics.

b. **OPSEC.** This technique can become an OPSEC problem if it is performed outside and the area around the rehearsal site is not secured. Another concern is that the collection of commanders and their vehicles that can bring attention from the enemy.

c. **Terrain.** The optimum location is overlooking the terrain on which the mission will be performed.

03106. **Radio Rehearsal.** The commander and his staff conduct radio rehearsals by interactively and verbally executing critical portions of the operation over established communications networks. This is accomplished in a general sequence of events which the commander establishes. Because of the obvious dangers involved in using this rehearsal, only the essential and most critical portions of the operation are rehearsed. When used, these rehearsals include all communications facilities and equipment necessary to conduct that portion of the operation. To be effective, all participants must have working communications equipment and a copy of the OPORD and overlays. The TOC can rehearse tracking the battle simultaneously. Radio rehearsal considerations are:

a. **Time.** This technique can be very time-consuming if the unit does not have a clear SOP for performing it and if all units have working communications and are up on the net.

b. **OPSEC.** This rehearsal can become an OPSEC problem because of the volume of the radio transmissions and potential compromise of information through enemy radio monitoring. Different frequencies should be used to protect the ones to be used for the operation. The use of wire systems is an option but does not exercise the radio systems, which is the strong point of the radio rehearsal.

## Rehearsal Responsibilities

03107. This discussion of rehearsal responsibilities addresses planning, preparation, and execution. The discussion is developed around the combined arms rehearsal. The responsibilities do not change for support rehearsals, only the position titles. For example, the G3 (S3) becomes the Security, Plans, and Operations officer (SPO).

03108. **Planning.** The armor commander provides the following information as part

of the commander's guidance during the initial mission analysis and re-evaluates it once he has selected a course of action:

a. The type of rehearsal

b. The rehearsal technique

c. The place

d. The attendees

e. The enemy COA to be portrayed

03109. The Chief of Staff or Ex-O ensures that all rehearsals are imbedded in the unit's time management system. He is responsible for:

a. Publishing the rehearsal time and location in the OPORD or in a warning order.

b. Completing any rehearsals with the staff.

c. Determining rehearsal products based on type, technique, and METT-TC.

d. Coordinating Liaison Officer (LNO) attendance from adjacent units.

03110. **Preparation.** Ideally, the armor commander prepares to rehearse the mission with events phased in proper order from start to finish. When time is short, this is not always possible. The commander:

a. Identifies and prioritizes key events to be rehearsed.

b. Allocates time for the events being rehearsed.

c. Conducts personal preparation, to include reviews of: 1) completeness of task organization, 2) readiness of personnel and material, 3) unit level of preparation for the assigned mission.

03111. The Chief of Staff or Ex-O, through war gaming and coordination with the commander:

a. Coordinates and allocates time for the key events requiring rehearsal.

b. Establishes rehearsal time limits according to the commander's guidance and METT-T.

c. Verifies rehearsal site preparation. A separate rehearsal site may be required for key rehearsal events such as an enlarged objective area or a possible obstacle site. The rehearsal site must be accurate and complete with: 1)

appropriate markings and associated training aids, 2) parking areas, 3) local security.

d. Determines the method for controlling the rehearsal and ensuring its logical flow.

03112. Subordinate leaders complete their planning process, to include:

a. Complete unit order and plans.

b. Identify issues derived from the parent unit order.

c. Provide copy of their unit order with graphics to the parent unit.

d. Conduct personal preparation similar to that of the senior commander.

03113. Higher headquarters responsibilities include:

a. Providing all subordinate unit graphics. These composite overlays are the first step for leaders to visualize the whole unit's plan.

b. Publish composite overlays at the rehearsal, to include, at a minimum: 1) Maneuver 2) Fire Support 3) Mobility and Survivability (M/S) 4) Sustainment Support.

03114. **Execution.** The armor commander must command the rehearsal, just as he will command the fight. He must maintain the focus and level of intensity, allowing absolutely no potential for subordinate confusion. Although the staff refined the plan, it belongs to the commander; he must use it to fight. The rehearsal cannot become his brief to subordinate commanders. The rehearsal's purpose is to validate synchronization: what, when, and where of subordinate units' tasks to execute the commander's intent.

03115. The Chief of Staff (Ex-O) should direct the rehearsal. If the G3 (S3) is the director, the Ex-O will not sense the intricacies necessary to synchronize the combined arms team. The TOC then becomes a mere site for situation maps, not the proactive agent that molds the force's effects to achieve the commander's vision for success. The C of S (Ex-O) must:

a. Conduct a formal roll call and ensure that everyone brings the necessary equipment, to include unit graphics and previously issued orders, to help adjust to the rehearsal plan.

b. Validate task organization for the mission. Linkups must be complete on schedule and required material and personnel must be on hand. The importance of this simple element cannot be overemphasized.

c. Rehearse the synchronization of combat power from flank, higher, and his

own units, which are often beyond communications' reach of the commander and the G3 (S3).

d. Synchronize the timing and contribution of each operating system by ensuring the rehearsal of the decisive points, by time or event that is connected to a decision. What are the conditions required to: 1) Commit the reserve? 2) Move a unit? 3) Close or set up an obstacle? 4) Fire a specific target? 5) Move a medical station; change a supply route; alert specific Observation Post (OP); and so on?

e. Discipline leader movements, enforce brevity, and ensure completeness at the rehearsal. The OPORD, Decision Support Template (DST), and synchronization matrix are his tools.

f. Keep within established time constraints.

g. Ensure that selected events receive appropriate attention.

h. Ensure that absentees and flank units receive changes. Changes must be transmitted by courier or radio immediately.

03116. The G3 (S3) assists the commander in the fight forward and should rehearse that task. The G3 (S3) must:

a. Portray his actions during the fight.

b. Ensure compliance with the plan.

03117. The G2 (S2) bases his execution actions on the enemy COA the commander selected during the planning process. The G2 (S2) must:

a. Portray his best assessment of the enemy COA.

b. Communicate the enemy commander's presumed concept of operation, desired effects, and intended end state.

03118. Using an established format subordinate unit leaders must:

a. Effectively articulate their units' actions and responsibilities.

b. Record changes on their copies of the graphics or OPORD.

03119. After the rehearsal is complete the recorder must:

a. Restate any changes, coordination, or clarifications directed by the commander.

b. Estimate the time that a written Fragmentary Order (FRAGO) codifying the

changes will follow.

03120. If the rehearsal is executed properly, leader participation should validate each leader's role as part of the whole force what is done, when relative to time and event, and where to achieve desired effects. The rehearsal ensures that commanders have a common visualization of the enemy, their own forces, and the terrain, and the relationship between them. It identifies specific actions requiring immediate staff resolution and informs the higher commander on critical issues or locations he or his C of S (Ex-O) or G2 (S2) must personally oversee.

## Rehearsal Script

03121. An effective technique for controlling the rehearsals is to use a script. It keeps the rehearsal on track and is a checklist to ensure all operating systems and outstanding issues are addressed during the rehearsal. The script has four major parts:

a. The Agenda

b. Response Sequence.

c. Actions Checklist.

d. Sequence of Events

03122. **The Agenda.** Units must rehearse using the tools they will use when fighting the battle: the OPORD, the synchronization matrix, and the DST. These can both drive and focus the rehearsal. During fire support or SS rehearsals, the fire support execution matrix and the logistics synchronization matrix can be used. The two products are tied directly to the corresponding level synchronization matrix and DST.

03123. **Response Sequence.** The players must respond in a logical sequence. The sequence might be by operating system, or by unit as the organization is deployed from front to rear. The sequence must be determined before the rehearsal and posted at the rehearsal site.

03124. **Actions Checklist.** The enemy force must be portrayed effectively and quickly without being distracted from the rehearsal. A technique is to establish a sequence much like that of the friendly units but coming from the enemy perspective.

03125. **Sequence of Events.** The following paragraphs provide a generic sequence of events for a rehearsal. Although developed for a combined arms rehearsal, with a few modifications, this sequence can be used for Fire Support (FS) and Sustainment Support (SS) rehearsals. This example fits brigade, task force/battalion, company/team level rehearsals and will support any rehearsal technique. The agenda's products depend on the rehearsal type. The response sequence must be determined up front and posted to keep the rehearsal moving.

a. **Step 1**. Ground Rules: 1) Roll Call, start on time. 2) Quickly review your SOP

if you have new players at the rehearsal. 3) Ensure a recorder is ready. 4) Provide an orientation to the rehearsal product and important graphic control measures. 5) Designate the rehearsal start time in relation to the operation. One event prior to the first event being rehearsed allows for proper deployment of forces. 6) Ensure everyone understands the parts of the plans to be rehearsed. 7) Update both friendly and enemy activities if necessary. Some units may already be executing activities. 8) Finish on time.

b. **Step 2**. Deployment of the Enemy. Deploy the enemy on the rehearsal product as they would look just prior to the first rehearsal event. Restating the enemy equipment may be a waste of time by this point.

c. **Step 3**. Deployment of the Friendly Forces/Units. Deploy the friendly forces, including adjacent units, to the point in time the rehearsal will start. As friendly units are initially posted to the rehearsal product, they should state their task and purpose, task organization, and strength. Other special units may brief their subordinate unit positions at the start time, as well as at any particular points of emphasis, to include forward arming and refueling points (FARPs), ROM, or communications checkpoints (CCPs). Restating the commander's intent may not be necessary at this point.

d. **Step 4**. Advancement of the Enemy. Begin to advance the force that has the initiative and continue in accordance with the war-gaming sequence of action, reaction, and counteraction. In Step 2 the enemy was deployed up to the point the rehearsal will start; the enemy continues to maneuver from there. Depiction must be definitive, tying enemy actions to specific terrain or friendly units' actions. An accurate portrayal of the situation template developed for the staff war-gaming process must be communicated. The enemy is uncooperative, but not invincible.

e. **Step 5**. Decision Point. Upon completion of the enemy action, conditions shall be assessed to determine if a decision point has been reached. These are the decision points taken directly from the DST: 1) Not at a Decision Point. If the unit is not at a decision point, and not at the end state, the unit continues by stating the next event on the synchronization matrix and, using the predetermined sequence, continues to act out and verbalize the actions. This is represented by the right side of the synchronization matrix. 2) At a Decision Point. If the unit has reached a decision point, the commander, having seen the conditions being established, states his decision to continue on the current course or to select a branch. If the commander selects to stay the current course of action, the next event from the matrix is stated and the friendly units have advanced. If a branch is selected, the commander states why he has selected that branch. The first event of that branch is stated, and the rehearsal continues from that point until all events of that branch are rehearsed. As decisive points are reached, the X-O states the conditions for success.

f. **Step 6.** Branch End State Reached. Terminate the initial phase of the rehearsal after the desired end state of the branch is achieved. In an attack, this

will usually be on the objective after consolidation and casualty evacuation are complete. In the defense, this will usually be after the decisive action, such as the commitment of the reserve, and the final destruction or withdrawal of the enemy and casualty evacuations are complete.

g. **Step 7**. Recock. After the initial phase, "recock" to the situation at the first decision point. State the criteria for a decision to change the plan. Assume these criteria have been met and then refight the fight from that point forward, all the way until the desired end state is attained. Complete any coordination to ensure understanding and requirements are met. Record any changes. Go to the next decision point and assume that the criteria have been met. Repeat the previous steps until all decision points and branches have been rehearsed.

03126. **Other Considerations.** Small issues that arise during the rehearsal are not normally addressed immediately; instead they are recorded. This ensures the rehearsal's flow is not interrupted. If the issue still stands at the end of the rehearsal, it must be resolved. However, if the issue can wait until the end of the rehearsal, it may not have been an issue after all. If the issue is a war stopper, the staff must stop and accomplish the coordination immediately. This coordination is one of the key points of the rehearsals. If it is not done immediately, it will be difficult to get the word to all the players later.

03127. Key MS and SS items need to be part of the rehearsal, including plans for casualty evacuation, routes, ambulance exchange point locations, ROM, Class IV and V resupply points, logistics release points, displacement times and locations for the BSA, Enemy Prisoners of War (EPW) collection points, aviation support, and military police actions. These items should be injected into the rehearsal at the appropriate times by the coordinating staff officer. Summarizing these actions at the end of the rehearsal lessens the value of the rehearsal as a coordination tool.

03128. The staff updates the decision support matrix and provides it to each leader prior to departure. An option is to provide it prior to the rehearsal and rely on individual pen and ink changes for each update. This is the final opportunity for subordinates to identify and resolve dangling issues. The staff must ensure all coordination done at the rehearsal is clearly understood by all players and captured by the recorder. All changes to the published order are, in effect, verbal FRAGOs. As soon as possible, the staff should collect the verbal FRAGOs into a written change to the order.

## Conducting a Rehearsal

03129. Participants can use the following sequential guidelines during rehearsals:

a. Orient participants to the training aid and the terrain.

b. Define the standard (what the commander will accept as satisfactory performance for the rehearsal).
c. Visualize and synchronize the concept of operations. Verbally walk through

the concept of operations. Subordinate commanders should interactively verbalize their units' actions, entering or leaving the discussion at the time they would expect to begin or end their tasks or activities on the battlefield. This will help the commander assess the adequacy of synchronization.

d. Focus on the key events and the synchronization required to achieve the desired effect on the enemy.

e. Consider the enemy's courses of action (COA) (using the possibilities derived from the war game during COA analysis).

f. Address any points in the operation where the execution of branches or sequels is likely to occur.

g. If the standard is not met and time permits, rehearse again.

h. For feedback, make the necessary changes to the decision support template (such as positions, synchronization, or decision point locations).

## Before the Rehearsal

03130. Once participants assemble at the rehearsal site, the rehearsal leader briefs them. The briefing must include an introduction and overview as well as an orientation.

03131. **Introduction and Overview.** The armor commander, C of S (Ex-O), or G3 (S3) appropriately introduces himself and all other participants. He then gives an overview of the briefing topics, the rehearsal subjects and sequence, and the time-line (specifying the no later than [NLT]) ending time). He explains participation in after-action reviews, how and when they will occur, and how he will incorporate changes into the existing order. He explains, in detail, restrictions imposed on the force, such as the use of pyrotechnics, light discipline, weapons firing, or radio transmissions. He ensures that all participants understand safety precautions and enforces their use. Last, he emphasizes results and what standards of task execution he expects.

03132. At this time, subordinate leaders relate any results of tactical planning or rehearsals they may have already conducted. If the subordinate unit recommends a change to the existing plan, the commander, C of S (Ex-O), or G3 (S3) acts on the recommendation before the rehearsal begins, if possible. If not, the recommendation will always be resolved by a decision before the rehearsal ends.

03133. **Orientation.** The rehearsal leader gives an orientation to familiarize participants with the terrain or scale model in use. He also issues supplemental materials, if appropriate. He identifies magnetic north on the terrain model or scaled terrain and points out objects and features representing actual terrain features. He also explains any graphic control symbols, obstacles, or fire support targets. The rehearsal leader always concludes the orientation with a call for questions.

## During the Rehearsal

03134. After the briefing, the rehearsal begins according to the rehearsal plan. Before the rehearsal occurs, the staff should develop the plan or order with at least the five basic paragraphs and necessary overlays. At this time, all annexes may not yet be published, however the details should be developed. Commanders must remember that the rehearsal is not a substitute for the war game.

03135. The commander, C of S (Ex-O), or G3 (S3) observes and critiques all portions of the rehearsal. Critiques center on meeting the commander's intent and coordination among units. The internal execution of tasks within the rehearsal is almost always left to the subordinate unit commander's judgment and discretion. Leaders at all levels conduct periodic after-action reviews. This ensures that units rehearse tasks to acceptable levels of competence and that substandard performance is not reinforced.

03136. After-action reports also provide an opportunity to incorporate lessons learned into the existing plan or into subsequent rehearsals. The rehearsal leader must emphasize integration of fire support, events which trigger different branch actions, and actions on contact. If units in reserve participate, they should rehearse all of their most likely branches.

03137. Rehearsals continue until units are competent or until the maximum time available expires. (Before the commander adjusts rehearsal times, he must consider how his decision will impact a subordinate commander's time plan.) Subsequent rehearsals may employ additional complexity and realism as time and the commander permit.

03138. The commander leads the rehearsal; his staff runs it. The director of the rehearsal is the C of S (Ex-O). As such, he rehearses his role during the operation. He ensures tasks are accomplished by the right unit at the right time and cues the commander to upcoming decisions. The C of S's (Ex-O's) script is the synchronization matrix and the decision support template. These are the foundations for the OPORD and are recorded in chronological order. A terrain model rehearsal takes a proficient brigade from one to two hours to execute to standard. The following steps outline the process for conducting a rehearsal:

a. **Step 1.** Start at the appointed time and conduct a formal roll call. Ensure everyone brings binoculars, maps, and necessary equipment.

b. **Step 2**. Ensure that the C of S (Ex-O) or the G3 (S3) orients the terrain model to the actual ground, the operations overlay, and the map. Describe and point out the overall area of operations and explain the markers used on the terrain model.

c. **Step 3.** Brief the time-line. The C of S (Ex-O) or the G3 (S3), in lieu of the C of S (Ex-O), should do this. Designate the rehearsal start time. For example, have the rehearsal begin by depicting the anticipated situation one hour before departure. Set the time interval to be used to begin and track the rehearsal. For

example, specify a 10-minute interval to equate to 1 hour of real time during the operation.

d. **Step 4.** Designate the recorder. This should be the G3 (S3), or his designated representative from the operations cell. Highlight the ground rules and incorporate ground rules into an SOP. The ground rules include who controls the rehearsal, who actually walks the terrain board, and when special staff officers will brief the commander. Special staff officers should brief by exception when a friendly or enemy event occurs within their operating system.

e. **Step 5.** The G3 (S3) reads the mission statement, the commander reads his commander's intent, and the G3 (S3) lays out the friendly situation as it currently exists, using the terrain model.

f. **Step 6.** The G2 (S2) briefs the current enemy situation. He then briefs the most likely enemy course of action. (The enemy situation should already be set up on the terrain model.) The G2 (S2) also briefs the status of the reconnaissance and surveillance plan (for example, citing any patrols that are still out or any observation post positions or combat outposts).

g. **Step 7.** The G3 (S3) briefs friendly maneuver unit dispositions at the rehearsal start time, including security forces. Other staff officers brief their subordinate unit positions at the start time, as well as at any particular points of emphasis. For example, the chemical officer briefs the Mission Oriented Protective Posture (MOPP) level and the fire support officer details the range of friendly and enemy artillery.

h. **Step 8.** The armor commander gives appropriate commands. Fire Support Officers and Fire Support Coordinator (FSCOORDs) tell when they initiate fires, which is firing, where the fire originates, the ammunition, and the desired target effect. Armor commanders tell when they initiate fire in accordance with their fire support plans. The C of S (Ex-O) talks for any staff section not present and ensures that all actions on the synchronization matrix or decision support template are addressed at the proper time or event. Everyone should avoid repeating the war-gaming except as absolutely necessary to ensure subordinate unit commanders understand the plan. If the staff has developed an order that addresses contingencies, they do not war-game the operation at the rehearsal site.

i. **Step 9.** The G2 (S2) section portrays the enemy and walks the enemy through the most likely course of action (situation template), stressing reconnaissance routes, objectives, security force composition and locations, initial contact, initial fires (artillery, air, attack helicopters), probable main force objectives or engagement areas, likely chemical attack times and locations, and likely commitment of reserves. The G2 (S2) section must be specific, tying enemy actions to specific terrain or friendly unit actions. The walk-through should be an accurate portrayal of the event template.

j. **Step 10.** Terminate the first phase of the rehearsal after achieving the desired end state (from the commander's intent). In the offense, this is usually on the objective after consolidation. In the defense, this is usually after the decisive action, such as the commitment of the reserve and the final destruction or withdrawal of the enemy.

k. **Step 11.** When it becomes obvious that the operation requires additional coordination to ensure success, the force must immediately try to accomplish it. This coordination is one of the key points of the rehearsal. The commander and staff must ensure that coordination is understood by all participants and captured by the recorder and that all changes to the published OPORD are in effect. However, this is not the time to make major changes. Changes are kept only to those that are vital. As soon as possible, the G3 (S3) should collect the verbal FRAGOs into a written change to the OPORD.

l. **Step 12.** After the initial walk-through of the base order, participants need to step back to the situation at the initial decision point. The commander should state the criteria for a decision to change the plan. The participants should assume these criteria have been met and then refight the fight from that point forward until they attain the desired end state. They complete coordination to ensure the plan is understood and requirements are met, and any changes are recorded.

m. **Step 13.** Participants then go to the next decision point and ensure that the criteria for that action have been met. Then they repeat step 12.

n. **Step 14.** The staff repeats step 13 until they have rehearsed all decision points.

o. **Step 15.** The staff briefs key MS and SS actions. These items should be integrated into the rehearsal at the appropriate times. Summarizing these actions at the end of the rehearsal adds to the value of the rehearsal as a coordination tool.

p. **Step 16.** After the rehearsal is complete, the recorder restates any changes, coordination, or clarifications the commander directs, and the recorder estimates how long it will take to codify changes in a written FRAGO.

q. **Step 17**. The armor commander should stress any points needing additional emphasis. He should consider reiterating his intent (purpose, method, end state).

## After the Rehearsal

03139. After the rehearsal, the commander reassembles participants to conduct an after-action review. The armor commander reviews lessons learned and makes only the absolute minimum-required modifications to the existing plan. (Normally, these changes are effected by issuing a FRAGO.) This meeting also allows the commander to issue any last minute instructions or reminders and to reiterate his intent.

03140. Subordinate commanders incorporate any changes the commander makes to the existing plan into their units' orders and plans. Such changes are also briefed to any key leader or unit that did not participate in the rehearsal. Changes to the plan should be refinements to that plan; they should not be radical or significant. Changes not critical to the operation's execution can confuse subordinates and desynchronize the plan.

# Chapter IV
# Offensive Operations

0401. The offense is the decisive form of war. The will to seize, retain, and exploit the initiative defines the spirit and purpose of the offense. It is essential to success in all operations (defensive, stability, and support) as well as offensive. Combined with demonstrated combat capability, it makes Armor forces credible in any situation. Circumstances may require defending; however, victory requires shifting to the offense as soon as possible. The offense ends when the force accomplishes the mission, reaches a limit of advance, or approaches culmination. It then consolidates, resumes the attack, or prepares for another operation.

0402. Because tactical offensive operations often expose the attacker, they normally require local superior combat power at the point of attack. This, and the need to have sufficient force available to exploit success, implies taking risk elsewhere. Armor Commanders build up sufficient combat power in their main effort to overwhelm the enemy at the critical time and place. At the point of their attack, commanders avoid the enemy's main strength, turning him out of his defensive positions, isolating his forces from sources of support, and forcing him to fight in an unintended direction over ground he has not prepared. By attacking this way, the disadvantages of fighting exposed shift to the defender.

## Fundamentals of Armor Offensive Operations

0403. Armor forces are employed using two distinct methods: as the direct or semi-indirect fire support element or as the maneuver element. Timing, protection, mobility, and speed are the key elements that must be synchronized between the mounted and the dismounted elements. Armor leaders must understand and employ the fundamentals of offensive operations:

a. **Fight as a combined arms team.** Armor always fights as part of a combined arms team. Capabilities and limitations of the armor and infantry make them complementary when they are employed as a team.

b. **Know the enemy.** The armor leader must know and understand the capabilities of the enemy’s weapons systems and defensive doctrine, including the enemy’s capabilities to conduct ambushes.

c. **See the battlefield.** The armor leader must know and identify key terrain. He should also learn to identify covered and concealed routes during movement. He must also anticipate how the enemy will use the terrain and determine how to counter it.

d. **Use weapons system to the best advantage.** The armor leader must know the capabilities and limitations of his weapon systems. Knowing the best kill probability ranges of all weapons (main gun, machine guns, special weapon

attachments, and light infantry weapons) plays a vital role. Leaders can improve kill probabilities by engaging enemy vehicle flanks.

e. **Concentrate combat power.** The armor leader must be able to control and concentrate weapon systems. To do this, he uses proper command and control techniques and trains his unit to move, shoot and communicate under all types of conditions. The leader must also make maximum use of available indirect fires.

f. **Use maneuver to best advantage.** Armor must move rapidly, strike first, and maintain the momentum until the enemy has been killed or captured.

g. **Suppress the enemy's defense fires.** As the attacking forces concentrate, they become increasingly vulnerable to enemy fires. The maneuver must therefore coincide with suppressive strikes by the enemy weapons system that could interfere with the movement. Priorities for suppressions should be made to ensure the proper allocation of resources. The suppression of enemy artillery by an attack or counter - battery is essential to avoid high losses. Suppressions of anti-tank missiles and tanks at the point of attack by direct fire, artillery and mortars will help maintain the momentum of the attack and reduce friendly casualties.

h. **Shock, overwhelm and destroy the enemy.** The attack maximizes on speed, suppression and violence. Since the attack is at the weakest point of the enemy, the attack must be narrow and in-depth. Suppressive fires, supporting attack, and close air support should coincide with the maneuver elements. Attacking units, which become stalled or disorganized, must be bypassed by fresh formation pressing on to deeper objectives. The attacker must be alert to changing situations and must fully exploit the weaknesses revealed to the attacker. Advancing units must bypass points of resistance, striking deep and fast.

i. **Attack the enemy rear.** The most decisive offense is that which strikes with overwhelming force into the enemy's rear, and destroys or captures his service support, combat support, and command and control elements. These elements are least likely to be protected and armed. Disrupting enemy command and control, interrupting the flow of fuel, ammunition, repair parts, and food, and interfering with enemy air defense and artillery support weakens or destroys the whole system of defense and makes it possible to the defeat the enemy with fewer forces.

j. **Coordinate continuous support.** Armor leaders must always be aware of their logistical status. Logistic support is extremely difficult in the austere environment of light forces. Armor leaders must understand the procedures for critical support, such as evacuation of personnel and equipment or resupply of repair parts, ammunition and fuel. This is important because organizational changes are very frequent in the realm of light forces operations.

k. **Be flexible.** Armor leaders achieve flexibility by ensuring units are properly trained, adhering to standing operating procedures (SOP) and battle drills, and becoming technically and tactically proficient. They must understand the commander's intent and anticipate changes in the fluid environment of war and thus ensure that the mission is accomplished.

# Purposes of Armor Offensive Operations

0404. Armor offensive operations seek to seize, retain, and exploit the initiative to defeat the enemy decisively. Armor forces attack simultaneously throughout the area of operations (AO) to throw enemies off balance, overwhelm their capabilities, disrupt their defenses, and ensure their defeat or destruction. The offense ends when the force achieves the purpose of the operation, reaches a limit of advance, or approaches culmination. Armor forces conclude a phase of an offensive by consolidating gains, resuming the attack, or preparing for future operations. Additional tasks offensive operations accomplish include:

a. Disrupting enemy coherence.

b. Securing or seizing terrain.

c. Denying the enemy resources.

d. Fixing the enemy.

e. Gaining information.

## Armor Offensive Operations at the Operational and Tactical Levels of War

0405. Armor operational commanders conduct offensive campaigns and major operations to achieve theater-level effects based on tactical actions. They concentrate on designing offensive land operations. They determine what objectives will achieve decisive results; where forces will operate; the relationships among subordinate forces in time, space, and purpose; and where to apply the decisive effort. Operational commanders assign AOs to, and establish command and support relationships among, tactical commanders. Tactical commanders direct offensive operations to achieve objectives (destroying enemy forces or seizing terrain) that produce the theater-level effects operational commanders require.

a. **Operational Offense.** At the operational level, offensive operations directly or indirectly attack the enemy is center of gravity. Armor Commanders do this by attacking enemy decisive points, either simultaneously or sequentially. Massed effects of combined arms operation allow attackers to seize the initiative. They deny the enemy freedom of action, disrupt his sources of strength, and create the conditions for operational and tactical success. To attain unity of effort, operational commanders clearly identify objectives and

reinforce the relationships among subordinate forces. By minimizing interoperability challenges and harnessing system capabilities, commanders tailor their forces to achieve decisive effects. They allocate sufficient combined arms forces to achieve their objectives.

b. **Tactical Offense.** Armor tactical commanders exploit the effects that combined arms forces contribute to the offense. They synchronize these forces in time, space, resources, purpose, and action to mass the effects of combat power at decisive points. Armor commanders direct battles as part of major operations. Battles are related in purpose to the operational commander's objectives.

0406. Battles may be linear or nonlinear, and are conducted in contiguous or noncontiguous AOs. Tactical commanders receive their AO, mission, objectives, boundaries, control measures, and intent from their higher commander. They determine the decisive, shaping, and sustaining operations within their AO. They direct fires and maneuver to attack and destroy the enemy and attain terrain objectives. Tactical commanders normally have clearly defined tasks—defeat the enemy and occupy the objective.

# Characteristics of Armor Offensive Operations

0407. Surprise, concentration, tempo, and audacity characterize the offense. Effective armor offensive operations capitalize on accurate intelligence and other relevant information regarding enemy forces, weather, and terrain. Armor commanders maneuver their forces to advantageous positions before contact. Force protection, including defensive Information Operations (IO), keeps or inhibits the enemy from acquiring accurate information about friendly forces. The enemy only sees what the friendly commander wants him to see. Contact with enemy forces before the decisive operation is deliberate, designed to shape the optimum situation for the decisive operation. The decisive operation is a sudden, shattering action that capitalizes on subordinate initiative and a common operational picture (COP) to expand throughout the AO. Commanders execute violently without hesitation to break the enemy's will or destroy him.

## Surprise

0408. Armor units achieve surprise when the enemy cannot react effectively to the commander's scheme of maneuver. It is achieved by conducting thorough reconnaissance and surveillance, striking the enemy from an unexpected direction at an unexpected time and by employing deception efforts.

0409. Operational and tactical surprise complements each other. Operational surprise creates the conditions for successful tactical operations. Tactical surprise can cause the enemy to hesitate or misjudge a situation. But tactical surprise is fleeting. Commanders must exploit it before the enemy realizes what is happening.

0410. Airborne, air assault, and special operations forces (SOF) attacks (combined

with strikes by Armor units and joint fires against objectives the enemy regards as secure) to create disconcerting psychological effects on the enemy.

0411. Surprise can come from an unexpected change in tempo. Tempo may be slow at first, creating the conditions for later acceleration that catches the enemy off guard and throws him off balance. Accelerated tempo results in operational and tactical surprise, despite increased publicity and heightened tensions beforehand.

0412. Armor commanders conceal the concentration of their forces. Units mask activity that might reveal the direction or timing of an attack. Armor commanders direct action to deceive the enemy and deny his ability to collect information.

## Concentration

0413. Concentration is the ability of armor forces to mass and synchronize overwhelming combat power and shock effects against an enemy weakness without unnecessarily massing large troops. Armor commanders designate a point of main effort and focus available resources to support it. Concentration is achieved by fixing the enemy to prevent his reaction to maneuver, rapidly massing forces and fires to overwhelm the enemy defense, and synchronizing maneuver with combat support.

0414. Concentration requires coordination with other services. At every stage of an attack, commanders integrate joint intelligence assets with joint fires. They capitalize on air superiority to deny the enemy the ability to detect or strike friendly forces from the air. Armor commanders direct ground, air, and sea resources to delay, disrupt, or destroy enemy reconnaissance elements or capabilities. They also direct security, IO, and counter fire to protect friendly forces as they concentrate.

## Tempo

0415. Armor forces move quickly to take advantage of enemy weaknesses. Speed in execution is the key to denying the enemy time to reposition or reorient to meet an attack. Armor commanders should strive for a tempo that would maintain relentless pressure on the enemy to prevent him from recovering from the shock and effects of the attack.

0416. Controlling or altering tempo is necessary to retain the initiative. At the operational level, a faster tempo allows attackers to disrupt enemy defensive plans by achieving results quicker than the enemy can respond. At the tactical level, a faster tempo allows attackers to quickly penetrate barriers and defenses and destroy enemy forces in depth before they can react.

0417. Armor commanders adjust tempo as tactical situations, Sustainment Support (SS) necessity, or operational opportunities allow to ensure synchronization and proper coordination, but not at the expense of losing opportunities to defeat the enemy. Rapid tempo demands quick decisions. It denies the enemy the chance to rest and continually creates opportunities.

## Audacity

0418. Audacity is the willingness to risk bold action to win. The armor commander is quick and decisive, and willing to take prudent risks. He bases his decisions on sound tactical judgment, personal observation of the terrain, and first-hand knowledge of the battle. He constantly seeks to attack the enemy on the flanks or rear and to rapidly exploit success. He shares the hazards of the battlefield with his troops, moving to critical places to lead by example. Armor commanders should understand when and where they are taking risks but must not become tentative in the execution of their plan.

# Armor Offensive Operations within the Operational Framework

0419. Armor commanders conduct offensive operations within the operational framework (AO, battle space, and battlefield organization). They synchronize their forces in time, space, resources, purpose, and action to conduct simultaneous and sequential decisive, shaping, and sustaining operations in depth (see Figure 4.1). In certain situations, commanders designate deep, close, and rear areas.

## Decisive Operations in the Offense

0420. Decisive armor offensive operations are attacks that conclusively determine the outcome of major operations, battles, and engagements. At the operational level, decisive operations achieve the goals of each phase of a campaign. Ground operations within campaigns may include several phases. Within each phase is a decisive operation. Its results substantially affect the course of the campaign. At the tactical level, decisive battles or engagements achieve the purpose of the higher headquarters mission. Armor commanders win decisive operations through close combat that physically destroys the enemy; overcomes his will to resist; or seizes, occupies, and retains terrain.

0421. Armor commanders weigh the decisive operation with additional resources and by skillful maneuver. For example, armor commanders may fix part of the enemy force with a frontal attack (a shaping operation), while the majority of the force envelops it to seize a decisive point. He decide when, where, and if additional supporting fires and reserves are to be committed. He shifts priority of fires as necessary. Maneuvering forces positions them to mass fires against the enemy.

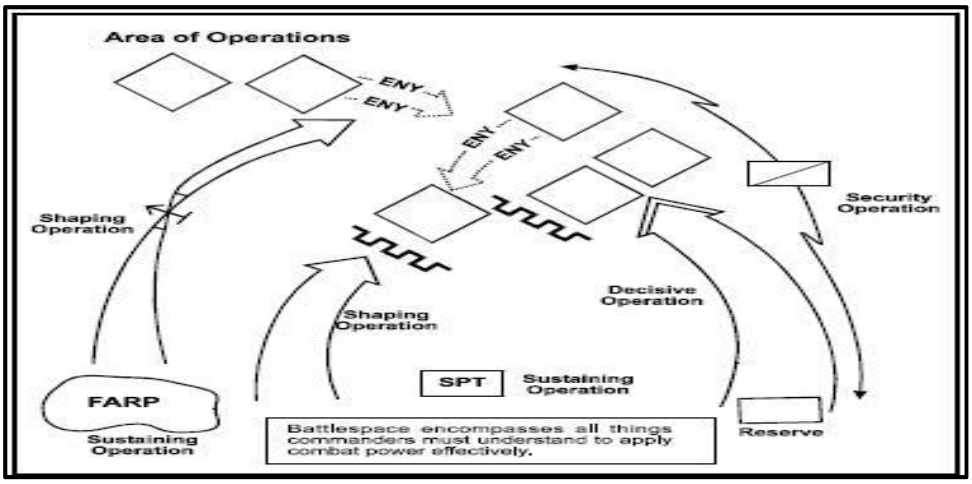



**Figure 4.1 Armor operational frameworks in the offense**

0422. Armor commanders designate a reserve to provide additional combat power at the decisive time and place. The more uncertain the situation is, the larger the reserve. Once the reserve is committed, the commander designates another. The initial strength and location of reserves vary with:

a. Potential missions, branches, and sequels

b. Form of maneuver

c. Possible enemy actions

d. Degree of uncertainty

0423. Reserves provide a hedge against uncertainty. Armor commanders assign them only those tasks necessary to prepare for their potential mission. Only the armor commander who designates the reserve can commit it, unless he specifically delegates that authority.

## Shaping Operations in the Offense

0424. Shaping operations create conditions for the success of the decisive operation. They include attacks in depth to secure advantages for the decisive operation and to protect the force. Armor commanders conduct shaping operations by engaging enemy forces simultaneously throughout the AO. These attacks deny the enemy freedom of action and disrupt or destroy the coherence and tempo of his operations. Attacking enemy formations in depth destroys, delays, disrupts, or diverts enemy combat power. They may expose or create vulnerabilities for exploitation. Shaping operations in the offense include:

a. Shaping attacks designed to achieve one or more of the following: 1) Deceive the enemy, 2) Destroy or fix enemy forces that could interfere with the decisive operation, 3) Control terrain whose occupation by the enemy would

hinder the decisive operation and 4) Force the enemy to commit reserves prematurely or into an indecisive area.

b. Reconnaissance and security operations

c. Passages of lines

d. Breaching operations

e. Unit movements that directly facilitate shaping and decisive operations.

f. Operations by reserve forces before their commitment.

g. Interdiction by ground and air movement and fires, singularly or in combination.

h. Offensive IO

0425. Other shaping operations include activities in depth, such as counter fire and defensive IO. These shaping operations focus on effects that create the conditions for successful decisive operations.

0426. The advance, flank, or rear security forces conduct security operations. These elements are:

a. Provide an early warning.

b. Find gaps in defenses.

c. Provide time to react and space to maneuver.

d. Develop the situation.

e. Orient the force or facility to be secured.

f. Perform continuous reconnaissance.

g. Maintain enemy contact.

0427. In extended and noncontiguous AOs, commanders secure or conduct surveillance of the gaps between subordinate units. Commanders secure gaps by assigning a force to secure the area, dedicating surveillance efforts to monitor it, designating a force to respond to an approaching enemy, or installing and overwatching obstacles.

## Sustaining Operations in the Offense

0428. Sustaining operations in the offense ensure freedom of action and maintain

momentum. They occur throughout the AO. SS unit locations need not be contiguous with those of their supported forces. An extended major operation may place tactical units far from the original support area. Armor commanders may separate attacking forces from the SS base, thus extending their lines of communication (LOCs). They provide security to SS units when operating with extended LOCs.

### Considerations for Nonlinear Offensive Operations

0429. Nonlinear offensive operations can occur in both contiguous and noncontiguous AOs. The size of an AO is normally very large compared to the number of soldiers deployed. The AO may also encompass diverse terrain. Enemy forces will be widely dispersed and may be numerically superior. Attacking forces must focus offensive actions against decisive points, while allocating minimal essential combat power to shaping operations. Reserves must have a high degree of tactical mobility. Forces conducting nonlinear operations require robust communications and sustainment capabilities. Armor commanders may dedicate forces for LOC security operations beyond that provided by available military police.

0430. The higher headquarters conducts security operations in those portions of the AO not allocated to subordinates. Flank security importance increases as operations extend and attacking forces expose their flanks. Linkup operations often occur in this environment. Linkup operations, particularly those involving vertical envelopments, require extensive planning and rehearsal. The potential for fratricide increases due to the fluid nature of the nonlinear battlefield and the changing disposition of attacking and defending forces. The presence of noncombatants in the AO further complicates operations. In this setting, armor commanders exercise prudent judgment in clearing fires, both direct and indirect.

## Forms of Maneuver

0431. The five forms of maneuver are the envelopment, turning movement, infiltration, penetration, and frontal attack. While normally combined, each form of maneuver attacks the enemy differently. Each poses different challenges for attackers and different dangers for defenders. Armor commanders determine the form of maneuver to use by analyzing the factors of Mission, Enemy, Troops, Terrain and Weather, Time available, and Civil Consideration (METT-TC).

### Envelopment

0432. The envelopment is a form of maneuver in which an attacking force seeks to avoid the principal enemy defenses by seizing objectives to the enemy rear to destroy the enemy in his current positions. At the tactical level, envelopments focus on seizing terrain, destroying specific enemy forces, and interdicting enemy withdrawal routes (see Figure 4.2). Envelopments avoid the enemy front, where he is protected and can easily concentrate fires. Single envelopments maneuver against one enemy flank; double envelopments maneuver against both. Either variant can develop into encirclement.

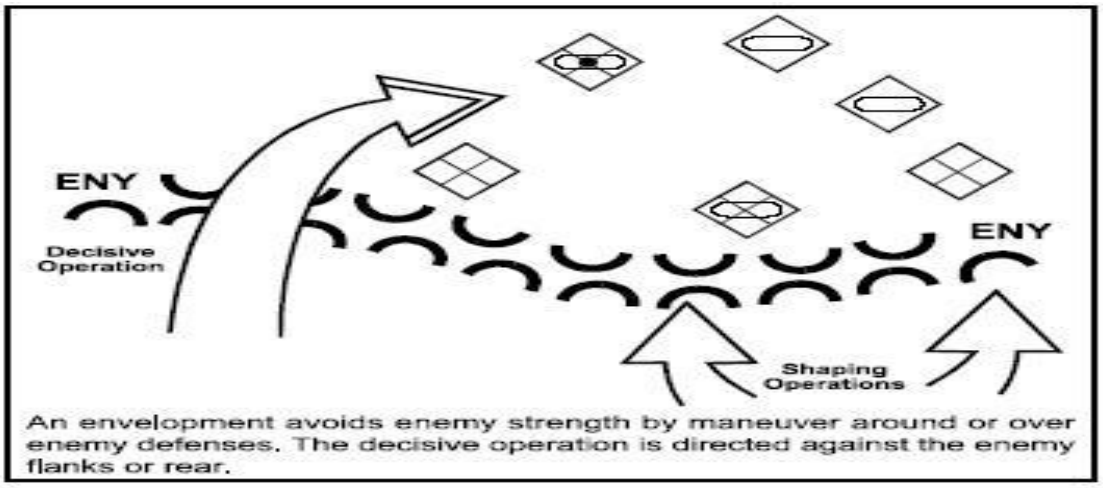



**Figure 4.2 - Envelopment**

## Turning Movement

0433. A turning movement is a form of maneuver in which the attacking force seeks to avoid the enemy's principal defensive positions by seizing objectives to the enemy rear and causing the enemy to move out of his current positions or divert major forces to meet the threat (see Figure 4.3). A major threat to his rear forces the enemy to attack or withdraw rearward, thus "turning" him out of his defensive positions. Turning movements typically require greater depth than other forms of maneuver. Deep fires take on added importance. They protect the enveloping force and attack the enemy.

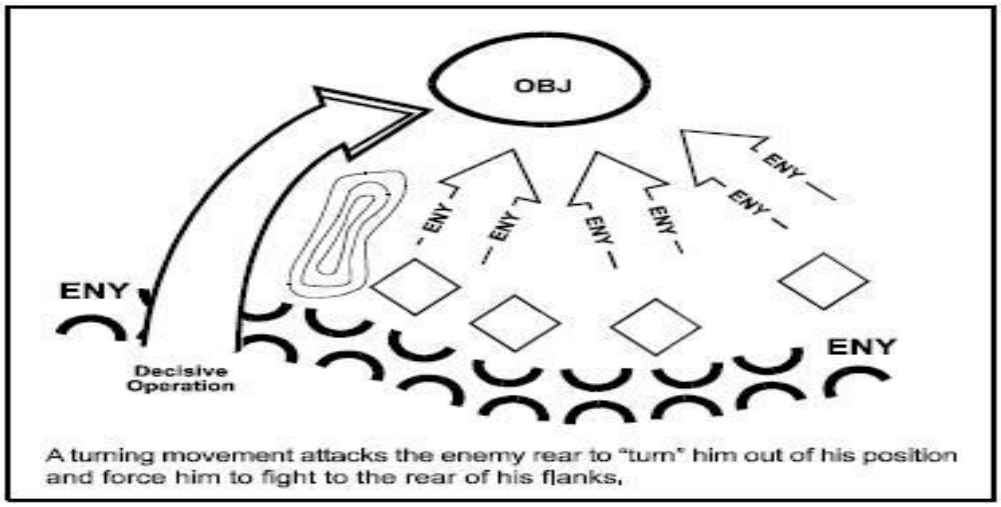



**Figure 4.3 - Turning movement**

## Infiltration

0434. An infiltration is a form of maneuver in which attacking forces conducts undetected movement through or into an area occupied by enemy forces to occupy a position of advantage in the enemy rear while exposing only small elements to enemy

defensive fires (see Figure 4.4). The need to avoid being detected and engaged may limit the size and strength of infiltrating forces. Infiltration rarely defeats a defense by itself.

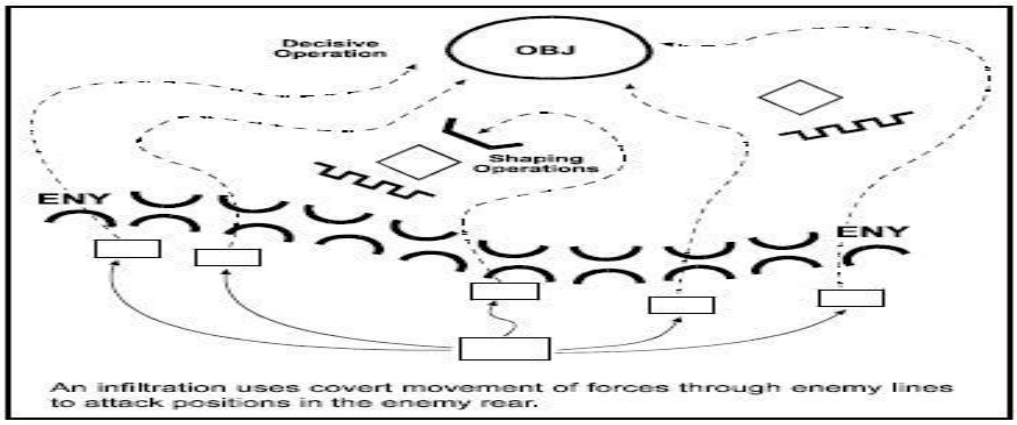



**Figure 4.4 - Infiltration**

## Penetration

0435. A penetration is a form of maneuver in which an attacking force seeks to rupture enemy defenses on a narrow front to disrupt the defensive system (see Figure 4.5). Armor commanders direct penetrations when enemy flanks are not assailable or time does not permit another form of maneuver. Successful penetrations create assailable flanks and provide access to enemy rear areas. Because penetrations are frequently directed into the front of the enemy defense, they risk significantly more friendly casualties than envelopments, turning movements, and infiltrations.

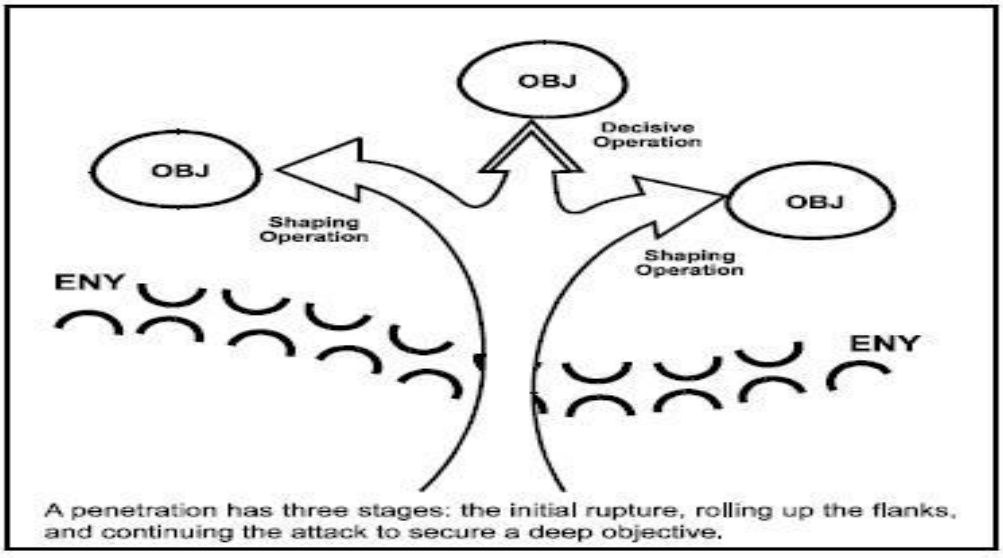



**Figure 4.5 - Penetration**

## Frontal Attack

0436. A frontal attack is a form of maneuver in which an attacking force seeks to destroy a weaker enemy force or fix a larger enemy force in place over a broad front (see Figure 4.6). At the tactical level, an attacking force can use a frontal attack to rapidly overrun a weaker enemy force. A frontal attack strikes the enemy across a wide front and over the most direct approaches. Armor commanders normally use it when they possess overwhelming combat power and the enemy is at a clear disadvantage. He masses the effects of direct and indirect fires, shifting indirect and aerial fires just before the assault. Success depends on achieving an advantage in combat power throughout the attack.

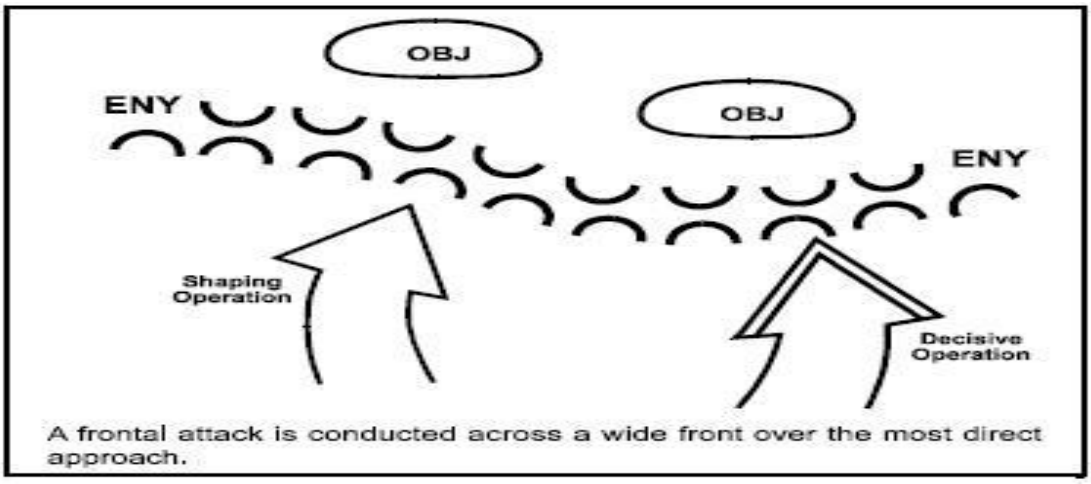



**Figure 4.6 - Frontal Attack**

# Types of Offensive Operations for Armor

## Movement to contact (MTC)

0437. Movement to contact is a tactical operation conducted to develop the situation and to establish or regain contact. The attacker, unsure of exactly where or when he will fight, moves aggressively toward the enemy. It is best when friendly forces discover the enemy with at least enough time to make hasty attack plans. Being surprised by inadvertently running into the enemy is always a possibility but not the preferred tactical option. Armor commanders adjust their tempo appropriately as they anticipate closing with the enemy force. They decide where and when their force can deploy into attack formation that would facilitate the initial contact and still provide freedom of action for the bulk of their force. The desired result of the MTC is to find the enemy. When this happens, the Commander fight a meeting engagement. To maintain their freedom of action once they make contact, commanders usually lead with a self-contained force that locates and fixes the enemy. The Armor Commander holds the bulk of his force back so when his lead forces make contact they can maneuver the majority of their force without becoming decisively engaged.

## Hasty attack

0438. The hasty attack is the most likely result of a meeting engagement. Armor commanders must launch the attack with the forces at hand and with minimum preparation to destroy the enemy before he is able to either concentrate or re-establish a defense. In the defense, hasty counterattack may recapture lost positions before the enemy has time to consolidate his success. There should be no pause in the momentum of force upon initial contact. The hasty attack must try to fix forward enemy elements in place with firepower. Find gaps, weak spots, or open flanks, and move through them rapidly. Speed is essential. Regardless of its purpose or echelon, a hasty attack enhances the risk of losing synchronization. To minimize this risk, units conducting hasty attacks should use standard formations and well-understood and rehearsed plans supported by sound intelligence preparation of the battlefield.

## Deliberate attack

0439. In contrast to the hasty attack, the deliberate attack is a fully synchronized operation that employs the effects of every available asset against the enemy defense. Because such synchronization requires careful planning and extensive coordination, a deliberate attack takes time to prepare. Armor Commanders should use deliberate attack when the enemy situation is known and when the combined arms team can be employed with sufficient combat power to defeat the enemy. A deliberate attack aims at a breakthrough on a narrow front, seeking penetration deep into the enemy's rear.

## Exploitation

0440. Exploitation is an operation undertaken to extend the destruction of the defending force by maintaining defensive pressure. The ultimate objective of the exploitation is the disintegration of the enemy until he has no other alternative except surrender or flight. Failure to exploit aggressively towards the success of the main effort may provide the enemy sufficient time to regain the initiative. Exploitation requires physical and mental aggressiveness to combat the friction of night, bad weather, dangers of fratricide and extended operations.

## Pursuit

0441. A pursuit is an offensive operation directed to a retreating enemy force. It follows a successful attack or exploitation and is ordered when the enemy cannot conduct an organized defense and attempts to disengage. It differs from the exploitation since its primary function is to complete the destruction of the enemy force and it cannot be anticipated.

# Other Armor Offensive Tasks and Missions

0442. The following describes other tactical tasks and missions that may be required of the armor forces during offensive operations.

0443. **Obstacle Breaching.** During offensive operations, armor forces may be required to assist in breaching obstacles. Armor forces play a key part in isolating the breach point and providing direct fire support to the breach and assault forces. The unit will be organized into support, breach, and assault forces. The breach operation is executed through suppression, obscuration, security, and reduction using all available infantry, engineers, artillery, and mortars.

0444. **Screen/Guard.** Armor forces may be given employment priority if the enemy threat is predominantly mechanized or is in wide-open terrain allowing long-range fires. The armor unit in this circumstance may be tasked to establish a screening force to provide security for the maneuver elements as it flows into the operational area.

0445. **Reconnaissance in Force.** A reconnaissance in force is a deliberate attack by armor forces to discover and test enemy disposition, composition, and strength. At times it may uncover a weakness that, if promptly attacked, would permit an immediate tactical success. Objectives are designated on terrain that will force the enemy to react.

0446. **Attacks from a Defensive Posture.** The types of attacks from a defensive posture are counterattack and spoiling attacks which may be hasty or deliberate in nature. Armor forces can conduct a counterattack or spoiling attack or as a part of a much larger force during operations.

a. **Counterattack.** The counterattack attempts to defeat an attacking enemy force or regain key terrain. Counterattacks may be conducted by a reserve or lightly committed maneuver element. Timing is critical and any delay or preparatory movement allows the enemy to react. Therefore, thorough counterattack planning and rehearsals should be done in advance. The counterattack is launched when the commander senses that the balance of power on the battlefield has changed and he can exploit the situation by counterattacking to seize the initiative.

b. **Spoiling Attack.** The spoiling attack attempts to strike the enemy when he is most vulnerable such as during preparations for attack in assembly areas, attack positions or on the move before crossing the line of departure. This breaks up and impedes the enemy's capability to conduct a synchronized offensive.

0447. **Follow and Support.** Armor forces may be employed as a follow and support force during exploitation and pursuit operations to facilitate and maintain the momentum of the attack. They may also be used in the conduct of a penetration.

0448. **Raid.** A raid is an attack into enemy territory to accomplish a specific purpose and with no intention of holding or gaining terrain. It may be conducted to capture prisoners, capture or destroy specific enemy material, destroy logistical installations, obtain information about the enemy, and disrupt enemy plans.

0449. **Feint.** A feint is a supporting offensive operation to draw the enemy attention

away from the area of the main attack and induce him to move his reserves or shift his fire support. Feints must appear realistic and true, thus, contact with the enemy is required.

0450. **Demonstration.** A demonstration is an operation to deceive the enemy about the main attack. Its purpose is similar to feint. However, no contact with enemy is made. Demonstrations support an operations plan of large forces.

## Planning for Armor Offensive Operations

0451. In planning for the armor offensive, the following factors should be considered:

a. **Intelligence**. Available avenues of approach should be considered in planning for offensive operations. Armor forces should move along an axis of advance that maximizes its firepower and maneuverability. This should be included in the IPB so that armor will not be limited during maneuver. Some of these IPB needs are number, armor protection level, armor-piercing capability of enemy forces, presence of antitank weapons, and terrain analysis for mobility corridors.

b. **Maneuver**. The rate of march of the armor forces can easily exceed the rate of march of the other arms. This should be planned carefully so as to synchronize all movement. Furthermore the rate of movement must be based on terrain and weather. Armor forces also rely on the attachment of light infantry for close-in security in closed and restrictive terrain. The armor unit's ground mobility is its greatest asset, as it can move faster than any other ground combat element.

c. **Fires.** The high mobility of the armor forces, when compared with the remainder of the other arms, may require additional fire support assets that can be positioned in direct support of the armored unit.

d. **Air Defense.** Armor forces require air defense augmentation in order to preserve its capabilities. Armor's presence in a predominantly dismounted operational area makes it easily identifiable from the air and a high-value target for enemy attack aircraft.

e. **Mobility, Counter-mobility and Survivability.** Armor forces should be augmented with units that have the capability to provide a full range of mobility, counter-mobility, and survivability tasks, to include assault gap crossing, complex obstacle breaching, preparation of fighting positions, emplacement of conventional and dynamic minefield, and construction of nonexplosive obstacles.

f. **Sustainment Support.** Armor forces normally do not possess the required logistical redundancy to sustain its units. These forces normally deploy with their own sustaining and unique support packages, which are either organic or are provided by other supported units. Special consideration is given to the

maintenance, repair, supply, transportation, and external communications requirements that augmentation forces provide to the unit.

g. **Mission Command.** Armor leaders must be willing to take responsibility and use initiative, guided by their commander's intent. Armor leaders must be able to think clearly and quickly. Speed and decisive action are critical. They must use terminology that is widely understood throughout the command. Above all, they must accurately communicate the commander's intent, which guides subordinate commanders in pursuit of the common goal in the absence of communications. Armor commanders must teach subordinates not what to think, but how to think. Conversely, when armor units are decentralized and attached to light infantry, armor leaders must be assertive in advising the infantry commander and his staff on the best means for armor employment and logistical requirements.

# Chapter V
# Defensive Operations

0501. Armor units defend until they gain sufficient strength to attack. Armor defensive operations defeat an enemy attack, buy time, economize forces, or develop conditions favorable for offensive operations. Alone, defensive operations normally cannot achieve a decision. Their purpose is to create conditions for a counteroffensive that allows friendly forces to regain the initiative. Although offensive operations are usually required to achieve decisive results, it is often necessary, even advisable at times, to defend. Commanders defend to buy time, hold terrain, facilitate other operations, preoccupy the enemy, or erode enemy resources.

## Purpose of Armor Defensive Operation

0502. The purpose of armor defensive operations is to defeat enemy attacks. (Defending armor forces await the attacker's blow and defeat the attack by successfully deflecting it.) Waiting for the attack is not a passive activity. Armor commanders seek out enemy forces to strike and weaken them before close combat begins.

0503. Operationally, armor defensive operations buy time, economize forces, and develop conditions favorable for resuming offensive operations. (Therefore, major operations and campaigns combine armor defensive operations with armor offensive operations.) Operational-level armor defensive operations normally include offensive, stability, and support operations.

0504. During force projection, armor defensive operations by in-theater or early arriving forces can maintain the operational initiative for joint forces. If conditions do not support offensive operations, initial-entry forces defend the lodgment while the joint force commander builds combat power. Initial-entry forces should include sufficient combat power to deter, attack, or defend successfully.

0505. In its broad concept, there are three main purposes of armor defensive operations in which armor units are capable of conducting:

a. Gain time while waiting for more favorable conditions to conduct offensive operations.

b. Economize forces in one area in order for other forces to concentrate for a decisive offensive action in other areas of operations.

c. Maintain control or possession of an objective

# Fundamentals of Armor Defensive Operations

## Understand the enemy

0506. Armor Commanders must be able to understand the enemy's doctrine in order to have a good defensive posture. It eliminates illogical conclusions on possible enemy courses of actions in a given time, space and condition.

## Analyze the battlefield

0507. Armor Commanders must be able to visualize the effects of terrain, weather, demography and other factors within the area of operations to the enemy and to his own unit as well.

## Use the advantages of the defender

0508. The defender should maximize the use of terrain in strengthening its defensive positions against enemy attacks. Obstacles must be made, reinforced and shaped to the desired outcome of the armor leader.

## Maximize the use of mobility and firepower

0509. Armor units have the edge on mobility and firepower against dismounted troops. Armor commanders must plan to concentrate fires to achieve decisive results. Direct and indirect fires should be delivered to the attacking enemy before it can bring effective fires on the defending friendly troops. Leaders should also consider counterattacking by fire to exploit enemy weaknesses as they arise.

# Characteristics of Armor Defensive Operations

0510. Successful defenses are *aggressive*; they use direct, indirect, and air-delivered fires; information operations (IO); and ground maneuver to strike the enemy. They maximize firepower, protection, and maneuver to defeat enemy forces. Static and mobile elements combine to deprive the enemy of the initiative. The defender resists and contains the enemy. Defending armor commanders seek every opportunity to transition to the offensive.

0511. While the fundamentals of the defense continue to apply to a modernized force, advanced technology systems modify the way commanders conduct defensive operations. Greater understanding of friendly and enemy situations and the fusion of command and control (C2); intelligence, surveillance, and reconnaissance (ISR); long-range precision fires; and sustainment support (SS) technologies make the mobile defense even more lethal and effective. Whenever practical, commanders of modernized forces use the mobile defense because it takes maximum advantage of Army force strengths.

0512. An effective defense engages the enemy with static and mobile forces. It combines the elements of combat power to erode enemy strength and create conditions for a counterattack. Defenders seek to increase their freedom to maneuver while denying it to the attacker. The enemy falters as losses increase and the initiative shifts to the defender, allowing counterattacks. Counterattack opportunities rarely last long; defenders strike swiftly to force the enemy to culminate. Preparation, security, disruption, massing effects, and flexibility all characterize successful defensive operations.

## Preparation

0513. The defense has inherent strengths. (The armor defender arrives in the area of operations (AO) before the attacker and uses the available time to prepare.) Armor defenders study the ground and select positions that allow massing fires on likely approaches. They combine natural and manmade obstacles to canalize attacking forces into engagement areas. Defending forces coordinate and rehearse actions on the ground, gaining intimate familiarity with the terrain. They place security and reconnaissance forces throughout the AO. These preparations multiply the effectiveness of the defense. Preparation ends only when defenders retrograde or begin to fight. (Until then, preparations are continuous.) Preparations in depth continue, even as the close fight begins.

0514. Preparation for defending armor forces takes longer compared to other forces. This is due to the complexity of the fighting positions for the armored vehicles and the dismounted troops. Time and engineer assets are of the essence in preparing the battle position for an armor unit. Because of this, armor commanders need to take into consideration the amount of time needed to establish a battle position in order to protect their assets during the enemy's attack.

## Security

0515. Armor commanders secure their forces principally through security operations, force protection, and IO. Security operations help deceive the enemy as to friendly locations, strengths, and weaknesses. They also inhibit or defeat enemy reconnaissance operations. These measures provide early warning and disrupt enemy attacks early and continuously. Force protection efforts preserve combat power. Offensive IO inaccurately portrays friendly forces and misleads enemy commanders through military deception, operations security, and electronic warfare. These measures contribute to the defender's security.

0516. Armor units employ passive security in their battle positions. In addition to the armored vehicles position, commanders also create hide positions to prevent enemy aerial observations. Effective use of tree lines and camouflaging nets will create a good hide position. This is to prevent the enemy from identifying the assets available on the defending force and prevents them from locating the bulk of the armored forces in the area of operation.

## Disruption

0517. Armor defenders disrupt attackers' tempo and synchronization with actions designed to prevent them from massing combat power. Disruptive actions attempt to unhinge the enemy's preparations and, ultimately, his attacks. Methods include defeating or misdirecting enemy reconnaissance forces, breaking up his formations, isolating his units, and attacking or disrupting his systems. Armor defenders never allow attackers to fully prepare. They use spoiling attacks before enemies can focus combat power, and counterattack before they can consolidate any gains. Defenders target offensive IO against enemy C2 systems and constantly disrupt enemy forces in depth.

## Massing Effects

0518. Armor defenders seek to mass the effects of overwhelming combat power where they choose and shift it to support the decisive operation. To obtain an advantage at decisive points, defenders economize and accept risk in some areas; retain and, when necessary, reconstitute a reserve; and maneuver to gain local superiority at the point of decision. Armor defenders may surrender some ground to gain time to concentrate forces.

0519. Armor commanders accept risk in some areas to mass effects elsewhere. Obstacles, security forces, and fires can assist in reducing risk. Since concentrating forces increases the threat of large losses from weapons of mass destruction (WMD), commanders use deception and concealment to hide force concentrations. They also protect their forces with air and missile defenses.

## Flexibility

0520. Armor defensive operations require flexible plans. Planning focuses on preparations in depth, use of reserves, and the ability to shift the main effort. Armor commanders add flexibility by designating supplementary positions, designing counterattack plans, and preparing to counterattack.

0521. Flexibility entails a rapid response to the highly fluid situation on the battlefield. Likewise, armor units need to plan and prepare for their sustainment support (SS) to be able to address changing situations during the defense. An armor unit's effectiveness relies greatly on logistics and how to sustain them, which is why a good SS plan will provide highly flexible defensive operation for armor forces.

# Armor Defensive Operations within the Operational Framework

0522. Armor Commanders use the operational framework (AO, battle space, and battlefield organization) to conduct defensive operations (see Figure 5.1). Armor Commanders base their framework on METT-TC and an understanding of their battle space. They conduct simultaneous and sequential decisive, shaping, and sustaining

operations in depth by synchronizing their forces in time, space, resources, purpose, and action. Armor Commanders may designate deep, close, and rear areas when conducting operations that are generally linear and contiguous.

0523. Armor Commanders deny enemy forces freedom of movement within the gaps formed by extended, noncontiguous AOs. They dominate their entire AO throughout the operation, assigning responsibility for unoccupied ground and allocating combat power. When armor commanders designate subordinate unit AOs that are noncontiguous, they retain responsibility for portions of their AO not assigned to subordinates. Regardless of the proximity or separation of its elements, commanders see their defense as a continuous whole. They fight decisive, shaping, and sustaining operations as one action, synchronizing simultaneous operations to accomplish a single purpose: defeating the attack and quickly transitioning to the offense.

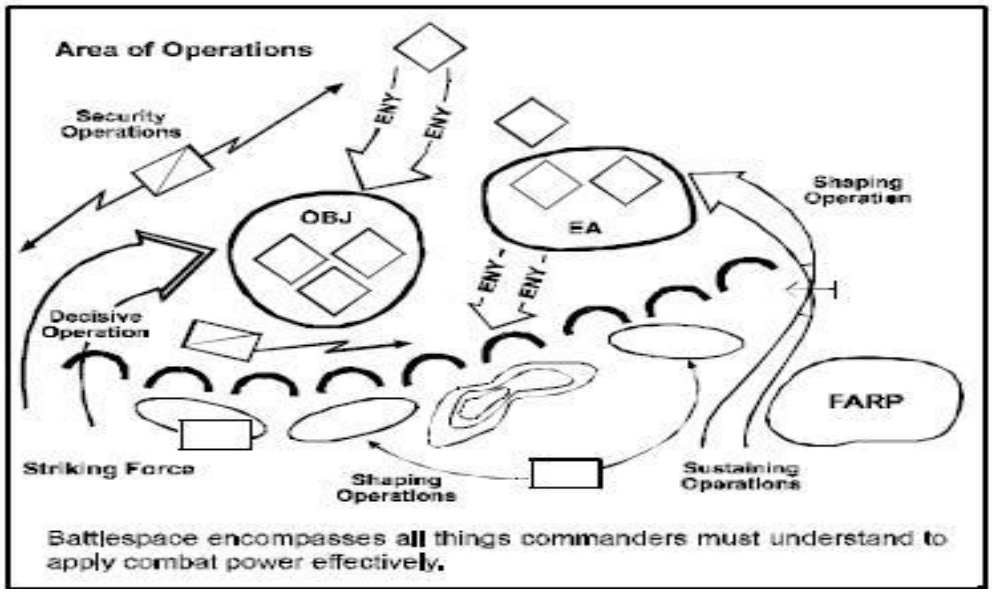



**Figure 5.1 – Armor Operational frameworks in the defense**

0524. The decisive operation in an armor defense defeats the enemy attack. In a mobile defense, the counterattack by the striking force is the decisive operation. The more commanders know about the enemy situation, the more they can weigh the counterattack. In area defenses, defeating the enemy attack within engagement areas is the decisive operation. Armor Commanders draw the enemy into engagement areas, where defenders destroy those using massed fires, obstacles, and other assets.

0525. Throughout the execution of mobile and area defenses, Armor Commanders designate a main effort and synchronize the War Fighting Functions (WFF) to support it. If necessary, they shift their main effort to concentrate forces and mass effects. Armor defending commanders may shift their main effort repeatedly to defeat an attack. If they correctly anticipate enemy actions, commanders can execute their plan for the decisive operation despite shifting the main effort. They always designate the decisive operation as the main effort at the decisive point.

0526. Reserves preserve commanders' flexibility and provide a hedge against uncertainty. Once, the reserve is committed, its operation becomes the main effort. If commanders commit their reserve, they should immediately designate another from uncommitted forces or forces in less threatened areas. Armor commanders may employ reserves throughout the operation. Typical reserve missions include counterattacking, reinforcing, blocking, and destroying a penetration. Armor commanders avoid assigning tasks to the reserve other than those required to support planning and preparing for their be-prepared missions. Reserves are best used to reinforce and expedite victory rather than prevent defeat. The concept of operations determines the reserve's primary mission. Unless otherwise delegated, the commander designating the reserve retains authority for its commitment.

## Shaping Operations in the Defense

0527. Shaping operations executed simultaneously throughout the AO support the conduct of the defender's decisive operation by upsetting the attacker's design. They selectively suppress or neutralize the enemy's WFF and disrupt his synchronization. IO shape enemy perceptions and can influence the decision to attack. Fires contribute to shaping operations by attacking high-payoff targets and create conditions for successful decisive operations. Shaping operations in the defense include:

a. Counter-mobility and mobility operations.

b. Reconnaissance and security operations.

c. Aerial-delivered and long-range precision indirect fires.

d. Passages of lines (forward and rearward).

e. Actions of fixing forces whose shape is to support the decisive operation.

f. Movements of units that directly facilitate other shaping operations and the decisive operation.

g. Actions by reserve forces before their commitment.

0528. Security forces perform critical functions in the defense. They secure gaps between defending units, protect the force from surprise, meet the leading enemy forces, strip away reconnaissance and security elements, report enemy strengths and locations, and help identify the enemy decisive operation. They harass and slow attacking forces to gain time and space for shaping enemy actions and protecting LOCs, headquarters, fire support units, and reserves.

## Sustaining Armor Operations in the Defense

0529. Sustaining operations in the defense occur throughout the AO. Armor Commanders ensure freedom of action and continuity of the defense by conducting

SS operations, rear area and base security, LOC maintenance, movement control, and terrain management.

0530. Security for sustaining armor operations is a primary concern. Armor Commanders organize forces and terrain to protect sustaining operations and retain freedom of action. Armor Commanders group forces that performing sustaining operations into bases and base clusters for protection and security. Base and base clusters organize for self-defense. Armor commanders designate response forces and tactical combat forces (TCFs) to augment base cluster self-defense capabilities.

0531. Force projection operations present distinct security challenges for sustaining operations. To protect combat power build up, combat, MS, and SS forces operate in the same area while establishing the initial lodgment. Forces conducting sustaining operations take increased active and passive self-protection measures until combat forces are available. Armor commanders assess threat capabilities, decide where and when to accept the risk, and assign units to protect and sustain the force.

### Considerations for Nonlinear Armor Defensive Operations

0532. Armor commanders may conduct nonlinear defensive operations in contiguous and noncontiguous AOs. In both cases, defenders focus on destroying enemy forces, even if it means losing physical contact with other friendly units. Successful nonlinear defenses require all friendly commanders to understand the higher commander's intent and share a current common operational picture (COP). They also favor use of a battlefield organization based on decisive, shaping, and sustaining operations rather than deep, close, and rear areas. Noncontiguous defenses are generally mobile defenses; however, some subordinate units may conduct area defenses to hold key terrain or canalize attackers into engagement areas. Even mobile defenses that begin as linear operations often evolve into nonlinear operations. Area defenses are typically more linear operations because of their orientation on terrain.

0533. Nonlinear defenses place a premium on reconnaissance and surveillance to maintain contact with the enemy, produce relevant information, and develop and maintain a COP. The defending force focuses almost exclusively on defeating the enemy force in depth rather than retaining large areas due to the size of the AO. All forces conducting nonlinear defenses require robust communications and sustainment capabilities. Noncombatants and the fluidity of nonlinear defensive operations require commanders to exercise judgments in clearing fires, both direct and indirect.

## Defensive Missions for Armor Units

0534. **Defense of a Battle Position.** A battle position is a general location and orientation of forces on the ground from which units defend. Armor units to concentrate direct fires at critical places and times to take advantage of the available terrain use this type of defense. The armor unit in battle positions orients its weapons on an enemy avenue of approach using designated engagement areas and sectors of fire. Units can maneuver in and outside of the BP as necessary to adjust fires or to seize opportunities for offensive action. Armor commanders may move his forces

freely within the assigned battle position. Defense in a battle position is done to accomplish one of the following tasks:

a. Destroy an enemy force in an engagement area

b. Control a key terrain by holding the battle position

c. Block an avenue of approach

d. Fix an enemy force to allow other friendly unit to maneuver

0535. **Defense in Sector.** An armor unit uses defense in sector to prevent an enemy force from passing the rear boundary of the sector. When defending in sector, armor uses the depth of the sector to defeat the enemy within his boundaries while maintaining flanks security and ensuring unity of effort. This type of defense is used when:

a. The enemy situation is vague.

b. Multiple avenues of approach exist within the area of operations.

c. Retention of terrain is not critical to the success of the defense.

d. Flexibility is desired.

0536. **Perimeter Defense.** Defense of a perimeter is conducted in the same manner as that of a battle position except that the perimeter defense orients on 360 degrees. Perimeter defenses are used to protect the force, hold specific terrain, or protect a key installation from destruction. While a battle position can allow some penetration, a perimeter cannot. Armor units are assigned to defend specific portions of the perimeter. Flanks of all units are tied in to provide mutual support. If the perimeter is penetrated, a reserve blocks the penetration or counterattacks to restore the defense. This type of defense is commonly used in the following:

a. Defense of an assembly area.

b. Defense of specific installations or equipment.

c. Defense of a position or key terrain.

d. When the unit has been bypassed or isolated by the enemy.

0537. **Strongpoint Defense.** A strongpoint is a defensive position that is fortified as extensively as time and materials allow. It may be part of any defensive plan. A strongpoint may be built to protect vital installations, as an anchor or anvil around which an armor unit maneuver, or as part of a trap designed to destroy enemy forces that attack. It may be in an urban or forested area. An armor unit required to defend a strongpoint will need a significant amount of time and engineer resources to construct

the position. Defense of a strongpoint is sometimes an extension of a defense of battle position if time, terrain and resources allow.

0538. **Reserve Force.** The primary purpose of the reserve is to retain flexibility, reinforce success or regain the initiative through counterattacks. When part of a larger unit is employed in a positional defense, such as the perimeter of defense of a battle position, an armor units can be used to conduct attacks against enemy penetrations by striking a decisive blow against an uncovered enemy flank. Should the enemy's attack fail, an armor unit in reserve could be used to reinforce success. The reserve is an active force, not reactive, and is not used to reinforce failures.

0539. **Counter Attack Force.** An armor unit participates in a counterattack to exploit enemy weaknesses in the area of operations. Counterattack is used to destroy enemy units, regain freedom of maneuver, regain the initiative, regain key terrain, or to relieve pressure on an engaged unit. Armor units execute two types of counterattacks:

a. **Counterattack by Fire.** This is executed to complete the destruction of exposed enemy elements, to free decisively engaged elements, and to regain the initiative. An armor unit executes a counterattack by fire by moving on a secured, concealed route to a predetermined battle position from which it can engage and maintain contact with the enemy.

b. **Counterattack by Fire and Maneuver** An armor unit uses this technique to destroy remaining enemy forces completely, relieve pressure on a friendly unit, or regain key terrain. The counterattack force hits the enemy on the flank if possible, using fire and maneuver to overwhelm him.

## Planning for Armor Defensive Operations

0540. Planning for the defense involves the integration of the battlefield operating system:

a. **Intelligence.** The key to successful intelligence planning in the defense is the conduct of a thorough IPB of the area. The armor commander should evaluate his area of operations and interest, conduct terrain and weather analysis, and have a detailed analysis of threat doctrine, tactics, weapons, and equipment. The commander likewise should try to relate how the threat force would fight in a specific terrain and weather scenario as a basis for determining how the threat force might have to fight.

b. **Maneuver.** The Armor Commander in the defense must learn how to counteract the threat initiative. Security, intelligent use of terrain, flexibility of defensive operations and timely resumption of offensive actions are the keys to a successful defense.

c. **Fires.** The armor commander should integrate combat support assets in his plan to maximize combat power. Synchronization of direct and indirect fires

with obstacles multiplies effects on the threat. Artillery fires make obstacle emplacements more effective.

d. **Mobility, Counter-mobility, and Survivability** Employment of obstacles during defense is a key activity. The engineer unit must make proper coordination with the armor commander regarding his obstacle plan. Direct and indirect fires should be coordinated with the obstacle plan.

e. **Air Defense.** In defensive operations, air defense assets are positioned to achieve both mass and gun missile mix. Generally, the priority of protection will begin with the C2 facilities. Logistical elements, reserve forces chokepoints, and fixed firepower positions should also be considered when making the air defense plan.

f. **Sustainment Support.** The armor commander should be able to create an effective logistics plan to ensure the supportability of the operations. Service support priorities should be carefully identified.

g. **Mission Command.** The concept of the defense and the commander's guidance and intent should be announced to the key members of the staff and subordinate unit commanders. The overall commander must ensure that all tasks are understood, rehearsed, and coordinated with the higher headquarters.

0541. When planning for the defense, the armor commander should also consider the following:

a. **Mission**.

b. **Enemy**. Armor should be placed in defensive positions that capitalize on its long-range firepower and mobility advantages. If an enemy mounted threat exists, armor units should be concentrated. When a dismounted enemy threat exists, armor should be augmented by infantry troops.

c. **Terrain**. In selecting positions for armored vehicles, the commander should consider the aspects of protection (cover and concealment), firepower (observation and fields of fire), and mobility (**obstacles** and avenues of approach). When the terrain allows, armor should be massed. Open terrain permits armor units to defend in battle positions to overwatch possible enemy mounted avenues of approach. In close terrain, armor units should be augmented by infantry troops for added protection.

d. **Troops Available**. When infantry, engineer, and field artillery units augment an armor unit, **the** commander should be able to plan for the coordinated use of these assets to strengthen its defense.

e. **Time Available**. **The** more time the commander has, the more detailed the defense should be.

f. **Supplies Available**. Ammunition and POL products are critical supplies for armor units. Commanders **should** be able to plan for the provision of said items in the conduct of the defense.

g. **Defensive Fire Planning.** Fire planning is a key activity in the armor defense. It enables the **commander** to effectively distribute and control fires in the defensive concept. To develop a fire plan, the Armor Commander should make the following: 1) Assign a location for each armor vehicle and sectors of fire. The position of each vehicle should complement each other. Sectors of fires should be overlapping. 2) Ask feedback from vehicle commanders. The overall commander should make sure that each vehicle clearly sees the target reference points assigned in their sectors. Likewise, he should be able to produce a sketch card for each of his units. 3) Coordinate with higher headquarters and adjacent units. A copy of the armor unit's sketch card for fires should be given to the immediate higher commander and to adjacent units for proper coordination. 4) Determine engagement priorities. The armor commander should give to his subordinate units his guidance on what target to engage first in their sector. a) Decide where to engage the enemy. b) Prepare for possible limited visibility engagements.

# Chapter VI
# Other Tactical Operations

0601. This chapter prescribes the general concept by which armor forces will conduct other tactical operations. It contains the different kinds of operations that require an armor unit such as Reconnaissance, Reserve, Security Operations, Relief in place Operations, Inland Waterways Operations, Link-Up Operation, Shore to Shore Operations, Operations on Forest or Wooded Areas, Operations at Defiles, Operations on Fortified Positions, Night and Limited Visibility Operations, Military Operations in Urban Terrain and Support to Civilian Authority.

## Reconnaissance Operations

0602. Reconnaissance operations provide the Armor Commander and staff with information about the terrain and enemy. Reconnaissance verifies or refutes analyzed intelligence information. Armor units conduct reconnaissance to obtain information by employing movement, observation and surveillance, fire and maneuver, and special equipment. They may be required to fight to gain intelligence through combined arms teamwork with ground or air cavalry or alone if cavalry is not present or available. Armor units, especially cavalry units, perform three distinct types of reconnaissance: route, zone, and area. Depending on the level performed, reconnaissance may be a separate mission or part of another operation.

## Reserve Operations

0603. Reserve operations play a vital role in any kind of armor operation whether it offensive or defensive in nature. Offensive action is sometimes necessary to gain the initiative, maintain the continuity of the defense and reposition forces to defeat following enemy echelons. A reserve force must be prepared to transition to attack whenever there is an opportunity for success. Ideally, reserve forces are composed of tank and mechanized infantry units that will conduct the following missions: counterattack, spoiling attack, and block, fix, or contain or reinforce.

### Types of Reserve Operations

0604. **Counterattack**. Counterattacks, like hasty attacks, are planned and executed rapidly. Counterattacking forces must complete their tasks and regain covered positions before the overwatching of following enemy echelons can interfere. Counterattacks destroy weakened enemy units to maintain freedom of maneuver within the area of operations and to eliminate threats to the rear area. Occasionally a reserve force must counterattack to regain terrain critical to the defensive system. When counterattacking, the reserve must mass enough combat power to overwhelm the enemy force.

0605. **Spoiling attack**. This is preemptive, limited objective attack aimed at preventing, disrupting, or delaying the enemy's ability to launch an attack. The objective of the spoiling attack is the enemy force and not to secure the terrain. The

reserve is often used to conduct spoiling attacks so that forward units can concentrate on defensive preparations within the area of operations.

0606. **Block, Fix or Contain**. The reserve may be ordered to establish hasty battle positions to block fix or contain enemy forces within a portion of the battlefield. An enemy force may be held in one area of the battlefield while he is defeated in another.

0607. **Reinforce**. Reserve armor forces may be committed to reinforce units that have sustained heavy losses or to build up stronger defenses in critical areas of the battlefield. Considerations must be given as to how they will be integrated into the overall defensive scheme.

## Security Operations

0608. Security operations are those operations undertaken by a commander to provide early and accurate warning of enemy operations to provide the force being protected with time and maneuver space that enables the force to respond to the enemy action, and to develop the situation to allow the commander to effectively use the protected force. The ultimate goal of security operations is to protect the force from surprise and reduce the unknowns in any situation. A commander may conduct security operations to the front, flanks, or rear of his force. The main difference between security operations and reconnaissance operations is that the former is oriented on the force or facility being protected, while the latter is enemy and terrain oriented. Security operations are shaping operations.

0609. There are the five forms of security operations that armor units will significantly play a vital role: screen, guard, cover, area security, and local security.

a. **Screen** is a form of security operations that primarily provides early warning to the protected force.

b. **Guard** is a form of security operations whose primary task is to protect the main body by fighting to gain time while also observing and reporting information and preventing enemy ground observation of and direct fire against the main body. Units conducting a guard mission cannot operate independently because they rely upon fires and combat support assets of the main body.

c. **Cover** is a form of security operations that protects the force from surprise, develops the situation and gives commanders time and space in which to respond to enemy's actions. Cover differs from Guard because it can sustain its own operation without support from the main body.

d. **Area security** is a form of security operations conducted to protect friendly forces, installations, routes, and actions within a specific area.

e. **Local security** consists of low-level security operations conducted near a unit to prevent surprise by the enemy.

0610. The armor units conduct independent security operations to accomplish their specific tasks, however in a combined arms operations, the cavalry units primarily perform these specific tasks or mission.

## Relief-In-Place Operations

0611. In relief-in-place operations, one unit is replaced in combat by another unit. It may be accomplished during offensive or defensive operations, preferably during periods of limited visibility. A relief-in-place is conducted when a unit needs to reconstitute rest or decontaminate. Units that have a change of mission may also require relief-in-place.

0612. The primary purpose for the relief-in-place is to maintain the combat effectiveness of committed elements. A relief-in-place may be conducted to: replace a combat-ineffective force, relieve a unit that has conducted prolonged operations, or replace a unit that requires medical treatment or decontamination as a result of exposure to nuclear, biological, or chemical munitions. Relief-in-place requires extensive planning. Security, secrecy, and speed are critical. Armor forces are employed to secure the area where relief in place operation will take place and clear the route where the relieving units and units to be relieved will travel. Besides, armor forces act as Quick Reaction Force (QRF) and as escort element.

## Inland Waterways Operations

0613. In an inland water crossing, the actual crossing is a means to extend the operation. The approach to the water obstacle is made rapidly on a broad front whenever possible. Armor forces in a combined arms operation push forward quickly to catch the enemy astride the obstacle and to seize and secure the crossing site where the opposition appears weakest. Crossing of opportunity, such as captured bridges or fords, are the product of rapid offensive action. Once the crossing means are captured, they are checked for the presence of demolition immediately after the lead elements have crossed. Air defense units should be immediately employed to protect the crossing means from enemy air attack. A crossing, however, is not predicated on the seizure of bridges or fords intact. In any case, armor forces cross first to establish a bridgehead on the far bank to protect the engineers as they establishes the crossing sites for the remainder of the command. Whenever possible, the plans for the crossing of every water obstacle which cannot be bypassed should be prepared well in advance by appropriate command echelons. This concept is applicable to operation across canals, lakes, and rivers bridgeable by tactical bridging or passable with swimming or fording vehicles.

## Types of Crossings

0614. The types of crossings discussed in this paragraph are those special operations required when crossings of opportunity fail to materialize.

0615. A **hasty crossing** is the crossing of an inland waterway using crossing means at hand or readily available without significant delay when the waterway is reached. It is a pre-planned operation, which is conducted as a continuation of the operation which is underway. Although the crossing is termed hasty, detailed prior planning assures that fire support and crossing means are available to armor force commanders on arrival at the water obstacle. A hasty crossing is characterized by minimum loss of momentum at the waterline, speed, surprise, minimum concentration of armor forces, and decentralization of control of specific crossing times for subordinate crossing forces. This type of crossing is preferred by armor units. Hasty crossings are feasible when the crossing areas are lightly held by the enemy or are undefended and when mobile task force are available to advance rapidly to the waterline.

0616. A **deliberate crossing** is the crossing of a waterway that requires extensive planning and detailed preparations. It is characterized by the detailed planning to support a selected course of action, deliberate preparation, the employment of specialized equipment, delay at the waterline to mass for breaching enemy defense, and the clearance of the near shore of enemy. The deliberate crossing is conducted when a unit is moving from a defense of the water obstacle to an offensive posture, a hasty crossing is not feasible, or a hasty crossing is unsuccessful. Overall planning and coordination are performed at division or higher levels. Tank units that are not involved in the initial assault are normally kept in reserve and pass through the crossing area after the far bank has been neutralized.

## Shore to Shore Operations

0617. Shore-to-shore operations use Army landing craft, amphibians, and LACV-30s to transfer cargo from one beach terminal to another along the same coastline. Water transport units may be called on to support combat forces conducting shore-to-shore assaults. They may also be requested to ferry cargo across or along rivers and between islands in routine resupply operations.

0618. During tactical shore-to-shore operations, the water transport unit commander does not select the final landing site. However, he does advise the tactical commander of the requirements for specific areas and indicates any conditions that would adversely affect the operational efficiency of the unit. Control points normally required for lighter age units in shore-to-shore operations are:

a. Lighter control center.

b. Loading area control point.

c. Near-shore beach control point.

d. Far-shore beach control point.

e. Discharge control point.

## Link-up Operations

0619. A linkup is a meeting of friendly ground forces, which occurs in a variety of circumstances. It happens when an advancing armor force reaches an objective area previously seized by other maneuver forces; when an encircled armor units breaks out to rejoin friendly forces or a force comes to the relief of an encircled force; and when converging maneuver forces meet. Both forces may be moving toward each other, or one may be stationary. Whenever possible, joining forces exchange as much information as possible before starting an operation. The headquarters ordering the linkup establishes:

a. A common operational picture.

b. Command relationship and responsibilities of each force before, during, and after linkup.

c. Coordination of fire support before, during, and after linkup, including control measures.

d. Linkup method.

e. Recognition signals and communication procedures to use, including pyrotechnics, armbands, vehicle markings, gun-tube orientation, panels, colored smoke, lights, and challenge and passwords.

f. Operations to conduct following linkup.

## Operations on Forest or Wooded areas

0620. Limitations restrictions on use of armor.

a. **Vegetation.** Densely forested areas, where tree trunks are close and heavy above-ground root systems exist, may make movement difficult. Thick stands of bamboo may slow or even stop tanks.

b. **Topography.** Many jungles exist in rugged mountainous areas which will impede tanks. Swamps, coastal river basins, and other areas intersected by waterways pose obstacles to tanks because of their soft soil and frequent deep channels.

c. **Weather.** Rainy seasons (monsoons) cause rivers and streams to rise and become unfold able. Heavy rains may also cause damage to roads.

0621. All of these factors limit but do not exclude the use of armor. An armor commander can overcome those limitations by using these techniques:

a. Know the terrain. Know where tanks can travel.

b. Avoid areas which are obviously impassable.

c. Know the weather. What were recent conditions? What is expected? What effect will these conditions have on the use of tanks?

d. Move dismounted infantry in front of tanks on unfamiliar and heavily vegetated terrain. They can check and verify conditions, act as guides, and provide security.

## Operations at Defiles

0622. A defile is a natural or man-made feature or obstacle which restricts traffic flow to one-way at a given time. Examples are narrow bridges or causeway, damaged stretch of roads, excessive debris, a path through from a woods or narrow valley, or a street on which structures restrict movements to one-way traffic. Since movement at the defile is restricted, it is an ideal opportunity for the enemy to attempt to disrupt armor operations.

0623. Controls used or prepared for use at a defile will include at least two of the following independent techniques:

a. Visual-the use of military police to observe the entire defile

b. Telephones should be used whenever possible between the ends of the defile and the turnouts. Defile control frequently generates a large volume of communication which, if radio were used exclusively, would become a source of intelligence to the enemy and would overload the military police radio net

c. Flag method

d. Rider method

e. Lead and/or follow method

f. Radio located at the ends and all turn- outs in the defile.

## Operation on Fortified Position

0624. Assaulting forces invariably encounter some type of fortified positions prepared by the defending force. These can range from field-expedient, hasty positions produced with locally available materials, to elaborate steel and concrete emplacements complete with turrets, underground tunnels, and crew quarters. Field-expedient positions are those most often encountered. Armor commander may have fortified weapons emplacements or bunkers, protected shelters, reinforced natural or constructed caves, entrenchments, and other obstacles.

## Night and Limited Visibility Operations

0625. Darkness during nighttime operations is just one condition that reduces the soldier's ability to see. On the battlefield, there are conditions other than darkness that limit visibility. The dust and smoke of combat often obscure parts of the battlefield. Rain, fog, smoke, and natural dust can also limit visibility. This section discusses night operations, but the techniques also apply in times when visibility is reduced by other conditions.

0626. Combat during night and conditions of limited visibility is an integral part of all armor operations. It offers excellent opportunities for achieving deception and surprise. Basically, night combat or limited visibility operations are conducted in the same manner as daylight operations.

## Operation in Urban Terrain

0627. Armor units may be deployed in a variety of battlefield scenarios. As such the ability of these units to fight in urban terrain is very critical to the success of any type of operations.

0628. Armor units will not normally fight as a lone force within built-up areas. Armor is employed as part of a combined arms force when committed to the seizure of built-up areas. The fight for urban terrain begins outside the built-up areas. The armor unit can be used to isolate the area and provide direct fire support for the assault force to gain a foothold.

0629. Armor units isolate the built up areas by enveloping it and seizing key terrain that dominates approach and escape routes. The isolation serves to prevent enemy reinforcement and/or contain the enemy within the built-up area.

0630. Because of the nature of the terrain, offensive operations in built-up areas are normally conducted by dismounted elements. Combat is characterized by house-to-house fighting; restricted observation, fields of fire, and maneuver space for armored vehicles; and difficulty in command and control.

0631. An attack of a built-up area comprises three phases namely: isolation of the BUA, gaining a foothold at the edge of the BUA and systematic clearance and seizure of objectives.

0632. In certain situations, armor units may be tasked to establish defensive positions in urban terrain. If terrain permits, armor establishes defenses as far forward of a built-up area as possible.

0633. Armor units are employed in the defense as part of an organized battle position. A small, well organized, determined force defending in a built-up area can hold off a much larger attacker for longer periods of time. Strongly constructed cities give the defender a decided advantage. Each building or group of buildings is a

potential strongpoint. When reconnoitering for covered routes of advance and withdrawal, armor unit commanders should not overlook the possibility of moving through ground floor lobbies and corridors of the larger buildings. This type of route requires careful marking, but has the advantage of being largely concealed from aerial observation.

0634. The urban defense may force the enemy into planned engagement areas that are covered by anti-armor or enemy battle positions. Forward battle positions or covering forces may be employed to disorganize and confuse the enemy as to the main defenses. Behind the battle positions, defenses are set up to protect the friendly support element and stop the deepest penetration of the attacking force. Armor may be located within this defense, on the flanks in battle positions, or as a mobile counterattack force. Armor units are employed in the urban defense to block the penetration of the enemy and to protect friendly logistics elements. The keys to success in this defense are surprise, effective use of terrain, protection, and coordinated massing of fires.

# Support to Civil Authorities Operations

0635. SCA operations may occur independently; however, more often they complement offensive, defensive, and stability operations. Many offensive, defensive, and stability operations often require complementary support operations before, during or after execution. These may range from transporting, feeding, and providing shelter for displace persons to providing medical care for people and livestock.

## Purpose

0636. Support operations are generally conducted to assist civil authorities in response to emergencies (natural or man-made disasters) or specified illegal activities, and to relieve or reduce suffering. Support operations meet the immediate needs of civil authorities or designated groups for a limited time until they can accomplish these tasks without military assistance. In extreme or exceptional cases, Army forces may provide relief or assistance directly to those in need.

## Types

0637. There are two (2) types of support operations that exist namely: Domestic Support Operations (DSO) and Foreign Humanitarian Assistance (FHA).

0638. **Domestic Support Operations (DSO).** DSOs are usually conducted after the president declares an emergency. However, a commander may also conduct them following a finding by the Secretary of National Defense, a general officer or an equivalent civilian or as an immediate response to an emergency under imminently serious conditions. Domestics support operations supplement the efforts of civil governments and voluntary organizations.

0639. **Foreign Humanitarian Assistance (FHA).** FHA operations apply Army forces to supplement the efforts of the host-nation civil authorities or agencies by conducting activities limited in scope and duration. FHA relieves or reduces the impact of natural or man-made disasters or other endemic conditions, such as human pain, disease, hunger or privation that might seriously threaten life or result in great damage to or loss of property.

0640. Army forces participate in FHA operations that may be unilateral, multinational, or coordinated by an international organization such as the United Nations.

## Forms of Support to Civil Authority Operations

0641. During DSO, Army forces perform relief operations, support to civil law enforcement, and community assistance. In FHA, Army forces *most* often conduct relief operations; however, FHA may also involve support to incidents involving Chemical, Biological, Radiological, Nuclear, and high yield Explosive Consequence Management (CBRNE-CM) and community assistance. Army forces involved in support operations execute overlapping activities. For purposes of this manual, the discussion will focus on Domestic Support Operations in which Armor forces may be involved.

0642. **Relief Operations.** In case of a disaster, state, local, and host nation authorities are responsible for restoring essential services. Army forces execute similar actions during relief operations. Disaster relief focuses on recovery of critical infrastructure after a natural or manmade disaster. Humanitarian relief focuses on the well-being of supported populations. Both normally occur simultaneously.

a. **Disaster Relief.** Disaster relief restores or recreates essential infrastructure. It includes establishing and maintaining the minimum safe working conditions, less security measures, necessary to protect relief workers and the affected population. Armor forces together with other units under a Task Force are tasked to conduct search, rescue, evacuation, and disaster relief operations throughout the country in the event of calamities. Some specific tasks of Armor units in these calamities are, but not limited to, the following: 1) Casualty evacuation 2) Transportation assistance 3) Towing assistance to stranded or mired vehicles

b. **Humanitarian Relief.** Humanitarian relief focuses on lifesaving measures that alleviate the immediate needs of a population in crisis. It often includes providing medical support, food, water, medicine, clothing, blankets, shelter, and heating or cooking fuels. In some cases, Armor vehicles are used in transporting affected people from a disaster area. Civilian relief agencies, governmental and nongovernmental, are best suited to provide humanitarian relief. Armor forces and other Army units conducting humanitarian relief usually facilitate civil relief efforts.

0643. **Support to Civil Law Enforcement.** Support to domestic civil law enforcement involves activities related to counterterrorism, counter-drug operations, and military assistance during civil disturbances, and general support. Army support involves providing resources, training, or augmentation. Military forces remain under the military chain of command while supporting civil law enforcement. The supported law enforcement agency coordinates Army force activities under appropriate civil laws and interagency agreements.

0644. Armor units do not perform law enforcement duties without authority from the national government. In the event of their assignment to law enforcement missions, Armor units are deployed as part of combined arms team with other forces. Armor units execute armor-related missions and tasks in support to the overall mission. Armor forces are employed by the task force in the conduct of the following law enforcement operations:

a. **Support to Counterterrorism.** Armor forces may provide transportation, equipment, and personnel. When terrorists pose an imminent threat, Armor forces may be used to counter it. The demonstrated capability to conduct these operations helps keep the territory from becoming a target.

b. **Civil Disturbance Operations.** The Army assists civil authorities in restoring law and order when state and local law enforcement agencies cannot control civil disturbances. Armor forces are often tasked to be the last line of defense when the mob becomes hostile and when armed groups join the melee to intentionally attack anti-riot forces, other civilians, or destroy public and private property. Armor forces are deployed in the immediate vicinity of the camp, government building, of the crowd's target area securing critical entry points. They are also utilized in a show of force role to demonstrate the state's resolve in stabilizing the situation.

c. **General Support.** The military may share information and provide equipment, facilities, and other services. Department of National Defense (DND) may direct Army forces to provide training to state, and local civilian law enforcement agencies. This training may include operation and maintenance of military equipment.

0645. **Community Assistance.** Community assistance is a broad range of activities that provide support and maintain a strong connection between the military and civilian communities. Community assistance activities provide effective means of projecting a positive military image, providing training opportunities, and enhancing the relationship between the Army and the public. They should fulfill community needs that would not otherwise be met.

0646. Community assistance activities can enhance individual and unit combat readiness. Projects should exercise individual soldier skills, encourage teamwork, and challenge leader's planning and coordination skills. They should result in measurable accomplishments and increase soldier proficiency.

# Considerations for Support Operations

0647. Although each support operation is different, the visualization process, military decision making process, and troop leading procedures apply. The following considerations supplement those processes and can help commanders develop tailored concepts for support operations.

0648. **Provide Essential Support to the Largest Number of People**. The principle commanders use to prioritize missions and allocate support is, essential support to the largest number. Commanders allocate finite resources to achieve the greatest possible good. Initial efforts usually focus on restoring vital services: food and water distribution, medical aid, power generation, search and rescue, firefighting, and community relations. It may be necessary to complete a lower priority task before accomplishing a higher priority one. For example, Army forces may have to restore limited electrical services before restoring hospital emergency rooms and shelter operations.

0649. Armor Commanders assess requirements to determine how and where to apply limited assets to benefit the most people. In some cases, war-fighting reconnaissance capabilities and techniques can be used. For example, unmanned aerial vehicles (UAV) can survey relief routes and locate civilian refugee groups. Civil affairs or dedicated disaster assessment teams, as well as interagency, host nation, and nongovernmental organization sources, can reinforce and supplement standard information collection methods. Combining traditional and nontraditional means of collecting information allows commanders to obtain a clear understanding of the situation and adjust plans accordingly.

0650. **Coordinate Actions with Other Agencies.** DSO and FHA operations are typically joint and interagency; FHA operations are also multinational. The potential for duplicating effort and working at cross-purposes is high. Unity of effort requires, at least, a common understanding of purposes and direction among all agencies. Ensuring unity of effort and efficient resource use requires constant coordination. In FHA operations, Army forces enhance unity of effort by establishing a civil-military operations center. In DSO, they provide liaison elements, planning support, advisors, and technical experts to the lead agency. Through these contacts, commanders determine where their objectives and plans complement or conflict with those of other agencies. Each participant's capabilities will be in constant demand.

0651. **Establish Measures of Effectiveness.** With supported agencies and governments, commanders establish measures of effectiveness to gauge mission accomplishment. Measures of effectiveness focus on the condition and activity of those being supported. Those that are discrete, measurable, and link cause and effect help commanders understand and measure progress and success. In famine relief, for example, it may be tempting to measure effectiveness by the gross amount of food delivered. In some cases, this may be an acceptable gauge. However, a better one may be the total nourishment delivered, as measured by the total number of calories delivered per person per day, or the rate of decline of deaths directly attributable to

starvation. Measures of effectiveness depend on the situation and require readjustment as the situation and guidance change.

0652. **Handover to Civilian Agencies As Soon As Feasible.** The timing and feasibility of transferring control from military to civil authorities depends on mission-specific considerations. The two most important are the ability of civil authorities to resume operations without armor assistance and the need to commit armor forces to other operations. Armor commanders identify and include other civil considerations as early in the planning process as possible. They continually consider the long-term goals of the civil leadership and communities they assist. While the immediate goal of support operations is relieving hardship and suffering, the ultimate goal is creating conditions necessary for civil follow-on operations. Transferring activities to civil authorities and withdrawing armor forces are positive signals to the supported population and the Army. It indicates that the community has recovered enough for civil agencies to resume control that life is returning to normal and that the Army has successfully completed its support mission.

# Chapter VII
# Armor Maneuver Support

0701. Armor Maneuver Support (AMS) encompasses critical combat functions provided by units and soldiers, in conjunction with Armor and other maneuver units and soldiers, to secure victory. Those functions include Field Artillery, Air Defense Artillery, Army Aviation, Combat Engineers, Army Intelligence, Army Signal, and CMO units. Refer to other PAM for the complete details and operations of the different Armor Maneuver Support units.

## Types of Armor Maneuver Support

### Field Artillery

0702. Field Artillery (FA) units provide indirect-fire support to Armor and other maneuver forces. Towed howitzers and Self-propelled Guns (SPG) are the most common artillery pieces used by FA units in delivering indirect fires. Field artillery can neutralize, suppress, or destroy enemy direct-fire forces, and attack enemy mortars to isolate and interdict enemy forces or protect friendly operations. The Armor Commander may use artillery fires to cover key terrain, flanks, obstacles, and dead space to reduce his risk when maneuver forces are not available. Ideally, the self-propelled mission of artillery can best support the armor operations.

0703. In Armor operations, usually combined arms, FA units attack the enemy throughout the depth of his formation to facilitate the movement of armored vehicles and other maneuver units. Artillery fires can provide simultaneous strikes to targets at long ranges thereby reducing the risk to both the armored vehicles and ground forces. As mobile as the maneuver force it supports, FA systems provide continuous fires in support of the commander’s scheme of maneuver.

### Air Defense Artillery

0704. Armor forces require air defense augmentation in order to preserve its capabilities. The presence of Armor units with its armored vehicles in predominantly dismounted operational area makes it easily identifiable from the air and a high-value target for enemy attack. It is in this aspect that Air Defense Artillery (ADA) units play a vital role. These units provide force protection against air and missile threats. ADA units protect Armor and other maneuver forces and their vital assets. ADA units provide short-range air defense (SHORAD) protection for units conducting tactical combat operations.

0705. Air defense artillery units also contribute to intelligence and information operations by gathering and disseminating information about the enemy. They also deny the enemy the ability to reconnaissance and gather information about friendly forces.

## Army Aviation

0706. Army Aviation units provide maneuver support to commanders of Armor units and other maneuver forces in the following areas:

a. C2, communications, and intelligence

b. Real-time reconnaissance and surveillance for use during Armor operations and other tactical missions.

c. Provide security operations and air cover to Armor units and other maneuver forces during troop movement.

d. Target acquisition for artillery providing indirect fire support to Armor and other maneuver units to facilitate movement of armored vehicles and ground forces.

e. Provide lift capability.

f. Provide close combat attack to armor units in offensive, defensive and other armor tactical operations.

## Combat Engineers

0707. Combat Engineers multiply the effectiveness of friendly forces on a lethal battlefield. Four of the five primary engineer functions are maneuver support. Combat engineers operate as an integral member of the combined arms force to provide a full range of mobility, counter mobility, survivability, and topographic capabilities. Engineers advise the Armor unit leaders and other maneuver ground commanders on the effective use of terrain; construct, improve, and maintain routes, bridges, and airfields; and reorganize to fight as infantry when required.

0708. In offensive operations, combat engineers concentrate their efforts to support Armor units and other maneuver forces by breaching and crossing obstacles, assisting in the assault of fortified positions, and emplacing obstacles to protect the flanks of friendly attacking forces.

0709. In defensive operations, engineers reinforce the terrain to anchor the defense in critical areas, maximize the effects of the defenders' fires, provide maximum protection to friendly fighting positions, and facilitate the movement of counterattack forces. Combat engineers also ensure that Armor units be placed in defensive positions that capitalize on its long-range firepower and mobility advantages.

## Army Intelligence

0710. Army Intelligence units provide Armor units and other maneuver forces commanders with early warnings of enemy intentions, intelligence preparation of the battlefield products, aid in the development of his situational understanding, and assist

in target development, force protection, and battle damage assessment. Intelligence units also assist in the intelligence gathering about all possible threat equipment against Armor, which usually includes anti-tank weapons, mines, and Improvised Explosive Devices (IEDs) that the enemies may have in their inventory. Intelligence units also participate in offensive information operations, as well as provide critical counterintelligence support to friendly command force protection programs.

## Army Signal

0711. Armor units do not have any command relationship with signal units. Armor commanders rely only to the personnel provided by the Signal Platoon organic to the Division. It is the Signal platoon that is responsible for establishing and operating the unit's radio and wire communication system. It also provides communications and electronic support for command and control of the Division.

0712. Other functions beyond the capability of the organic Signal Platoon shall be provided by the Army Signal Regiment.

## Civil Military Operations

0713. Similar to other maneuver units, CMO for Armor units includes three main pillars, namely: Civil Affairs, Public Affairs, and Information Support Affairs. CMO support the Armor units and other maneuver forces to establish legitimacy of operations, gain popular support, and destroy the morale of the enemies.

# Chapter VIII
# Sustainment Support to Armor Operations

0801. This chapter provides guidance to commanders and staffs of armor units below division level in sustainment support operations, including supply, sustainment support aspects of transportation, maintenance, and services in an active theater of operations. It also outlines command and staff sustainment support responsibilities.

0802. Discussion of Sustainment Support (SS) in this chapter refers to the brigade, battalion and company, unless otherwise specified. It is also applicable to the armored cavalry squadron or troop.

## Principles of Sustainment Support

0803. Sustainment Support is a part of administration that provides for and manages supply, medical and dental, transportation, and maintenance and service. Although procedures may vary at different echelons, the principles of sustainment support remain constant and are applicable to all forms of warfare. However, the mass destruction of personnel, supplies, and equipment that may result from nuclear attacks, and the mass casualties and contamination of supplies and equipment that may result from chemical and biological attacks requires that additional consideration be given to applying these principles. All echelons in the sustainment support structure adhere to four basic principles.

0804. **Sustainment Support is a Function of Command.** Armor Commanders at all echelons are responsible for the sustainment support of organic and attached elements of their commands. From the tank commander or squad leader upward, each commander assures himself that sustainment support is adequate by anticipating his requirements, making his needs known, and employing properly the sustainment support elements available to him. The logistics officer (S4) assists the commander in performing his sustainment support responsibility. To ensure an active, well-coordinated effort, the commander makes his policies and decisions known to his logistics officer.

0805. **The Impetus of Sustainment Support is From Rear to Front.** All sustainment support echelons must be impressed with the necessity for constantly pushing the sustainment support forward to the user to provide him the means to accomplish his mission. At all echelons, sustainment support agencies must be in reasonable reach of the organic transportation of lower units. This is especially important in armor operations, which normally extend over great distances.

0806. **Advanced Planning is Essential to Successful Sustainment Support.** The logistical plan is based upon and provides adequate and timely support to the tactical operation. It is complete, simple, and flexible. It must be prepared with foresight and coordinated with all appropriate staff officers. Sustainment support and tactical

planning are concurrent. The logistics officer advises the commander on all sustainment support matters pertaining to anticipated operations.

0807. **Reserves of Supplies are maintained at All Echelons.** Certain reserve supplies, including rations, ammunition, fuel and lubricants, and repair parts, are carried by all echelons. These reserves ensure that the combat units can continue to operate if the supply chain should be broken temporarily; they must be replenished when supplies become available.

0808. **Role of ASCOM to Armor Units.** Armor units can operate independently for a short period of time. In terms of logistics, the armor units and ASCOM have no particular relation. However, all the logistical requirements beyond the level of armor units are requested to higher headquarters and the logistical items will be provided by Army Support Command (ASCOM) through the nearest Forward Service Support Unit (FSSU).

## Logistical Characteristics of Armor Units

0809. Tank Battalions, Mechanized Infantry Battalions, Cavalry Squadrons, and separate troops are logistically capable of operating independently for limited periods of time. Companies of Tank Battalions and Troops of Cavalry Squadrons are dependent upon the parent unit for sustainment support.

0810. Headquarters and headquarter companies of the Mechanized Infantry Brigades, Battalions, Tank Battalions or Cavalry Squadron contain only the personnel and equipment required to establish and maintain the headquarters and the company logistically. These units do not have organic sustainment support elements required to support attached combat battalions or combat support units.

0811. Logistically, armor units are characterized by their large consumption of fuels, heavy expenditures of ammunition, and support requirements. The success of the armor unit depends on how well these requirements are satisfied.

0812. Armor units may be logistically supported by air for limited periods. Supplies and equipment are delivered by parachute, by low-level extraction, and by utility or cargo helicopter. Terrain and tactical situation permitting, low-level extraction is preferred due to accuracy of delivery.

## Sustainment Support Personnel

0813. At brigade and battalion level, the Armor Commander is provided a logistical staff officer (S4) and other key sustainment support to assist him. At company level, the Executive Officer acts as the logistic officer and is being assisted by trained enlisted men whose primary duties are concerned with sustainment support matters.

## Brigade Sustainment Support Personnel

0814. The brigade headquarters is primarily a tactical headquarters and normally enters sustainment support channels only to coordinate and establish priorities for critical items of supply. However, the armor commander must ensure adequate sustainment support for the subordinate elements. Higher headquarters attaches or places in support of the sustainment support means to support operations. The commander provides a staff to monitor the sustainment support situation, control and coordinate sustainment support elements, and recommend adjustment of sustainment support means, as necessary.

## Battalion or Cavalry Squadron Sustainment Support Personnel

0815. Within the division, the normal chain of sustainment support is direct from divisional support command elements to the battalion or squadron. The cavalry squadron of the normal chain of sustainment support comes from its supported units. The battalion commander is provided with a logistics staff officer (S4) and other key sustainment support personnel to assist him.

## Company Sustainment Support Personnel

0816. The key sustainment support personnel at the company level include supply, communication, and maintenance noncommissioned officers who perform or supervise the functions of supply or maintenance.

0817. Normally the company executive officer is responsible to the commander for the supervision of all sustainment support matters of the unit. This does not preclude the assignment of other unit officers as assistant supply or maintenance officers.

## Separate Company or Troop Sustainment Support Personnel

0818. The separate company-level armor units are organized to operate without battalion-level support. These units are capable of performing unit level maintenance. Like other companies, its executive officer is responsible to the commander for the supervision of all sustainment support matters of the unit.

# Armor Operations Classes of Supplies

0819. **Class I.** Class I is rations and gratuitous issue, health, morale and welfare items.

0820. **Class II.** Class II is clothing, individual equipment, bivouac equipment (tent), tool sets and administrative and housekeeping supplies and equipment.

0821. **Class III.** Class III is petroleum, oil and lubricants.

0822. **Class IV.** Class IV is construction materials.

0823. **Class V.** Class V is ammunition.

0824. **Class VI.** Class VI is personal demand items sold through Post Exchange.

0825. **Class VII.** Class VII is major end-items.

0826. **Class VIII.** Class VIII is medical supplies.

0827. **Class IX.** Class IX is repair parts and documents for equipment maintenance.

## Glossary

**Area of Influence**
Is a geographical area in which a commander can directly influence operations by maneuver or fire support systems normally under the commander's command or control.

**Area of Interest**
Is that area of concern to the commander, including the area of influence and areas adjacent to it. It extends into enemy territory and to the objectives of current or planned operations.

**Armor**
Is a combined arms organization composed of tank, cavalry and mechanized infantry. It can be organized into a tank unit or mechanized unit with cavalry.

**Army Doctrine**
Is a common language and a common understanding of how Army forces conduct operations. It is also concise expression of how Army forces contribute to unified action in campaigns, major operations, battles, and engagements. It touches all aspects of the Army. It facilitates communication among soldiers no matter where they serve, contributes to a shared professional culture, and serves as the basis for curricula in the Army Education System.

**Army Power**
Is the summation of the potency that any total Army force generates. It consists of eight warfighting functions tied together by leadership and enhanced by information.

**Battle Command**
Is the art and science of understanding, visualizing, describing, directing, leading, and assessing forces to impose the commander's will on a hostile, thinking, and adaptive enemy.

**Battle Position**
Is a general location and orientation of forces on the ground from which units defend. The BP can be used for units from battalion- to platoon-size.

**Civil Affairs**
Is a CMO pillar undertaken independent of, or in coordination with, civilian entities that promotes public trust through cooperation, collaboration, and the conduct of sociological and developmental activities to gain popular support.

**Civil Military Operations**
Civil Military Operations are planned activities undertaken independently or in coordination with civilian entities to gain popular support and weaken the will of the enemy to fight

**Close Air Support**
Close air support (CAS) the use of air assets to directly support the ground force. CAS is flown against targets that are in close proximity to friendly forces; that proximity requires detailed integration between CAS missions and the fire and movement of surface forces.

**Combined Arms**
Is the synchronized or simultaneous application of several Army-specific capabilities represented by military occupational specialties to achieve an effect on the enemy or on a situation that is greater than if each capability was used separately or in sequence.

**Command and Control (C2)**
It is the exercise of authority and direction by a properly designated commander over assigned and attached forces to accomplish the mission. Command and control functions are performed through an arrangement of personnel, equipment, communications, facilities and procedures employed by a commander in planning, directing, coordinating and controlling forces and operations in the accomplishment of the mission.

**Command**
Is the authority that a commander in the armed forces lawfully exercises over subordinates by virtue of rank or assignment. Command includes the authority and responsibility for effectively using available resources and for planning the employment of, organizing, directing, coordinating, and controlling military forces for the accomplishment of assigned missions.

**Commander's Intent**
Is a clear, concise statement of what the force must do and the conditions the force must establish with respect to the enemy, terrain, and civil considerations that represent the desired end state.

**Concept of Operations**
Is a statement that directs the manner in which subordinate units cooperate to accomplish the mission and establishes the sequence of actions the force will use to achieve the end state. It is normally expressed in terms of decisive, shaping, and sustaining operations.

**Conflict**
Is defined as a politico-military struggle short of prolonged conventional armed action between organized parties within a nation or between nations. It is often protracted and usually confined to a geographic region but may have regional or global implications.

**Contingency Operations**
Involve military forces to achieve operational objectives or protect national interests. They are usually in response to a sudden or short-notice crisis or emergency and occur across the scope of operations

**Contingency Planning** - is a type of planning activity conducted principally in peacetime to develop joint operation plans for contingencies identified in strategic planning documents.

**Control**
Is the regulation of forces and warfighting functions to accomplish the mission in accordance with the commander's intent.

**Counter-reconnaissance**
Is an inherent task, either active or passive in nature, conducted to thwart enemy reconnaissance and surveillance (R&S) efforts. Its purpose is to deny the enemy commander his eyes and ears, impeding his ability to determine the disposition of friendly forces

**Cover** - is a form of security operations that protects the force from surprise, develops the situation and gives commanders time and space in which to respond to enemy's actions

**Crisis**
Refers to any sudden or unforeseen situation that requires immediate action or the urgency of response. It is an event or series of events, which have direct strategic impact and may be regarded as a turning point, which will decisively determine an outcome.

**Decisive Operation**
Is the operation that directly accomplishes the mission. It determines the outcome of a major operation, battle, or engagement. It is the focal point around which commanders design the entire operation.

**Defensive Operations**
Are combat operations conducted to defeat an enemy attack, gain time, economize forces, and develop conditions favorable for offensive or support to civil governance operations.

**Depth**
Is the extension of operations in time, space, and resources. Commanders use depth to obtain space for effective maneuver, time to conduct operations, and resources to achieve and exploit success.

**Essential Elements of Friendly Information**
Are critical aspects of a friendly operation that, if known by the enemy, would subsequently compromise, lead to failure, or limit success of the operation, and therefore should be protected from enemy detection. EEFI

**Guard**
Is a form of security operations whose primary task is to protect the main body by fighting to gain time while also observing and reporting information and preventing enemy ground observation of and direct fire against the main body.

**Indirect Fire**
Fire delivered at a target that cannot be seen by the aimer.

**Initiative**
Is setting or dictating the terms of action throughout an operation.

**Insurgency**
Is a condition where an organized movement aims to overthrow a constituted government through the use of subversion and armed conflict. It is a condition of politically motivated conflict involving significant intra- or interstate violence but usually short of large-scale operations by opposing forces.

**Intelligence**
In broad terms, intelligence is knowledge about an enemy, adversary or competitor that is not apparent or evident from the public record or by overt observation. Intelligence activities are designed to elicit information that an enemy, adversary or competitor would prefer not to disclose or to have discovered.

**Intelligence Preparation of the Battlefield** (IPB)
Is a four-step, systematic, continuous process of analyzing the threat and environment in a specific geographic area. It is designed to support staff estimates and military decision making.

**Link-up Point**
The point where two infiltrating elements in the same or different infiltration lanes are scheduled to meet to consolidate before proceeding on with their missions

**Main Effort**
Is the designated subordinate unit whose mission at a given point in time is most critical to overall mission success. It is usually weighted with the preponderance of Army power.

**Mass**
Is to bring the fire of two or more ships on the same target.
**Military** Decision **Making Process -** is a single, established and proven analytical process. It helps the commander and his staffs examine a battlefield situation and reach a logical decision.

**Movement to contact** (MTC)
Is an offensive operation designed to gain contact with the enemy.

**Neutralization Fire**
Fire that is delivered to hamper and interrupt movement and/or the firing of weapons. Destruction of enemy personnel and weapons is not the primary consideration.

**Offensive Operations**
Are combat operations conducted to defeat and destroy enemy forces and seize terrain, resources, and population centers.

**Operational Control (OPCON)**
Transferable command authority that may be exercised by commanders at any echelon at or below the level of combatant command. It may be delegated and is the authority to perform those functions of command over subordinate forces involving organizing and employing commands and forces, assigning tasks, designating objectives, and giving authoritative direction necessary to accomplish the mission.

**Phase**
Is a specific part of an operation that is different from those that precede or follow. A change in phase usually involves a change of task. Phasing assists in planning and controlling.

**Phase Line**
A line utilized for control and coordination of military operations, usually an easily identified feature in the operational area

**Plan**
Is a design for a future or anticipated operation. This continuous, evolving frame work of anticipated actions guides subordinates through each phase of the operation. A plan is a frame work from which to adapt rather than a script to follow.

**Planning**
Is the process by which commanders (and the staff, if available) translate the commander's visualization into a specific course of action for preparation and execution, focusing on the expected results.

**Planning Guidance**
Is a broad description of when, where, and how the commander intends to employ Army power to accomplish the mission within the higher commander's intent.

**Priority Intelligence Requirements**
Are intelligence requirements stated as a priority for intelligence support that the commander and staff need to understand the adversary or the operational environment.

**Public Affairs**
Is the CMO pillar that conducts information-related activities directed at the general public to ensure clear, accurate, and timely dissemination of information.

**Rear Area**
Is provided primarily for the performance of support functions and where the majority of the echelon's sustaining operations occur. Operations in rear areas assure freedom of action and continuity of operations, sustainment, and C2.

**Reconnaissance**
Reconnaissance complements surveillance in obtaining, by visual observation or other detection methods, specific information about the activities and resources of an enemy

or potential enemy; or in securing data concerning the meteorological, hydrographic, or geographic characteristics of a particular area.

**Retrograde operations**
Are movements to the rear or away from the enemy. The movement may be forced or voluntary, but it must be with the higher commander's approval.

**Rules of Engagement**
Rules of Engagement (ROE) are directives to military forces and individuals that define the circumstances, conditions, degree and manner in which the use of force or other action may or may not be applied.

**Screen**
Is a form of security operations that primarily provides early warning to the protected force

**Search and Attack**
Technique for conducting a movement to contact that shares many of the characteristics of an area security mission

**Search and Rescue**
Search and Rescue (SAR) is the use of aircraft, surface craft, submarines, specialized rescue teams and equipment to search for and rescue personnel in distress on land or at sea.

**Security operations**
Are those operations undertaken by a commander to provide early and accurate warning of enemy operations to provide the force being protected with time and maneuver space within which to react to the enemy, and to develop the situation to allow the commander to effectively use the protected force.

**Shaping Operation**
Is an operation at any echelon that creates and preserves conditions for the success of the decisive operation. Shaping operations establish conditions for the decisive operation through effects on the enemy, population (including local leaders), and terrain.

**Support by Fire**
A tactical mission task in which a maneuver force moves to a position where it can engage the enemy by direct fire in support of another maneuvering force.

**Surveillance**
Surveillance is the function of systematically observing air, surface, or subsurface areas, places, persons, or things, by visual, aural, electronic, photographic, or other means.

**Sustaining Operation**

Is an operation at any echelon that enables the decisive operation or shaping operations by generating and maintaining Army power. Sustaining operations differ from decisive and shaping operations in that they are focused internally (on friendly forces) rather than externally (on the enemy or environment).

**War**

Declared and undeclared, is defined as sustained arm conflict between nations or organized parties within a nation. It involves conventional, unconventional, or, in advanced countries, nuclear/chemical armed action in a series of connected battles and campaigns.

**Abbreviation and Acronym List**

| | | |
|---|---|---|
| **A/L** | - | Administration/Logistics |
| **ADA** | - | Air Defense Artillery |
| **AFP** | - | Armed Forces of the Philippines |
| **AIFV** | - | Armored Infantry Fighting Vehicle |
| **AMS** | - | Amor Maneuver Support |
| **AO** | - | Area of Operations |
| **AOR** | - | Area of Responsibility |
| **AV** | - | Armored Vehicle |
| **BMO** | - | Battalion Maintenance Officer |
| **BN** | - | Battalion |
| **BP** | - | Battle Position |
| **BSA** | - | Brigade Support Area |
| **BUA** | - | Build Up Area |
| **C2** | - | Command and Control |
| **C3** | - | Command, Control, and Communications |
| **CA** | - | Civil Affairs |
| **CBRNE-CM** | - | Chemical, Biological, Radiological, Nuclear, and high yield Explosive Consequence Management |
| **CCIR** | - | Commanders Critical Information Requirements |
| **CCP** | - | Communication Checkpoints |
| **COA** | - | Courses of Action |
| **COIN** | - | Counter Insurgency |
| **COP** | - | Common Operational Picture |
| **CP** | - | Command Post |
| **CSM** | - | Command Sergeant Major |
| **CTCP** | - | Combat Trains Command Post |
| **CVC** | - | Combat Vehicle Crewman |
| **DND** | - | Department of National Defense |
| **DP** | - | Decision Point |
| **DSO** | - | Domestic Support Operations |
| **DST** | - | Decision Support Template |
| **EEFI** | - | Essential Elements of Friendly Information |
| **EPW** | - | Enemy Prisoner of War |
| **FA** | - | Field Artillery |
| **FARP** | - | Forward Arming and Refueling Points |
| **FDC** | - | Fire Direction Center |
| **FHA** | - | Foreign Humanitarian Assistance |
| **FRAGO** | - | Fragmentary Order |
| **FS** | - | Fire Support |
| **FSCOORD** | - | Fire Support Coordinator |
| **FSE** | - | Fire Support Element |
| **G1/S1** | - | Personnel Officer |
| **G2/S2** | - | Intelligence Officer |
| **G3/S3** | - | Operations Officer |
| **G4/S4** | - | Supply Officer |
| **HHC** | - | Headquarters and Headquarters Company |
| **IO** | - | Information Operation |
| **IPB** | - | Intelligence Preparation of the Battlefield |
| **ISO** | - | Internal Security Operations |

| | | |
|---|---|---|
| **ISR** | - | Intelligence, Surveillance and Reconnaissance |
| **LD** | - | Line of Departure |
| **LNO** | - | Liaison Officer |
| **LOA** | - | Line of Advance |
| **LOC** | - | Lines of Communication |
| **LOGPAC** | - | Logistics Package |
| **M/S** | - | Mobility and Survivability |
| **MBG** | - | Misuari Breakaway Group |
| **MDMP** | | -Military Decision Making Process |
| **METT-T** | - | Mission, Enemy, Troops, Terrain and weather, and Time available |
| **METT-TC** | - | Mission, Enemy, Troops, Terrain and weather, Time available and Civil Consideration |
| **MOPP** | - | Mission-Oriented Protective Posture |
| **NGO** | - | Non-Governmental Organization |
| **NLT** | - | No Later Than |
| **NRT** | - | Near-Red-Time |
| **OP** | - | Observation Post |
| **OPCON** | - | Operational Control |
| **OPLAN** | - | Operation Plan |
| **OPORD** | - | Operation Order |
| **OPSEC** | - | Operations Security |
| **PEO** | - | Peace Enforcement Operation |
| **POL** | - | Petroleum, Oil and Lubricants |
| **QRF** | - | Quick Reaction Force |
| **R&S** | - | Reconnaissance and Surveillance |
| **ROM** | - | Refuel on the Move |
| **SFV** | - | Simba Fighting Vehicle |
| **SHORAD** | - | Short-Range Air Defense |
| **SITTEMP** | - | Situation Template |
| **SOF** | - | Special Operations Forces |
| **SOI** | - | Signal Operation Instructions |
| **SOP** | - | Standard Operating Procedure |
| **TAC** | - | Tactical |
| **TCF** | - | Tactical Combat Forces |
| **TF** | - | Task Force |
| **TOC** | - | Tactical Operations Center |
| **TOE** | - | Table of Organization and Equipment |
| **UAV** | - | Unmanned Aerial Vehicles |
| **UMCP** | - | Unit Maintenance Collection Point |
| **VIP** | - | Very Important Person |
| **WMD** | - | Weapons of Mass Destruction |